Call: HORIZON-CL4-2021-RESILIENCE-01

(A DIGITISED, RESOURCE-EFFICIENT AND RESILIENT INDUSTRY 2021)

Topic: HORIZON-CL4-2021-RESILIENCE-01-12

Type of Action: HORIZON-RIA

Proposal number: 101058699

Proposal acronym: FreeMe

Type of Model Grant Agreement: HORIZON Action Grant Budget-Based

Table of contents

Section	Title	Action
1	General information	
2	Participants	
3	Budget	
4	Ethics and security	

Proposal ID **101058699**

Acronym **FreeMe**

1 - General information

Fields marked * are mandatory to fill.

Topic	HORIZON-CL4-2021-RESILIENCE-01-12	Type of Action	HORIZON-RIA
Call	HORIZON-CL4-2021-RESILIENCE-01	Type of Model Grant Agreement	HORIZON-AG

Acronym FreeMe

Proposal title Toxic Free metallization process for plastic surfaces

Note that for technical reasons, the following characters are not accepted in the Proposal Title and will be removed: < > " &

Duration in months 48

Fixed keyword 1 Coating and films

Free keywords *Plating on Plastics, Etching, Activation, Metallization, Bio-based resins, adhesion*

Abstract *

Technological advancement has significantly transformed plating methods by introducing metal coatings on plastic substrates. Plating on plastics (PoP) has emerged as an effective solution in a wide range of applications in the automotive, aerospace and home appliances industry. There is a high preference for plating on plastic in the automotive industry, owing to its light weight, cost effectiveness and corrosion resistance. In the automotive industry, it not only adds aesthetic value but also helps in increasing fuel efficiency by reducing the overall vehicle weight.
Decorative chrome, copper and nickel plating is quite popular on various interior and exterior plastic parts such as knobs, handles, logos, emblems, trims, etc. ABS, PC, PET, Nylon, and PEEK are widely used polymers in plastic plating.
However, the current processes for PoP present some significant disadvantages related to environmental sustainability, occupational safety and quality of coatings i.e the use of hexevalent chromium in the etching process and the use of Palladium in the activation step.
FreeMe project proposes two revolutionaries approaches for the metallization of polymeric (plastic or resins) surfaces without the use of hexavalent chromium and palladium.
The first technology is based on the use of resins (e.g. epoxy-acrylate resins) with the additions of suitable nickel based precursors. This composite resin can be deposited uniformly in the surface of the plastic object either by spray deposition methods or other simple deposition methods. The nickel precursors are afterwards reduced utilizing appropriate reduction agents (e.g. NaBH4) creating the necessary nuclei for the next step of electroless deposition for the formation of the conductive layer.

Remaining characters 235

Has this proposal (or a very similar one) been submitted in the past 2 years in response to a call for proposals under any EU programme, including the current call? ○ Yes ◉ No

Please give the proposal reference or contract number.

Administrative forms

Proposal ID **101058699**

Acronym **FreeMe**

Declarations

Field(s) marked * are mandatory to fill.

1) We declare to have the explicit consent of all applicants on their participation and on the content of this proposal. * ☒

2) We confirm that the information contained in this proposal is correct and complete and that none of the project activities have started before the proposal was submitted (unless explicitly authorised in the call conditions). ☒

3) We declare:
- to be fully compliant with the eligibility criteria set out in the call
- not to be subject to any exclusion grounds under the EU Financial Regulation 2018/1046
- to have the financial and operational capacity to carry out the proposed project.

☒

4) We acknowledge that all communication will be made through the Funding & Tenders Portal electronic exchange system and that access and use of this system is subject to the Funding & Tenders Portal Terms and Conditions. ☒

5) We have read, understood and accepted the Funding & Tenders Portal Terms & Conditions and Privacy Statement that set out the conditions of use of the Portal and the scope, purposes, retention periods, etc. for the processing of personal data of all data subjects whose data we communicate for the purpose of the application, evaluation, award and subsequent management of our grant, prizes and contracts (including financial transactions and audits). ☒

6) We declare that the proposal complies with ethical principles (including the highest standards of research integrity as set out in the ALLEA European Code of Conduct for Research Integrity, as well as applicable international and national law, including the Charter of Fundamental Rights of the European Union and the European Convention on Human Rights and its Supplementary Protocols. Appropriate procedures, policies and structures are in place to foster responsible research practices, to prevent questionable research practices and research misconduct, and to handle allegations of breaches of the principles and standards in the Code of Conduct. ☒

7) We declare that the proposal has an exclusive focus on civil applications (activities intended to be used in military application or aiming to serve military purposes cannot be funded). If the project involves dual-use items in the sense of Regulation 428/2009, or other items for which authorisation is required, we confirm that we will comply with the applicable regulatory framework (e.g. obtain export/import licences before these items are used). ☒

8) We confirm that the activities proposed do not
- aim at human cloning for reproductive purposes;
- intend to modify the genetic heritage of human beings which could make such changes heritable (with the exception of research relating to cancer treatment of the gonads, which may be financed), or
- intend to create human embryos solely for the purpose of research or for the purpose of stem cell procurement, including by means of somatic cell nuclear transfer.
- lead to the destruction of human embryos (for example, for obtaining stem cells)

These activities are excluded from funding. ☒

9) We confirm that for activities carried out outside the Union, the same activities would have been allowed in at least one EU Member State. ☒

The coordinator is only responsible for the information relating to their own organisation. Each applicant remains responsible for the information declared for their organisation. If the proposal is retained for EU funding, they will all be required to sign a declaration of honour.

False statements or incorrect information may lead to administrative sanctions under the EU Financial Regulation.

2 - Participants

List of participating organisations

#	Participating Organisation Legal Name	Country	Action
1	GASER OSSIDO DURO SRL	IT	
2	POLITECNICO DI MILANO	IT	
3	CREATIVE NANO PC	EL	
4	ARISTOTELIO PANEPISTIMIO THESSALONIKIS	EL	
5	IDENER RESEARCH & DEVELOPMENT AGRUPACION DE INTERES ECONOMICO	ES	
6	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	
7	UNIVERSIDAD DE BURGOS	ES	
8	EXELISIS IKE	EL	
9	INNOVACION Y CONSULTING TECNOLOGICO DE LEVANTE, S.L.	ES	
10	CENTRO RICERCHE FIAT SCPA	IT	
11	ARCELIK A.S.	TR	
12	ISRAEL AEROSPACE INDUSTRIES LTD.	IL	

Organisation data

PIC	Legal name
952145793	*GASER OSSIDO DURO SRL*

Short name: GASER OSSIDO DURO SRL

Address

Street	VIA PO 0027
Town	ROZZANO
Postcode	20089
Country	Italy
Webpage	www.gruppogaser.com

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2013	- yes
SME self-assessment	31/12/2013	- yes
SME validation	unknown	

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Enrico**

Last name* **Galliani**

E-Mail* **enrico.galliani@gruppogaser.com**

Position in org. General Manager

Department GASER OSSIDO DURO SRL

☒ Same as organisation name

☒ Same as proposing organisation's address

Street VIA PO 0027

Town ROZZANO

Post code 20089

Country Italy

Website *Please enter website*

Phone *+xxx xxxxxxxxx*

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Gianni Luca	Franzosi	gianni.franzosi@gruppogaser.com	+xxx xxxxxxxxx
Alessio	Carminati	alessio.carminati@gruppogaser.com	+xxx xxxxxxxxx
Ilias	Gkotsis	ilias.g@exelisis.gr	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mr	Alessio	Carminati	Man	Italy	alessio.carminati@gruppogaser.com	Category A Top grad	Leading		

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Other achievement	*Gaser Ossido Duro has certified its production according to Certification NSF 51, Certification ISO 9001, Certification ISO 14001 and Certification IATF 16949*
Service	*Gaser Ossido Duro is involved in the development of innovative surface treatment as nanoXHAM that increases surface anticorrosive resistance and avoids the adherence of organic and inorganic substances to the surface.*
Service	*Duralti surface treatment for aluminum and its alloys. Provides maximum protection against corrosion, wear, non-stick and friction.*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
PureNano	*The merit of PureNano novel method it is based in the timely treatment of plating baths with functionalized magnetic nanoparticles that can absorb in their surface the contaminants and chemical species that are accumulated in the bath.Gaser is responsible for delivering all the necessary specifications for the pilot scale process purification. Following, the company is actively involved in the preliminary work that has to be conducted for the design and manufacturing of the process.*
E-WASTE ? the smart cycle	*Gaser was responsible for bulding a pilot plant according to the directives imposed by the reference legislation. The activities carried out in Gaser aimed at transferring lab research developed by the research partners in a semi-industrial pilot plant capable of validating the results obtained in the recycling of electronic boards for the recovery of the metals contained.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Premises	*Several pilot lines and production lines for plating processes including plating on plastic. Full equipped characterization laboratory wih state of the art characterization methods for testing adhesion, corrosion, mechanical properties, structure. Waste treatment systems.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
999879881	*POLITECNICO DI MILANO*

Short name: POLIMI

Address

Street	PIAZZA LEONARDO DA VINCI 32
Town	MILANO
Postcode	20133
Country	Italy
Webpage	www.polimi.it

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	10/05/2016	- no
SME self-assessment	unknown	
SME validation	25/02/2009	- no

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Prof.

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Luca**

Last name* **Magagning**

E-Mail* **luca.magagnin@polimi.it**

Position in org. Professor

Department Dipartimento di Chimica, Materiali e Ingegneria Chimica "Giulio Natta?

☐ Same as organisation name

☒ Same as proposing organisation's address

Street PIAZZA LEONARDO DA VINCI 32

Town MILANO

Post code 20133

Country Italy

Website www.polimi.it

Phone *+xxx xxxxxxxxx*

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Chrysavgi	Kostoula	chrysavgi.kostoula@polimi.it	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Prof	Luca	Magangin	Man	Italy	luca.magagnin@polimi.it	Category A Top grad	Leading	0000-0001-5553-6441	Orcid ID
Prof	Luca	Nobili	Man	Italy	luca.nobili@polimi.it	Category A Top grad	Team member	0000-0002-9300-2405	Orcid ID
Mr	Eugenio	Gibertini	Man	Italy	eugenio.gibertini@polimi.it	Category D First stag	Team member		
Dr	Chrysavgi	Kostoula	Woman	Greece	chrysavgi.kostoula@polimi.it	Category C Recognis	Team member	55897236200 36163245500	Researcher ID

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☐
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Wang, Qianzhi; Callisti, Mauro; Miranda, Alberto; McKay, Brian; Deligkiozi, Ioanna; Milickovic, Tatjana Kosanovic; Zoikis-Karathanasis, Alexandros; Hrissagis, Kostas; Magagnin, Luca; Polcar, Tomas. ?Evolution of structural, mechanical and tribological properties of Ni?P/ MWCNT coatings as a function of annealing temperature? Surface & coatings technology, 2016-09-25, Vol.302, p.195-201.*
Publication	*Sansotera, Maurizio; Baggioli, Alberto; Ieffa, Simona; Tedesco, Mariella; Sacchi, Benedetta; Bianchi, Claudia L; Navarrini, Michele; Migliori, Massimo; Magagnin, Luca; Navarrini, Walter. ? Catalytic microreactor with electrodeposited hierarchically nanostructured nickel coatings for gas-phase fluorination reactions? Journal of fluorine chemistry, 2018-01, Vol.205, p.22-29*
Publication	*Panzeri, Gabriele; Accogli, Alessandra; Gibertini, Eugenio; Varotto, Sara; Rinaldi, Christian; Nobili, Luca; Magagnin, Luca. ?Electrodeposition of cobalt thin films and nanowires from ethylene glycol-based solution?. Electrochemistry communications, 2019-06, Vol.103, p.31-36*
Publication	*Panzeri, Gabriele; Accogli, Alessandra; Gibertini, Eugenio; Rinaldi, Christian; Nobili, Luca; Magagnin, Luca. ?Electrodeposition of high-purity nanostructured iron films from Fe(II) and Fe(III) non-aqueous solutions based on ethylene glycol?. Electrochimica acta, 2018-05-01, Vol.271, p.576-581*
Other achievement	*Patent (US20170092396) AMPHIPHILIC MAGNETIC NANOPARTICLES AND AGGREGATES TO REMOVE HYDROCARBONS AND METAL IONS AND SYNTHESIS THEREOF*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
LightMe	*LightMe ecosystem targets to boost the introduction of new composite lightweight materials that provide new functionalities and features. PoliMi is the coordinator of the project and is responsible for the setup of the LightMe Non-Profit Organization (NPO) that will be established and operate as a Single Entry Point (SEP). Additionally, PoliMi leads the Testing activities and services of the Ecosystem including the environmental assessment of the new materials.*
PureNano	*PureNano proposes a novel method for an in-situ purification/regeneration process of plating baths from contaminants and undesirable chemical species hindering the operation of the bath. The concept is based on a real industrial need with a high market demand and will be implemented through the utilization of appropriate functionalized magnetic nanoparticles (MNPs), leading to a significant extension of the lifetime of the baths, up to 10 times.*
PROCETS	*PROCETS focused on delivering a new generation of environmentally-friendly and health-conscious coatings based on reinforced nano-particles. PoliMi in collaboration with other partners have modified a Pilot Line for integration of nanoparticles, used for the production of nanocomposite Ni-P coatings. Based on the results of the project the main barriers for obtaining excellent coatings have been recognized (e.g. inhomogeneous current density, agglomeration of particles with time).*
HARDALT	*HardAlt aimed at the restriction or even elimination of the use of hexavalent chromium in electrodeposition industry and at providing coatings with equal or even enhanced functional properties compared to hard chrome. PoliMi collaborated on the creation of the section on the generation of specifications related to the electroplating equipment and all operational parameters as well as on the specifications on Ni-P and Ni-P composites electroplating baths, chemicals, additives and other components*

ENABLE	*The ENABLE project made an important contribution to the advancement of environmentally friendly and cost-effective coating technologies for painted metals. The consortium provided resources previously lacking, process control experience and accelerated testing required by SMEs to begin evaluating the new pre-treatments in a standardised way to draw relevant conclusions regarding feasibility and market potential.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	**Short description (Max 300 characters)**
Facilities hosted in the Department	*The PoliMi unit will be hosted at the Department of Chemistry, Materials and Chemical Engineering (CMIC) "Giulio Natta".*
Material science Section	*Coating synthesis, deposition and material characterization facilities are available*
Clean facilities	*Glove box for controlled atmosphere reaction, small scale lab reactor for controlled pressure and temperature synthesis, chemical hoods, ovens for controlled coating densification*
Analytical tools	*Characterization techniques and equipment for electrolytic, physical and chemical vapour deposition. Measuring instruments and equipment for material/process characterisation include the following: X-ray Fluorescence.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
950342951	*CREATIVE NANO PC*

Short name: CNANO

Address

Street	LEVENTI 4
Town	PERISTERI
Postcode	12132
Country	Greece
Webpage	http://www.creativenano.gr

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2013	- yes
SME self-assessment	unknown	
SME validation	19/04/2013	- yes

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Dr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Alexandros**

Last name* **Zoikis Karathanasis**

E-Mail* **a.karathanasis@creativenano.gr**

Position in org. CEO

Department CREATIVE NANO PC

☒ Same as organisation name

☒ Same as proposing organisation's address

Street LEVENTI 4

Town PERISTERI

Post code 12132

Country Greece

Website *Please enter website*

Phone *+xxx xxxxxxxxx*

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Thanasis	Lazarou	info@creativenano.gr	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Alexandros	Zoikis Karathanasis	Man	Greece	a.karathanasis@creativenano.gr	Category A Top grad	Leading	0000-0002-8701-6907	Orcid ID
Dr	Alexios	Grigoropoulos	Man	Greece	a.grigoropoulos@creativenano.gr	Category B Senior re	Team member	0000-0002-3108-7052	Orcid ID
Mrs	Katerina	Mavronasou	Woman	Greece	k.mavronasou@creativenano.gr	Category D First stag	Team member		
Dr	Evanthia	Papadaki	Woman	Greece	e.papadaki@creativenano.gr	Category C Recognis	Team member	0000-0002-8347-2112	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Evolution of structural, mechanical and tribological properties of Ni?P/MWCNT coatings as a function of annealing temperature* *June 2016 Surface and Coatings Technology 302* *DOI: 10.1016/j.surfcoat.2016.06.011*
Publication	*Nickel and nickel-phosphorous matrix composite electrocoatings* *August 2009Transactions of Nonferrous Metals Society of China 19(4):800-804* *DOI: 10.1016/S1003-6326(08)60353-2*
Service	*Provision of research services to industry and academia for upscaling reearch concepts utilizing the laboratory and pilot lines of the company*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
3DPLATE	*3DPlate project focuses on the design, printing, and finally electroplating, of a wide range of 3D structures and complex geometries, based on client?s desire (customized objects). Moreover, a prototype vertical pilot line will be set up and operate to fabricate custom 3D-printed metallic-coated objects. Cnano will advance its knowledge on the utilization of appropriate design software, on 3D printing technologies, as well as on plating on plastics processes.*
SAByDOMA (GA#: 862296, ending on 30.09.2023)	*SABYDOMA addresses developments in the safety by design (SbD) paradigm by examining four industrial case studies in detail where the Technology Readiness Levels (TRLs) will advance from 4 to 6. Cnano will redesign and modify their pilot line based on SbD principles developed in the project, in order to include a reflow system and incorporate high through-put testing. A new SbD concept will be developed for the electroplating of nano-enabled composite coatings, with Ni-P/SiC and/or TiO2 nanomater*
PROCETS	*PROCETS main target is to deliver protective coatings covering a wide range of applications such as automotive, aerospace, metal-working, oil and gas and cutting tools industries via thermal spray and electroplating methods by utilizing more environmental friendly materials, compared to the ones currently used. Cnano has a major role in the project aiming to deliver a sustainable electroplating method for applying hard protective coating, alternative to hard chromium, in applications.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Fully equipped chemistry labs	*Fully equipped laboratories (fumecupboards, ovens and furnaces, centrifuge, hot plates, glovebox, rotary evaporator) for the synthesis and characterization of materials which can accommodate up to 10 users.*
Electroplating pilot line for composite coatings	*Cnano's electroplating services cover a wide range of experimental scales, from lab-scale processes to semi-industrial pilot line production. Cnano?s electroplating line is specially designed for fabrication of nanocomposite coatings by pulse current electroplating method.*
3D Printer Flashforge Creator 3	*Flashforge creator 3 is an FFF/FDM printer with the ability to combine two different materials, as it contains two nozzles.*
Electrochemical station	*Potentiostat/galvanostat by MetroOhm (Autolab PGSTAT 302N) with Booster and flat corrosion cell for measuring electrochemical properties of materials (e.g. corrossion behaviour, impedance).*
Contact Angle goniometer	*Attension Theta Lite Optical Tensiometer for the measurement of the wettability, the hydrophilicity and the free energy of a solid-state surface or the surface tension of a liquid.*

Vickers Micro Hardness Tester	*Shimazu HMV-G20 micro hardness tester for the measurement of the surface hardness, essential in the areas of manufacturing and quality control of materials.*
UV-Vis chamber	*Agilent Carry 60 UV-Vis spectrophotometer equipped with a Diffuse Reflectance Accessory Fibre Optic couple for measurements of liquid and solid samples.*
TG/DTA	*Perkin Elmer Pyris Diamond Analyzer for carrying out simultaneously Thermogravimetric (TG) and Differential Thermal Analysis (DTA) under a specific gas stream (air, nitrogen, CO2 etc) to determine the chemical composition, the thermal stability and the adsorption properties of materials.*
Optical digital microscope	*The Leica DM 2700 M optical microscope is an ideal inscpection tool for materials quality control. A combination of high-quality optics with state-of-the-art universal white light LED illumination allows us to acquire images up to 100X magnification.*
Digital Viscometer	*Measures the viscosity of a fluid at a given shear rate. Typical examples include creams, food, paints, oils, emulsions, and dispersions.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
999895692	*ARISTOTELIO PANEPISTIMIO THESSALONIKIS*

Short name: ARISTOTLE UNIVERSITY OF THESSALONIKI

Address

Street	KEDEA BUILDING, TRITIS SEPTEMVRIOU, ARISTOT
Town	THESSALONIKI
Postcode	54636
Country	Greece
Webpage	www.auth.gr

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2015	- no
SME self-assessment	31/12/2015	- no
SME validation	20/08/2008	- no

Departments carrying out the proposed work

Department 1

Department name	Department of Chemistry	☐ not applicable
	☐ Same as proposing organisation's address	
Street	University Campus, P.O. Box 116	
Town	Thessaloniki	
Postcode	54124	
Country	Greece	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Prof.

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Konstantinos**

Last name* **Triantafyllidis**

E-Mail* **ktrianta@chem.auth.gr**

Position in org. Professor - Group Head

Department Department of Chemistry

☐ Same as organisation name

☐ Same as proposing organisation's address

Street University Campus, P.O. Box 116

Town Thessaloniki

Post code 54124

Country Greece

Website http://ktrianta.webpages.auth.gr/

Phone +302310997730

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Dimitrios	Bikiaris	dbic@chem.auth.gr	+302310997812

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Prof	Konstantinos	Triantafyllidis	Man	Greece	ktrianta@chem.auth.gr	Category A Top grad	Leading	0000-0001-8658-8500	Orcid ID
Prof	Dimitrios	Bikiaris	Man	Greece	dbic@chem.auth.gr	Category A Top grad	Team member	0000-0001-8458-4952	Orcid ID
Dr	Alexandra	Zamboulis	Woman	Greece	azamboulis@gmail.com	Category C Recognis	Team member	0000-0003-1608-0773	Orcid ID
Dr	Zoi	Terzopoulou	Woman	Greece	terzoe@gmail.com	Category C Recognis	Team member	0000-0001-7935-2145	Orcid ID
Dr	Dimitrios	Gkiliopoulos	Man	Greece	dgiliopo@chem.auth.gr	Category C Recognis	Team member	0000-0002-1871-2458	Orcid ID
Mrs	Christina	Pappa	Woman	Greece	x.pappa@yahoo.com	Category D First stag	Team member	0000-0001-6636-3769	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☐
IPR management incl. technology transfer	☒
Public procurer of results	☒
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Cresnar K.P., Zemljic L.F., Papadopoulos L., Terzopoulou Z., Zamboulis A., Klonos P.A., Bikiaris D.N., Kyritsis A., Pissis P. Effects of Ag, ZnO and TiO2 nanoparticles at low contents on the crystallization, semicrystalline morphology, interfacial phenomena and segmental dynamics of PLA. Materials Today Communications 27, 102192, 2021. https://doi.org/10.1016/j.mtcomm.2021.102192*
Publication	*Merachtsaki, D., Xidas, P., Giannakoudakis, P., Triantafyllidis, K. & Spathis, P. Corrosion Protection of Steel by Epoxy-Organoclay Nanocomposite Coatings, Coatings 7, 2017, 84, https://doi.org/10.3390/coatings7070084*
Publication	*L. Papadopoulos, M. Kluge, D.N. Bikiaris, T. Robert, Straightforward Synthetic Protocol to Bio-Based Unsaturated Poly(ester amide)s from Itaconic Acid with Thixotropic Behavior, Polymers 2020, 12, 980 https://doi.org/10.3390/polym12040980*
Publication	*Terzopoulou Z., Klonos P., Kyritsis A., Tziolas A., Avgeropoulos A., Papageorgiou G., Bikiaris D.N. Interfacial interactions, crystallization and molecular mobility in nanocomposites of poly(lactic acid) filled with new hybrid inclusions based on graphene oxide and silica nanoparticles. Polymer, 166; 1-12: 2019. https://doi.org/10.1016/j.polymer.2019.01.041.*
Publication	*Sanusi O.M., Papadopoulos L., Klonos P.A., Terzopoulou Z., Hocine N.A., Benelfellah A., Papageorgiou G.Z., Kyritsis A., Bikiaris D.N. Calorimetric and dielectric study of renewable poly(hexylene 2,5-furan-dicarboxylate)-based nanocomposites in situ filled with small amounts of graphene platelets and silica nanoparticles. Polymers 12, 1239, 2020, http://doi:10.3390/polym12061239.*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
BIOMAC	*BIOMAC aims to establish an Open Innovation Test Bed (OITB), a collaborative ecosystem where technologies and solutions utilizing nano-enabled bio-based nano-additives (i.e. nanocellulose and nanolignin), polymers and composites (NBMs) will be upscaled and prepared for market applications within Circular Bioeconomy concepts. The BIOMAC ecosystem consists of 17 Pilot Lines and supported by various technical and non-technical services for the upscaling of new bio-based eco-friendly polymers.*
SafeMarble	*The aim of the proposed project is to improve the properties of commercial protective agents/epoxy polymers using (nano)additive materials (e.g. nanosilica, natural layered materials, nanotitania), to develop new systems of epoxy resins/hardeners for marble protection and restoration applications, and to develop new biopolymers (e.g. polyesters) and their composites for the production of polymer composites to be used as marble protection agents.*
Poly Nano	*Several nanocomposites based on polypropylene for pipe production, polyacrylates as wall colors and urea-formaldehyde crosslinked resins for construction applications have been prepared and studied in this project. Graphene oxide, TiO2 and SiO2 were used to enhance the properties of all studied polymeric matrices. Emphasis was given to the enhancement of their mechanical and fire-retardant properties.*
GEOPIPE	*The target of the project was to use various additives with high thermal conductivity like multiwalled carbon nanotubes (MWCNTs) and nanodiamonds at several concentrations to enhance the thermal conductivity of high-density crosslinked polyethylene (HDPE) and polypropylene (PP) pipes used for heating and cooling applications. Emphasis was given to the production of pipes for geothermal applications combining the selected nano-additives.*
CELL4GLUE	*The project CELL4GLUE aims to isolate the cellulose fraction of lignocellulosic biomass, convert it to nanocellulose and utilize it as nano-additive for the development of novel polymer (resins) nanocomposites with advanced properties.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Premises	*This instrumentation is appropriate for the production of thermoset, including epoxy-based polymers, and thermoplastic polymers and composites*
Reactors for synthesis of polymers	*High pressure solvothermal reactor, Microwave reactor (CEM), Small-scale polyesterification apparatus (50g), UV curing apparatus, Quantum yield reactor (Photochemical Reactors Ltd.), Test light fastness apparatus (Suntest), Pressure reactor (Parr), Brabender Haake-Bachler Reomixer.*
Extruders	*Double screw and single screw extruders*
Thermal analysis instruments for polymers	*Differential Scanning Calorimeter (Puris-1, Perkin-Elmer), Thermomechanical analyser (TMS-2, Perkin-Elmer),*
Thermal analysis instruments for polymers	*Thermogravimetric analyser (TGA-Perkin Elmer), Dynamic Thermomechanical analyser (DMA, Perkin Elmer, Diamond), Hot stage polarizing microscope (Nikon- Linkam).*
Spectrometers, Electron Microscopes	*Single crystal X-ray diffractometer, Elemental (C, H, N) analyser, FT-IR spectrophotometers, UV-Vis spectrophotometers, Atomic Force Microscopy (AFM), Electron microscopy (SEM, TEM).*
Mechanical testers of polymers	*Surface and Interface tensiometer (KSV), Universal testing instrument 5kN (Instron 1122), Impact tester (Tinious Olsen), Heated-plate hydraulic press (Otto Weber, Type PW30).*
Viscosity, MW, density measurements	*Electronic and melt viscometer, Gel Permeation Chromatography (GPC) (Polymer Laboratories), Density gradient column (Davenport), Contact angle measuring device (Sinterface), Melt flow index device*
Chromatography, Permeability	*High Pressure liquid chromatography (HPLC, Shimadzu), pyrolysis-gas chromatography/mass spectrometry (Py-GC/MS), Permeability apparatus (Davenport), Electronic membrane osmometer (Knauer).*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
901322643	*IDENER RESEARCH & DEVELOPMENT AGRUPACION DE INTERES ECONOMICO*

Short name: IDENER RESEARCH & DEVELOPMENT AIE

Address

Street	CALLE EARLY OVINGTON 24-8
Town	LA RINCONADA SEVILLA
Postcode	41300
Country	Spain
Webpage	www.idener.es

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is no (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

Department 1

Department name: IT & Digital Applications

☐ not applicable

☒ Same as proposing organisation's address

Street: CALLE EARLY OVINGTON 24-8

Town: LA RINCONADA SEVILLA

Postcode: 41300

Country: Spain

Department 2

Department name: Chemical Applications

☐ not applicable

☒ Same as proposing organisation's address

Street: CALLE EARLY OVINGTON 24-8

Town: LA RINCONADA SEVILLA

Postcode: 41300

Country: Spain

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ◉ Woman ○ Man ○ Non Binary

First name* **Carlos**

Last name* **Leyva**

E-Mail* **carlos.leyva@idener.es**

Position in org. Head of IT & Digital Applications department

Department IT & Digital Applications department

☐ Same as organisation name

☒ Same as proposing organisation's address

Street CALLE EARLY OVINGTON 24-8

Town LA RINCONADA SEVILLA

Post code 41300

Country Spain

Website https://www.idener.es

Phone +34 954460278

Phone 2 *+xxx xxxxxxxxx*

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mr	Carlos	Leyva	Man	Spain	carlos.leyva@idener.es	Category A Top grad	Leading	0000-0003-2599-2694	Orcid ID
Dr	Maria	Lopez	Woman	Spain	maria.lopez@idener.es	Category A Top grad	Team member	0000-0002-3547-0323	Orcid ID
Mr	Maria	Tripiana	Woman	Spain	maria.tripiana@idener.es	Category B Senior re	Team member	0000-0002-5676-6895	Orcid ID
Mr	Maria	Rueda	Woman	Spain	maria.rueda@idener.es	Category B Senior re	Team member	0000-0003-3271-094X	Orcid ID
Dr	Alvaro	Cabeza	Man	Spain	alvaro.cabeza@idener.es	Category C Recognis	Team member	0000-0002-7735-5059	Orcid ID

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Otero-Gonzalez, L., Sierra-Alvarez, R., Boitano, S., & Field, J. A. (2012). Application and validation of an impedance-based real time cell analyzer to measure the toxicity of nanoparticles impacting human bronchial epithelial cells. Environmental Science and Technology, 46(18), 10271?10278.*
Publication	*Otero-Gonzalez, L., Mikhalovsky, S. V., V?clav?kov?, M., Trenikhin, M. V., Cundy, A. B., & Savina, I. N. (2020). Novel nanostructured iron oxide cryogels for arsenic (As(III)) removal. Journal of Hazardous Materials, 381, 120996. https://doi.org/10.1016/j.jhazmat.2019.120996*
Publication	*Hidalgo J, Tripiana M, Sanchez-Cupido L et al. Development of an innovative process involving the use of ionic liquids for the recovery and purification of rare earths from permanent magnets and NIMH batteries [version 1; peer review: awaiting peer review]. Open Research Europe 2021, 1:89 (https://doi.org/10.12688/openreseurope.13833.1)*
Publication	*"Cabeza, A., Piqueras, C. M., Sobr?n, F., & Garcia-Serna, J. (2016). Modeling of biomass fractionation in a lab-scale biorefinery: Solubilization of hemicellulose and cellulose from holm oak wood using subcritical water. Bioresource Technology, 200. Retrieved from https://doi.org/10.1016/j.biortech.2015.09.063*
Publication	*Lopez-Abelairas, M., Pereira de Oliveira, L., Verstraete, J.J. (2016). Application of Monte Carlo techniques to LCO gas oil hydrotreating: molecular reconstruction and kinetic modeling. Catal. Today., 271, 188?198 DOI : 10.1016/j.cattod.2016.02.041*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
BIOMAC (H2020)	*BIOMAC project will create an OITB ecosystem for nanomaterials openly accessible including services for regulation, safety, sustainability, circularity and market potential with modelling, process control, standardisation and characterisation. IDENER's role relates to overall process modelling and optimisation*
LIGHTCOCE (H2020)	*LightCoce project aims in offering an open access Ecosystem for upscaling and testing multifunctional lightweight concrete and ceramic materials through 5 upgraded Pilot Lines (PLs) as well as characterisation facilities, process modelling, quality assurance and monitoring, and standardisation, safety and innovation management services. IDENER focuses on process modelling and optimisation*
PURENANO (H2020)	*PureNano proposes an innovative, low cost, sustainable method for purification of plating baths which will lead to a significant extension of their lifetime. IDENER's role focus on process control and optimisation*
REE4EU (H2020)	*The REE4EU project will develop, validate and demonstrate in 2 industrially relevant Pilots an innovative Rare Earth Alloys (REA) production route from permanent magnets PM and Ni metal hydride (NiMH) Battery waste. IDENER's role is related to process modelling, design and optimisation.*
REVOLUTION (H2020	*REVOLUTION focuses on overcoming the challenges hindering the use of recycled materials, but more broadly, restricting the widespread adoption of circular economy principles in the automotive industry. By using mathematical modelling and AI methods, a platform is being built to develop a range of recycled formulations to deliver innovative components. IDENER's role is related to process and material modelling.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
998993786	*IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA*

Short name: IRIS

Address

Street	CALLE VELAZQUEZ, NO 94 PRIMERA PLANTA
Town	MADRID
Postcode	28006
Country	Spain
Webpage	www.iris.cat

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	unknown
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2019	- yes
SME self-assessment	31/12/2019	- yes
SME validation	24/02/2009	- yes

Departments carrying out the proposed work

Department 1

Department name	IRIS Innovation Lab	☐ not applicable
	☐ Same as proposing organisation's address	
Street	Carretera d'Esplugues, 39-41	
Town	Cornella de Llobregat	
Postcode	08940	
Country	Spain	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ◯ Woman ◉ Man ◯ Non Binary

First name* **Ioannis**

Last name* **Kakogiannos**

E-Mail* **ikakogiannos@iris-eng.com**

Position in org. Project and Innovation Manager

Department IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA

☒ Same as organisation name

☐ Same as proposing organisation's address

Street Carretera d'Esplugues, 39-41

Town Cornella de Llobregat

Post code 08940

Country Spain

Website https://www.iris-eng.com/

Phone *+xxx xxxxxxxxx*

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Albert	Torres	albert.torres@iris-eng.com	+xxx xxxxxxxxx
Alexandra	Poch	apoch@iris-eng.com	+xxx xxxxxxxxx
Joan	Puig	jpuig@iris-eng.com	+xxx xxxxxxxxx
Colm	Digby	cdigby@iris.cat	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mr	Johannes	Ziegler	Man	Germany	jziegler@iris-eng.com	Category D First stag	Leading	0000-0002-2022-4314	Orcid ID
Dr	Sara	Sanchez	Woman	Spain	ssanchez@iris-eng.com	Category B Senior re	Team member	000-0002-1154-165X	Orcid ID
Mr	Manel	Sanchez Velando	Man	Spain	manel.sanchez@iris-eng.com	Category D First stag	Team member	0000-0001-6736-5838	Orcid ID
Ms	Diana	Galilea	Woman	Spain	diana.galilea@iris-eng.com	Category D First stag	Team member	0000-0003-4905-8687	Orcid ID
Dr	Laurent	Philipet	Man	France	lphilippet@iris-eng.com	Category C Recognis	Team member	000-0001-6296-6669	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Elodie Bugnicourt,Timothy Kehoe , Marcos Latorre, Cristina Serrano, Saverine Philippe and Markus Schmid, Recent Prospects in the Inline Monitoring of Nanocomposites and Nanocoatings by Optical Technologies, Nanomaterials 2016, Review, 6(8), 150; doi:10.3390/nano6080150* *Year: 2016*
Publication	*E. Bugnicourt, M. Schmid, D. Kainz, P.Lafortune, L. Rodriguez-Turienzo & P. Cinelli, Simulation and experimental validation of the denaturation of a whey protein-based coating during convection and/or infrared drying on a plastic film and influence on its oxygen barrier properties. Polymer-Plastics Technology and Engineering, 2016.* *Year: 2016*
Software	*Scadalytics is a Process Analytical Technology data management software. Machine Learning & Chemometrics tools for complex process control. Ability to intelligently interpret & connect metadata. Communication drivers for main PAT sensors on market, simplifying integration in plant & consolidation of real-time monitoring information for optimal running of processes & production plant. Connectivity with ERP, MES & SCADA; elevate chain of communication to plant managers & directors.*
Good	*VISUM HSI: Industrial in-line NIR HyperSpectral Imaging (HSI) analyzer that can be integrated into any type of production line for real-time monitoring of product quality when the spatial information (shape and position) is relevant. Designed for: (1) Non-invasive installation; (2) Continuous monitoring of Quality Attributes; (3) Fast and efficient detection of foreign bodies on surfaces; (4) In-line monitoring of product distribution; (5) Anomaly Detection*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
OptiNanoPro	*The OptiNanoPro project developed different approaches for the introduction of nanotechnology into packaging, automotive and photovoltaic materials production lines. In particular, the project focused on the development and industrial integration of tailored online dispersion and monitoring systems to ensure a constant quality of delivered materials.*
OLEDSOLAR	*OLEDSOLAR aims to tackle these challenges by developinginnovative manufacturing processes for critical steps in the production of opto-electronic devices including OLEDs, OPVs and CIGs solar cells. The proposed activities include reconfigurable high yield (>10% improvement) processes to be scaled up, tested at pilot lines and implemented in production line for validation.*
BIONANOPOLYS	*The EU-funded BIONANOPOLYS project aims to launch an open, innovative test bed for the design, development, testing, regulatory and environmental assessment, and upscaling of promising nano-enabled bio-based materials. With easy access to users across Europe, it will foster collaboration and innovation to secure Europe's lead in a field with growing influence in areas from packaging and coatings to biomedicine.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Spectroscopic Laboratory	*Equipped with several instruments, namely, a FTIR-ATR (Interspectrum), 2 UV-VIS (PerkinElmer and Ocean Optics), an Extended-SWIR (StellarNet) and a NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics).*

Optical photonics-based technologies and PAT syste	*HSI and NIR based cameras suitable for the surface and composition analysis of different materials and for the optimization and monitoring of numerous kinds of processes. Also, the described tools can be applied for the digitalisation of bio-waste valorisation processes, adding value to the circular*
Electronics, mechanical anSpectroscopic Laboratory	*Equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
998959642	*UNIVERSIDAD DE BURGOS*

Short name: UBU

Address

Street	HOSPITAL DEL REY
Town	BURGOS
Postcode	09001
Country	Spain
Webpage	www.ubu.es

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	20/01/2009	- no
SME self-assessment	unknown	
SME validation	20/01/2009	- no

Departments carrying out the proposed work

Department 1

Department name	ICCRAM	☐ not applicable
	☐ Same as proposing organisation's address	
Street	Plaza Misael Banuelos s/n	
Town	Burgos	
Postcode	09001	
Country	Spain	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Dr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **JUAN ANTONIO**

Last name* **TAMAYO RAMOS**

E-Mail* **jatramos@ubu.es**

Position in org. Head of Toxicology Group

Department ICCRAM

☐ Same as organisation name

☐ Same as proposing organisation's address

Street Plaza Misael Banuelos s/n

Town Burgos

Post code 09001

Country Spain

Website www.ubu.es/iccram

Phone +34947258709

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Rocio	Barros	rbarros@ubu.es	+34947499113
Carlos	Rumbo	crumbo@ubu.es	+34947258709
Sonia	Martel	smartel@ubu.es	+34947499113

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Juan Antonio	Tamayo Ramos	Man	Spain	jatramos@ubu.es	Category B Senior re	Leading	0000-0002-7071-002X	Orcid ID
Prof	Javier	Garcia Tojal	Man	Spain	qipgatoj@ubu.es	Category A Top grad	Team member	0000-0003-3816-0483	Orcid ID
Dr	Roberto	Quesada Pato	Man	Spain	rquesada@ubu.es	Category B Senior re	Team member	0000-0003-2764-7157	Orcid ID
Dr	Carlos	Rumbo	Man	Spain	crumbo@ubu.es	Category C Recognis	Team member	0000-0002-5038-0334	Orcid ID
Dr	Rocio	Barros	Woman	Spain	rbarros@ubu.es	Category B Senior re	Team member	0000-0002-0392-8504	Orcid ID
Ms	Sonia	Martel	Woman	Spain	smartel@ubu.es	Category D First stag	Team member	0000-0001-9080-0877	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Martel S, Barros R, Domi B, Rumbo C, Poddighe M, Aparicio S, Suarez-Diez M, Tamayo-Ramos JA (2021) Low Toxicological Impact of Commercial Pristine Multi-Walled Carbon Nanotubes on the Yeast Saccharomyces cerevisiae. Nanomaterials 11(9), 2272 https://doi.org/10.3390/nano11092272* *Toxicological assessment of nanomaterials*
Publication	*Domi B, Bhorkar K, Rumbo C, Sygellou L, Yannopulos S, Barros R, Quesada R, Tamayo-Ramos JA (2021) Assessment of Physico-Chemical and Toxicological Properties of Commercial 2D Boron Nitride Nanopowder and Nanoplatelets. International Journal of Molecular Sciences, 22(2), 267. https://doi.org/10.3390/ijms22020567* *Application of different physico-chemical and toxicology techniques to perform the safety assessment of nanomaterials.*
Publication	*M Suarez-Diez, S Porras, F Laguna-Teno, PJ Schaap, JA Tamayo-Ramos (2020) Toxicological response of the model fungus Saccharomyces cerevisiae to different concentrations of commercial graphene nanoplatelets. Scientific Reports, 10, 3232 https://doi.org/10.1038/s41598-020-60101-7*
Publication	*Rumbo C, Espina CC, Gassmann J, Tosoni O, Barros R, Martin SM, Tamayo-Ramos JA (2021) In vitro safety evaluation of rare earth-lean alloys for permanent magnets manufacturing. Scientific Reports 11 (1) 10.1038/s41598-021-91890-0*
Publication	*Domi B, Bhorkar K, Rumbo C, Sygellou L, Martin SM, Quesada R, Yannopoulos SN, Tamayo-Ramos JA (2021) Toxicological assessment of commercial monolayer tungsten disulfide nanomaterials aqueous suspensions using human A549 cells and the model fungus Saccharomyces cerevisiae Chemosphere 272 10.1016/j.chemosphere.2021.129603*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
DIAGONAL. H2020-NMBP-TO-IND-2018-953152.GA:953152	*Development and scaled implementation of safe by design tools and guidelines for multicomponent. Project coordinator: Universidad de Burgos- ICCRAM. DIAGONAL will develop strategic business development knowledge and tools to implement in industries working with multicomponent (advanced) nanomaterials (MCNMs) and high aspect ratio nanomaterials (HARNs). The results will be useful for risk management guidelines, tools and strategies to increase nanomaterials safety. https://diagonalproject.eu*
BIOMAC. H2020. NMBP-04-2020. GA: 952941	*European Sustainable Biobased nanomaterials community. BIOMAC aims to create an Open Innovation Test Bed (OITB) ecosystem openly accessible by SMEs and the industry to a single-entry point. OITB services will assess regulation, safety, sustainability, circularity and market potential with modelling, process control, standardisation and characterisation. Project Web: https://cordis.europa.eu/project/id/952941. https://www.biomac-oitb.eu/en/normal/home*
LIGHT-ME. H2020-NMBP-01-2018. GA: 814552	*An Open Innovation Ecosystem for upscaling production processes of lightweight metal alloys composites. University of Burgos-ICCRAM is in charge of sustainability services and the evaluation of environmental impacts of different lightweight metal alloys for industrial applications. https://cordis.europa.eu/project/id/814552/es https://www.lightme-ecosystem.eu/*

SURFBIO. H2020. EU.4.b. Twinning of research	*Innovation Hub for surface and colloid biology research. The projec aims to understand the interactions at the interface between two different materials plays a critical role in material design in fields ranging from aerospace to biomedicine. Whether the material interacts with a solid, a liquid or a gas or with biological systems such as microbes or tissues, understanding and controlling surface interactions is paramount to the material's performance, safety and reliability.*
ICARUS. H2020. FET-OPEN-2015. GA: 713514	*ICARUS proposes a new thermodynamic methodology able to identify the elements and the relative chemical composition allowing a nanocrystalline state to occupy a relative minimum of the Gibbs free energy, which makes the nanostructure reasonably stable against coarsening. The alloys selected were evaluated from the point of view of toxicology and sustainability.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	**Short description (Max 300 characters)**
Last generation complete laboratory on toxicology	*Biosecurity laboratory Level 2+. Analysis of the safety of products, waste and leachate. State-of-the art in vitro assays are applied in eukaryotic and prokaryotic systems to perform nanosafety evaluations and explore nanomaterials mechanistic toxicity.*
ICCRAM_Molecular and Microbiology laboratory	*A fully equipped molecular and microbiology laboratory (autoclave, freezer, fridge, biosecurity level II flow-hood, culture incubators, ELISA reader, fluorescence- microscope, biowaste, protein/DNA/RNA related experimnets equipments, etc.).*
Science and Technology Park-University of Burgos	*Its main equipments and services are: (i) Last generation equipment and instrumental techniques for chemical analysis; (ii) Last generation equipments for physical and structural characterization; (iii) Equipment for scientific computing. UBU-PCT is member of the INFRARED network*
Licence for LCA and LCC - Gabi and Simapro	*UBU-ICCRAM counts with the requested database licenses updates.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
897674376	*EXELISIS IKE*

Short name: EXELISIS

Address

Street	XANTHIS 19
Town	GALATSI
Postcode	111 47
Country	Greece
Webpage	www.exelisis.gr

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	01/08/2019 - yes
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

Department 1

Department name	EXELISIS IKE	☐ not applicable
	☒ Same as proposing organisation's address	
Street	XANTHIS 19	
Town	GALATSI	
Postcode	111 47	
Country	Greece	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Dr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Dimitrios**

Last name* **Zoikis Karathanasis**

E-Mail* **info@exelisis.gr**

Position in org. Project Manager

Department EXELISIS IKE

☒ Same as organisation name

☒ Same as proposing organisation's address

Street XANTHIS 19

Town GALATSI

Post code 111 47

Country Greece

Website www.exelisis.gr

Phone *+xxx xxxxxxxxx*

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Ioanna	Katsavou	ioanna.k@exelisis.gr	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Nikos	Tsiantis	Man	Greece	nikos.t@exelisis.gr	Category C Recognis	Team member	0000-0003-4199-6126	Orcid ID
Dr	Ioanna	Katsavou	Woman	Greece	ioanna.k@exelisis.gr	Category C Recognis	Leading		
Mr	Ilias	Gkotsis	Man	Greece	ilias.g@exelisis.gr	Category D First stag	Team member		
Dr	Kostas	Pyrgakis	Man	Greece	kostas.p@exelisis.gr	Category C Recognis	Team member	0000-0002-0670-9630	Orcid ID

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☒

Exploitation and business planning

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Tsiantis, N., Banga, J.R., 2020, Using optimal control to understand complex metabolic pathways* *BMC Bioinformaticsthis link is disabled, 21(1), 472*
Publication	*Pyrgakis, K.A., Kokossis, A.C., 2019, Systematic synthesis and integration of multiple?effect distillation into overall processes: The case of biorefineries, AICHE Journal. 65, 7, e16631*
Service	*EXE is specialized in providing Innovation Management service, including exploitation and business planning, as well as intellectual property rights management for the consortium members and the exploitable project results.*
Service	*AXIA is organizing and carrying out a number of internal and external training workshops. For this purpose, a set of materials, including training manuals and guidelines are developed and prepared.*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
AGRICORE	*The AGRICORE modelling tool will be released as open-source with the overall aim to improve policy design, impact assessments and monitoring. EXE is contributing to the communication, dissemination, exploitation and IPR management of the project.*
MultiCycle	*MultiCycle will deliver an industrial recycling pilot plant for fossil and bio-based thermoplastic multilayer packaging and fibre reinforced composites using a novel solvent based selective extraction process which allows recovering pure plastics and additives in mixed wastes for their later reprocessing into value-added applications.*
LightCoce	*LightCoce project aims in offering an open access Ecosystem for upscaling and testing multifunctional lightweight concrete and ceramic materials through 5 upgraded Pilot Lines (PLs) (targeting to Concrete, Conventional and Advanced Ceramics), as well as characterisation facilities, process modelling, quality assurance and monitoring, and standardisation, safety and innovation management services that will be accessible to the interested customers at fair conditions and cost.*
SECRETed	*SECRETed is a multidisciplinary project aiming to fully exploit aquatic biotechnology for the production of novel compounds (biosurfactants and siderophores) with tailor-made properties for the agrochemical, pharmaceutical, cosmetic and chemistry sectors*
BIOMAC	*BIOMAC project will create an Open Innovation Test Bed (OITB) ecosystem for nanomaterials openly accessible by SMEs and the industry to a single-entry point. OITB services include regulation, safety, sustainability, circularity and market potential with modelling, process control, standardisation and characterisation.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Premises	*EXE owns licenses for various software tools that will deliver high quality innovation management services, including market research, IPR management, business planning and more (e.g. PatSnap software tool used for high-end patent landscape analysis as part of both IPR and business management*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
949251604	*INNOVACION Y CONSULTING TECNOLOGICO DE LEVANTE, S.L.*

Short name: INCOTEC

Address

Street	RONDA NARCISO MONTURIOL, 6 - CENTRO EMPR
Town	Paterna
Postcode	46980
Country	Spain
Webpage	http://www.incotec.es

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is yes (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Laura**

Last name* **Zacares**

E-Mail* **laura.zacares@incotec.es**

Position in org. Head of the technology transfer department

Department Technology transfer

☐ Same as organisation name

☐ Same as proposing organisation's address

Street Calle Alvaro de Bazin, 10

Town Valencia

Post code 46010

Country Spain

Website https://www.incotec.es/en/

Phone +34961318394

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Alejandro	Ferrero	alejandro.ferrero@incotec.es	+34961318394
Asier	Sorli	asier.sorli@incotec.es	+34946792288

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Alejandro	FERRERO	Man	Spain	alejandro.ferrero@incotec.es	Category A Top grad	Leading		
Mrs	Paula	Cervera Chasan	Woman	Spain	paula.cervera@incotec.es	Category C Recognis	Team member		
Mrs	Beatriz	Perez	Woman	Spain	beatriz.perez@incotec.es	Category C Recognis	Team member		
Dr	Laura	Zacares	Woman	Spain	laura.zacares@incotec.es	Category B Senior re	Team member		

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Service	*management of tax credits*
Service	*innovative public procurement processes*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
LightMe (GA 814552)	*Open innovation ecosystem for efficient, low cost and fast upscaling of lightweight metal alloys and nanocomposites*
Lee Bed (GA 814485)	*Open Innovation Test Bed for development and production of nanomaterials for lightweigth embedded electronics*
INN PRESS M (GA 952972)	*Open Innovation Test Bed (OITB) for developing new market-ready products and goods, from bio-based materials, by feedstock conversion. Nano-enabling will make these materials reach and exceed performances of current fossil-based materials*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
999992886	*CENTRO RICERCHE FIAT SCPA*

Short name: CENTRO RICERCHE FIAT

Address

Street	STRADA TORINO 50
Town	ORBASSANO
Postcode	10043
Country	Italy
Webpage	www.crf.it

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	27/10/2008	- no
SME self-assessment	unknown	
SME validation	27/10/2008	- no

Departments carrying out the proposed work

Department 1

Department name	Materials Engineering	☐ not applicable
	☐ Same as proposing organisation's address	
Street	Corso Settembrini 40	
Town	Torino	
Postcode	10100	
Country	Italy	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Dr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Renzo**

Last name* **Costa**

E-Mail* **renzo.costa@crf.it**

Position in org. Material Specialist

Department Materials Engineering & Sustainability

☐ Same as organisation name

☐ Same as proposing organisation's address

Street Corso Settembrini 40

Town Torino

Post code 10100

Country Italy

Website www.crf.it

Phone +39 3355239864

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Gabriele	Ciaccio	gabriele.ciaccio@crf.it	+xxx xxxxxxxxx
Elena	Verna	elenaelisabettamaria.verna@crf.it	+xxx xxxxxxxxx
Li Pira	Nello	nello.lipira@crf.it	+xxx xxxxxxxxx
Giancarlo	Marchiaro	giancarlo.marchiaro@crf.it	+xxx xxxxxxxxx
Alessandra	Bugaudi	alessandra.bugaudi@crf.it	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Renzo	Costa	Man	Italy	renzo.costa@crf.it	Category B Senior re	Leading		
Dr	Gabriele	Ciaccio	Man	Italy	gabriele.ciaccio@crf.it	Category B Senior re	Team member	0000-0002-7940-8257	Orcid ID
Dr	Elena	Verna	Woman	Italy	elenaelisabettamaria.verna@crf.it	Category B Senior re	Team member		
Dr	Nello	Li Pira	Man	Italy	nello.lipira@crf.it	Category A Top grad	Team member	0000-0003-0710-3355	Orcid ID
Dr	Giancarlo	Marchiaro	Man	Italy	giancarlo.marchiaro@crf.it	Category B Senior re	Team member		

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Albini, G., Brunella, V., Placenza, B., Martorana, B., & Guido Lambertini, V. (2019). Comparative study of mechanical characteristics of recycled PET fibres for automobile seat cover application. Journal of Industrial Textiles, 48(6), 992-1008. https://doi.org/10.1177/1528083717750887*
Publication	*M. Lattore, S. Lazzari, Life Cycle Assessment of bio-based materials for automotive applications; BiPoCo Conference 2012, Siofok*
Other achievement	*Pipino; The evolution of biobased materials in automotive; BiPoCo Conference 2012, Siofok*
Publication	*Grasso, A. Pipino New tailor-made biopolymers produced from lignocellulosic sugars waste as an example of bio-based materials applied in transport sectors; BiPoCo Conference 2014, Visegrad*
Publication	*TOWARDS GREEN ELECTRONICS IN EUROPE, Strategic Research Agenda, Organic & Large Area Electronics Dec 2009*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
INN-PRESSME	*INN-PRESSME will support the development of new bio-based products, their scaling-up towards industrial production and their introduction to the market by using a model of cooperation to speed up bioeconomy and CE solutions. To that end, the project will focus on (nano)cellulose, bioplastics and natural fibres together with nanotechnology approaches to develop bio-based materials with properties and functionalities that equal or outperform their fossil counterparts at competitive prices.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Microscopy	*Contact angle, AFM, FIB, SEM, confocal microscopy*
Automotive validation: aging and reliability	*Reliability chambers: UV , ? C, %RH cycling, salt spray, cyclic corrosion, CASS test Mechanical tests: abrasion, adhesion, stone chipping, dynamometer, Charpy, Izod*
Optical characterization	*Spectrophotometers, gonioreflectometer, X Ray Fluorescence (XRF and micro-XRF) X Ray Diffraction (Micro XRD Spectroscopy)*
Chemical characterization	*DSC, FTIR, TGA, GC-MS, HDT, Vicat*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
998821708	*ARCELIK A.S.*

Short name: ARCELIK

Address

Street	KARAAGAC, SUTLUCE BEYOGLU CAD. NO:2-6
Town	ISTANBUL
Postcode	34445
Country	Turkey
Webpage	http://www.arcelikas.com.tr/Cultures/en-US/

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	29/10/2008	- no
SME self-assessment	unknown	
SME validation	29/10/2008	- no

Departments carrying out the proposed work

Department 1

Department name	ARCELIK A.S.	☐ not applicable
	☒ Same as proposing organisation's address	
Street	KARAAGAC, SUTLUCE BEYOGLU CAD. NO:2-6	
Town	ISTANBUL	
Postcode	34445	
Country	Turkey	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title: Mrs

Gender: (•) Woman ( ) Man ( ) Non Binary

First name*: **Merve**

Last name*: **Metin Yurt**

E-Mail*: **merve.metin@arcelik.com**

Position in org.: Senior Specialist - R&D - Metal & Surface

Department: Material Technologies

☐ Same as organisation name

☒ Same as proposing organisation's address

Street: KARAAGAC, SUTLUCE BEYOGLU CAD. NO:2-6

Town: ISTANBUL

Post code: 34445

Country: Turkey

Website: www.arcelikglobal.com

Phone: *+xxx xxxxxxxxx*

Phone 2: *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Evrim	Ozgul	evrim.ozgul@arcelik.com	+xxx xxxxxxxxx
Cansu	Musaoglu	cansu.musaoglu@arcelik.com	+xxx xxxxxxxxx
Alper	Yesilcubuk	alper.yesilcubuk@arcelik.com	+xxx xxxxxxxxx
Onur	Cetin	onur.cetin3@arcelik.com	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mrs	Merve	Metin	Woman	Turkey	merve.metin@arcelik.com	Category B Senior re	Leading		Orcid ID
Dr	Alper	Yesilcubuk	Man	Turkey	alper.yesilcubuk@arcelik.com	Category A Top grad	Team member		
Mr	Onur	Cetin	Man	Turkey	onur.cetin3@arcelik.com	Category C Recognis	Team member		

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Good	*Corensis Medical Kiosk offers an automation-based health platform that performs basic health measurements with the help of high precision medical sensors and advanced algorithms.*
Good	*Asista, uses Natural Language Understanding (NLU) solutions and processing technologies that Arcelik has developed. It can listen, understand, and responds to human questions in Turkish.*
Good	*Arcelik, we are working on computer vision applications particularly focusing on model compression in deep learning models running on edge devices as well as on cloud to scale up Vision Intelligence Platforms in a most efficient way.*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
Prometheus	*Within the scope of the European Union Horizon 2020 Framework Program, " Pulsed Rapid ultra-short laser surface texturing for Manufacture of FlexiblE and CusTomisEd ProdUctS " project called as "Prometheus", Laser-based surface modification studies are ongoing for high contact angle for large metal surfaces.*
FormPlanet	*" Sheet metal forming testing hub " project called as " FormPlanet", Minimizing errors during metallic sheet forming processes to increase production efficiency studies are ongoing.*
NanoHybrids	*" New generation of nanoporous organic and hybrid aerogels for industrial applications: from the lab to pilot scale production " project called as "NanoHybrids ", the development of hygroscopic aero-gels and odor-absorbing aero-gels have been successfully completed.*
Bio4Self	*"Biobased self-functionalised self-reinforced composite materials based on high performance nanofibrils PLA fibers" project called as "Bio4Self", Arcelik carried out the studies for improvement moisture resistant of PLA materials. Owing to this, formulations that show durability for many years suitable for use in durable consumer products were developed and put into practice.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Coating Technologies	*Automatic film applicator (Dr. Blade, Meyer Rod)* *Dip coating* *Spray pyrolysis* *Spraying system* *Plasma coating* *Physical vapor deposition (PVD)* *Chemical vapor deposition (CVD)*
Characterization	*Raman Spectrophotometer* *Atomic Force Microscopy (AFM)* *Field emission scanning electron microscope (FESEM)* *Fourier-transform infrared spectrometer* *Contact Angle* *Optic microscope* *Colorimeter Move 100* *UV-Vis-NIR Spectrophotometer*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
999969315	*ISRAEL AEROSPACE INDUSTRIES LTD.*

Short name: ISRAEL AEROSPACE INDUSTRIES LTD.

Address

Street	BEN GURION INTERNATIONAL AIRPORT
Town	LOD
Postcode	70100
Country	Israel
Webpage	

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	19/08/2008	- no
SME self-assessment	unknown	
SME validation	19/08/2008	- no

Departments carrying out the proposed work

Department 1

Department name	Chemistry Department/Engineering Division	☐ not applicable
	☒ Same as proposing organisation's address	
Street	BEN GURION INTERNATIONAL AIRPORT	
Town	LOD	
Postcode	70100	
Country	Israel	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title: Mr

Gender: ○ Woman ◉ Man ○ Non Binary

First name*: **Ron**

Last name*: **Cohen**

E-Mail*: **rocohen@iai.co.il**

Position in org.: Chemistry Department Manager

Department: Chemistry Department/Engineering Division

☐ Same as organisation name

☒ Same as proposing organisation's address

Street: BEN GURION INTERNATIONAL AIRPORT

Town: LOD

Post code: 70100

Country: Israel

Website: www.iai.co.il

Phone: +972 39357404

Phone 2: +972 526093488

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mr	Ron	Cohen	Man	Israel	rocohen@iai.co.il	Category A Top grad	Leading		
Dr	Daniel	Amara	Man	Israel	damara@iai.co.il	Category A Top grad	Leading		
Mrs	Nofar	Azulay	Woman	Israel	nazulay@iai.co.il	Category B Senior re	Team member		

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Good	*Ground defend systems RBY MK1, RAM MK3*
Good	*Military Air Systems , Rafael Python 5, Phalcon*
Good	*Military Aircraft 377M Anak, Lavi, Kfir*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
DIMAG	*Development and implementation of Magnesium (2010-2012), during this project we developed a concept for design aircraft parts with Magnesium. The demonstrator part was a removable shelf in the radio rack section. In EAAL the knowledge on Magnesium sheet forming will be used*
COMAG	*Coating for Magnesium (2011-2013), during this project we developed coatings and surface treatment to comply with aviation standards. The demonstrator part from DIMAG (the shelf) was used to apply the varied coatings. In EAAL, the experience for surface treatment for the door will be applied*
ecoTECH	*Clean Sky 2 consortium ecoTECH (2016-2024)) https://www.cleansky2ecotech.eu/. During this project we re-designed and test small interior parts made of Magnesium. The vast experience gained in this program will be used in EAAL*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Chemistry lab	*Ability to perform solution analysis, coating testing, salt fog testing and more.*
Analytical lab	*Ability to perform FT-IR analysis, SEM analysis and more.*
Environmental condition capabilities	*Large humidity and temp. chambers, large thermal shock test chamber, UV and rain test chambers and more. All conforming to MIL-STD-810*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

3 - Budget

?

No.	Name of beneficiary	Country	Role	Personnel costs/€	Subcontracting costs/€	Purchase costs - Travel and substistence /€	Purchase costs - Equipment/€	Purchase costs - Other goods, works and services/€	Internally invoiced goods and services/€ (Unit costs-usual accounting practices)	Indirect costs/€	Total eligible costs	Funding rate	Maximum EU contribution to eligible costs	Requested EU contribution to eligible costs/€	Max grant amount	Income generated by the action	Financial contributions	Own resources	Total estimated income
1	Gaser Ossido Duro Srl	IT	Coordinator	454,500		10,000	12,000	40,000		129125.00	645625.00	100	645625.00	645,625	645625.00				645625.00
2	Politecnico Di Milano	IT	Partner	380,000		8,000		31,000		104750.00	523750.00	100	523750.00	523,750	523750.00				523750.00
3	Creative Nano Pc	EL	Partner	330,000		8,000	35,000	19,000		98000.00	490000.00	100	490000.00	490,000	490000.00				490000.00
4	Aristotelio Panepistimio Thessalonikis	EL	Partner	228,000		15,500	25,000	23,000		72875.00	364375.00	100	364375.00	364,375	364375.00				364375.00
5	Idener Research & Development Agrupacion De Interes Economico	ES	Partner	364,000		8,000	0	38,000		102500.00	512500.00	100	512500.00	512,500	512500.00				512500.00
6	Iris Technology Solutions, Sociedad Limitada	ES	Partner	197,400		8,000		71,000		69100.00	345500.00	100	345500.00	345,500	345500.00				345500.00
7	Universidad De Burgos	ES	Partner	264,000		8,250	0	40,000	5,000	78062.00	395312.00	100	395312.00	395,312	395312.00				395312.00
8	Exelisis Ike	EL	Partner	268,800		8,000	0	18,000		73700.00	368500.00	100	368500.00	368,500	368500.00				368500.00

Proposal ID **101058699**

Acronym **FreeMe**

9	Innovacion Y Consulting Tecnologico De Levante, S.l.	ES	Partner	231,000		8,000	0	9,000		62000.00	310000.00	100	310000.00	310,000	310000.00				310000.00
10	Centro Ricerche Fiat Scpa	IT	Partner	202,800		8,000	0	20,000		57700.00	288500.00	100	288500.00	288,500	288500.00				288500.00
11	Arcelik A.s.	TR	Partner	183,600		8,000	0	19,000		52650.00	263250.00	100	263250.00	263,250	263250.00				263250.00
12	Israel Aerospace Industries Ltd.	IL	Partner	236,824		8,000	0	17,000		65456.00	327280.00	100	327280.00	327,280	327280.00				327280.00
			TOTAL	3,340,924	0	105,750	72,000	345,000	5,000	965918.00	4834592.00		4834592.00	4,834,592	4834592.00	0	0	0	4834592.00

Proposal ID **101058699**

Acronym **FreeMe**

4 - Ethics & security

Ethics Issues Table

1. Human Embryonic Stem Cells and Human Embryos			Page
Does this activity involve Human Embryonic Stem Cells (hESCs)?	○ Yes	◉ No	
Does this activity involve the use of human embryos?	○ Yes	◉ No	

2. Humans			Page
Does this activity involve human participants?	○ Yes	◉ No	
Does this activity involve interventions (physical also including imaging technology, behavioural treatments, etc.) on the study participants?	○ Yes	◉ No	
Does this activity involve conducting a clinical study as defined by the Clinical Trial Regulation (EU 536/2014)? (using pharmaceuticals, biologicals, radiopharmaceuticals, or advanced therapy medicinal products)	○ Yes	◉ No	

3. Human Cells / Tissues (not covered by section 1)			Page
Does this activity involve the use of human cells or tissues?	○ Yes	◉ No	

4. Personal Data			Page
Does this activity involve processing of personal data?	○ Yes	◉ No	
Does this activity involve further processing of previously collected personal data (including use of preexisting data sets or sources, merging existing data sets)?	○ Yes	◉ No	
Is it planned to export personal data from the EU to non-EU countries? Specify the type of personal data and countries involved	○ Yes	◉ No	
Is it planned to import personal data from non-EU countries into the EU or from a non-EU country to another non-EU country? Specify the type of personal data and countries involved	○ Yes	◉ No	
Does this activity involve the processing of personal data related to criminal convictions or offences?	○ Yes	◉ No	

5. Animals			Page
Does this activity involve animals?	○ Yes	◉ No	

6. Non-EU Countries			Page
Will some of the activities be carried out in non-EU countries?	○ Yes	◉ No	
In case non-UE countries are involved, do the activities undertaken in these countries raise potential ethics issues?	○ Yes	◉ No	
It is planned to use local resources (e.g. animal and/or human tissue samples, genetic material, live animals, human remains, materials of historical value, endangered fauna or flora samples, etc.)?	○ Yes	◉ No	
Is it planned to import any material (other than data) from non-EU countries into the EU or from a non-EU country to another non-EU country? For data imports, see section 4.	○ Yes	◉ No	
Is it planned to export any material (other than data) from the EU to non-EU countries? For data exports, see section 4.	○ Yes	◉ No	
Does this activity involve low and/or lower middle income countries, (if yes, detail the benefit-sharing actions planned in the self-assessment)	○ Yes	◉ No	
Could the situation in the country put the individuals taking part in the activity at risk?	○ Yes	◉ No	

7. Environment, Health and Safety	Page

Proposal ID **101058699**

Acronym **FreeMe**

Does this activity involve the use of substances or processes that may cause harm to the environment, to animals or plants.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	◯ Yes	◉ No	
Does this activity deal with endangered fauna and/or flora / protected areas?	◯ Yes	◉ No	
Does this activity involve the use of substances or processes that may cause harm to humans, including those performing the activity.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	◯ Yes	◉ No	

8. Artificial Intelligence			Page
Does this activity involve the development, deployment and/or use of Artificial Intelligence? (if yes, detail in the self-assessment whether that could raise ethical concerns related to human rights and values and detail how this will be addressed).	◯ Yes	◉ No	

9. Other Ethics Issues			Page
Are there any other ethics issues that should be taken into consideration?	◯ Yes	◉ No	

I confirm that I have taken into account all ethics issues above and that, if any ethics issues apply, I will complete the ethics self-assessment as described in the guidelines How to Complete your Ethics Self-Assessment ☒

Ethics Self-Assessment

Ethical dimension of the objectives, methodology and likely impact

Explain in detail the identified issues in relation to:
- objectives of the activities (e.g. study of vulnerable populations, etc.)
- methodology (e.g. clinical trials, involvement of children, protection of personal data, etc.)
- the potential impact of the activities (e.g. environmental damage, stigmatisation of particular social groups, political or financial adverse consequences, misuse, etc.)

Remaining characters 4594

Compliance with ethical principles and relevant legislations

Describe how the issue(s) identified in the ethics issues table above will be addressed in order to adhere to the ethical principles and what will be done to ensure that the activities are compliant with the EU/national legal and ethical requirements of the country or countries where the tasks are to be carried out. It is reminded that for activities performed in a non-EU countries, they should also be allowed in at least one EU Member State.

Remaining characters 5000

Proposal ID **101058699**

Acronym **FreeMe**

Security issues table

1. EU Classified Information (EUCI)[2]			Page
Does this activity involve information and/or materials requiring protection against unauthorised disclosure (EUCI)?	◯ Yes	◉ No	
Does this activity involve non-EU countries?	◯ Yes	◉ No	

2. Misuse			Page
Does this activity have the potential for misuse of results?	◯ Yes	◉ No	

3. Other Security Issues			Page
Does this activity involve information and/or materials subject to national security restrictions? If yes, please specify: (Maximum number of characters allowed: 1000)	◯ Yes	◉ No	
Are there any other security issues that should be taken into consideration? If yes, please specify: (Maximum number of characters allowed: 1000)	◯ Yes	◉ No	

[2]According to the Commission Decision (EU, Euratom) 2015/444 of 13 March 2015 on the security rules for protecting EU classified information, "European Union classified information (EUCI) means any information or material designated by an EU security classification, the unauthorised disclosure of which could cause varying degrees of prejudice to the interests of the European Union or of one or more of the Member States".

[3]Classified background information is information that is already classified by a country and/or international organisation and/or the EU and is going to be used by the project. In this case, the project must have in advance the authorisation from the originator of the classified information, which is the entity (EU institution, EU Member State, third state or international organisation) under whose authority the classified information has been generated.

[4]EU classified foreground information is information (documents/deliverables/materials) planned to be generated by the project and that needs to be protected from unauthorised disclosure. The originator of the EUCI generated by the project is the European Commission.

Topic	HORIZON-CL4-2021-RESILIENCE-01-12: Safe- and sustainable-by-design metallic coatings and engineered surface (RIA)
Title	Toxic Free metallization process for plastic surfaces
Acronym	FreeMe

List of participants

Participant No.	Participant organisation name	Country	Type
1. (Coordinator)	**Gaser Ossido Duro srl – Gaser**	IT	SME
2	Politecnico di Milano – PoliMi	IT	Univ
3	Creative Nano PC – Cnano	GR	SME
4	Aristotelio Panepistimio Thessalonikis – AUTH	GR	Univ
5	Idener research and development agrupacion de inters economico – IDE	ES	SME
6	Iris Technology Solutions, Sociedad Limitada – IRIS	ES	SME
7	University of Burgos – ICCRAM	ES	Univ
8	Exelisis PC – EXE	GR	SME
9	Innovacion Y Consulting Technologico de Levante S.L. – INC	ES	SME
10	Centro Ricerche FIAT SCPA – CRF	IT	LE
11	Arcelic A.S. – ARC	TR	LE
12	Israel Aerospace Industty – IAI	IS	LE

1. Excellence

1.1 Objectives and ambition

1.1.1 FreeMe objectives

The **industrial use of plastics** worldwide has increased remarkably over the last 30 years by virtue of their exceptional properties such as **low density, lightweight, low cost, versatility, flexibility, strength, and easy machinability**. Many **plastic parts require a protective metallic coating** to enhance their **wear and corrosion resistance and to provide a shiny, polished, aesthetically pleasing final surface**. The **electrodeposition** of metals on solid substrates (**electroplating**) is widely considered as the most efficient strategy for producing metal coatings. However, **plating on plastics (PoP)** via the application of an external electric potential is practically not feasible because most plastics are non-conductive. PoP of common, non-conductive plastics (polymers) such as ABS, PET, PLA, Nylon blends and 3D printed resins requires the initial deposition of a thin metal film **(metallization)** via a process called **"electroless plating**" [1] during which metal cations are chemically reduced to metal inside a liquid bath. Therefore, metal deposition on the polymer surface proceeds in the absence of an external source of electrical power. Importantly, the surface of the polymeric substrate must undergo a specific pre-treatment before the metallization step that includes two crucial steps:

- **etching to increase the hydrophilicity and roughness of the surface that will have a vital role in the adhesion of the metallic film on plastic**
- **surface activation to "seed" the surface with a catalyst which in turn creates nucleation sites for the final electroless metallization step.**

Substrate pre-treatment is key to obtain a homogeneous and dense metal coating during the metallization step. **Surface etching** enhances metal adhesion whereas **surface activation** provides finely dispersed nucleation sites for ensuing electroless metal deposition. Currently, **industrial PoP processes** utilize **Cr^{6+} based solutions** (e.g., chromic acid) for surface etching. With respect to surface activation, **Sn-Pd colloidal baths** are currently employed. Palladium is an exceedingly rare, precious metal which significantly raises costs. Concluding the **current industrial processes for PoP** present the following **significant disadvantages** in respect to **environmental sustainability, occupational safety, and quality of coatings**: [2]

a. *A mixture of Chromic (Cr^{6+}) and sulfuric acids bath solution is used as the SoA process for **plastic etching**. The bath should operate at a temperature between 50-70°C making the **process of high risk in relation to the occupation safety** and environment as Cr^{6+} is a recognized carcinogenic and highly toxic compound.*
b. ***Palladium** (in solutions with tin salts) is used to catalyze the reduction of copper or nickel to form a continuous conductive layer. This layer is necessary for the deposition of the final electroplated coating in the surface of*

1 A. Equbal, and A. K. Sood, Investigation on metallization in FDM build ABS part using electroless deposition method, Journal of Manufacturing Processes **19**, 22-31 (2015)

2 J. Margolis, "Metallic Plating and Coating on Plastics", Conductive Polymers and Plastics. Springer, Boston, MA. https://doi.org/10.1007/978-1-4613-0851-5_3

the plastic component. ***Pd is a Critical Raw material*** *(CRM) as its use is strongly depended in imports (i.e.from Russia). Replacement of Pd from processes with more abundant materials is highly desirable.*

c. *The plating result is* ***highly dependent on the composition of the plastic component****. Inhomogeneities in the final coating or bad adhesion may occur, degrading the aesthetics as well as its functionalities.*

Figure 1.1 a: Gasers PoP baths based on SoA technology

Figure 1.1b: Metal coated plastic articles by Gaser baths

Figure 1.1c: Coated and uncoated 3D printed artifact (Cnano) [3]

FreeMe project will investigate **two revolutionaries approaches** for the metallization of polymeric (plastic or resins) surfaces, alternative to the SoA, **without the use of hexavalent chromium and palladium**.
The **first approach is based on the use of sprayable resins** (e.g. epoxy-acrylate resins) with the additions of suitable nickel based precursors (i.e. anhydrous nickel salts). This composite resin can be deposited uniformly in the surface of the plastic object by spray deposition methods. The nickel precursors are afterwards reduced utilizing appropriate reduction agents creating the necessary nuclei for the next step of electroless deposition and the formation of the conductive layer. The **second technology is based on a Cr^{6+} & Pd free pre-treatment of the polymer surface** followed by incorporation of nickel active sites via reduction of nickel salts in the surface of the polymer. Similarly, to the first technology the electroless step follows. For both cases the final step is to apply the desired metal coating for the final metal finishing (i.e. nickel, copper, chromium or even precious metals). The proposed methodologies are based on results of four partners (Gaser, PoliMi, AUTH and Cnano) obtained by in-house, national and EC funded research project. The **FreeMe project main objectives** are:

Elimination of use of Cr^{6+} and Pd from the PoP process. The FreeMe technologies will be based on **REACH compliant chemicals**, avoiding toxic compounds, and thus ensuring the long-term sustainability of the process.

To increase the maturity of the proposed FreeMe technology to TRL 5 and bring it closer to market applications the following **specific objectives** should be achieved.

1. ***Deployment and implementation of a safe and sustainable by design (SSBD) strategy*** *concerning the development of the new FreeMe technologies for PoP that will cover the whole lifecycle (from raw materials to recycling of the plated component). This will be based on selection of REACH compliant chemicals, evaluation of processes and materials related to safety in operational environment, end application and end-of life scenarios (e.g. recycling) and assessment of the overall sustainability of the technology.*
2. ***Development of a metallization technology of plastic surfaces (i.e. ABS, PET, PLA, Nylon blends and 3D printed resins,) with a technology based on green REACH compliant sprayable UV-curable composite bio-based resins****: Sprayable bio-based epoxy or epoxy-acrylate resins containing anhydrous nickel salts, i.e. $NiCl_2$ particles with size lower than 1,5μm, in concentrations of 0.5-2%wt will be produced. The Ni salt particles in the resin's surface will be chemically reduced forming Ni nuclei, acting as catalysts for the following electroless process.*
3. ***Development of Cr^{6+} free etching technologies for plastics (i.e. ABS, PET, PLA, Nylon blends and 3D printed resins) pretreatment.*** *The technology is based on the use acids or alkali mixtures to micro-etch the surface and prepare it for the activation steps utilizing Ni-based salts and reduction of them For the 3D printed resins to used mixtures of alkali to micro-etch the surface, followed by the activation step*

For both the objectives 2 and 3, the ultimate target is to achieve excellent adhesion of the coatings in the plastic surface (4,9 N/25mm) based on peel tests

4. ***Development of in-silico techniques (simulation and modelling****) facilitating both the development and the synthesis and application of the sprayable composite resins and on the optimization of plastic surface etching based on multi-scale models, ranging from atomic to mesoscopic scales.*
5. ***Development of a fast and accurate inspection method*** *based on spectroscopic imaging ellipsometry and optical coherence tomography for micro-scale monitoring (up to 3μm) for the QC of the process (KPI: to monitor up to 3μm scale resolution, with 99% probability of defect detection)*

3 https://www.3dplate.gr/electroless-metallization-of-a-3d-printed-resin-object/

6. ***Demonstration of the FreeMe technologies*** *in automotive, aerospace and home appliances demonstrators by applying metallic coatings on plastic components and proceed in validation of them*
7. ***Assessment of the recyclability of coated polymers*** *based on the current SoA technologies for recycling (i.e. abrasive blasting, cryogenic embrittlement and mechanical crushing).*
8. ***Development of a data driven Decision Support Tool*** *combining data from SSBD assessment, in-silico approaches, experimental/manufacturing results (i.e. operational parameters, structural and morphological characterization, properties), LCA/LCC assessment and specification/requirement of test cases, for serving industrial end users, including SMEs, on the selection, development and application of the most suitable PoP technology.*
9. ***To facilitate the acceptance of the new technologies by the market*** *defining the standardization landscape and contribution to future standards development.*
10. ***To prepare business plans and define exploitation routes*** *for the partners of the project based on the results and to enhance their impact by undertaking targeted dissemination and exploitation activities, including training course for the PoP industry*

1.1.2 Relation of FreeMe objectives with the topic and realistic feasibility

The proposed technologies of FreeMe project aims on revolutionizing the PoP surface treatment industry. By eliminating the use of **Cr^{6+}** (a toxic and carcinogenic compound) and the use of **Pd** (a platinum group metal), FreeMe will offer a **safe and sustainable solution** that can dominate the market. The novel processes will deliver high quality coating systems that will provide protection of plastic surfaces from corrosion and wear, increase the surface hardness, promote multifunctionality (i.e. electromagnetic shielding) and high aesthetics (appearance). A **basic requirement** to achieve these properties is the **excellent adhesion of the coating on the plastic surface**. [4] Poor adhesion will lead to premature failure of the coating and consequently of the coated polymer components. A main part of the research activities of the project focuses on the achievement of excellent bonding of the coating systems with the plastic substrate. Finally, in respect to the targets of the European Green Deal, a SSbD strategy will be set up and implemented across the project so that:

(a) **chemicals, materials and products** designed, produced and used in the FreeMe project will not harm people and the environment (i.e. **decrease pollution, achieve toxic-free productions and promote circular economy**)

(b) **give a boost to the European surface finishing industry** that struggles to survive between the stricter EC regulatory framework and the increased competition from Asian countries (e.g. China, India). [5]

The involvement of industrial partners covering the supply chain of the PoP sector enhances the market orientation of the research activities, based on industrial standards and methodologies. Finally, the FreeMe technology will be validated in demonstrators covering industries such as **automotive, aerospace and home appliances.**

1.1.3 Advancements in relation to the SoA

The last decades a series of new technologies were oriented towards **novel metallization routes**. The following paragraph briefly presents the most important of them.

1. **Vacuum Metallization (Physical Vapor Deposition)**. In this process a metal coating is deposited on prepared surfaces (washed, etched). A metal (usually aluminum) is evaporated in a vacuum chamber and the vapor condenses onto the surface of the plastic part. The main disadvantages of the process are the high cost of the technology, the thorough pre-treatment process, and the use of aluminum layer, a difficult to plate metal for providing the final metal finishing of the component. [6]
2. **Flame/arc spraying**. A low-cost metallization process in which liquid metal is sprayed onto a plastic part. The produced layer is thick compared to other metallization methods. Main drawbacks of the technology are that the coatings prepared by these methods are quite porous, coating adhesion is lower, and the surface roughness is relatively high. [7]
3. **Sputtering metallization**. It is a process in which metal atoms are transferred from a metal target via inert gas plasma to the plastic surface. The technology presents a better adhesion compared to the previous mentioned due to higher kinetic energy of the metal atoms. Main disadvantages are the high cost of the method and the low production rate that makes it not suitable for mass applications [8]
4. **Direct electroplating in conductive polymers**. The use of conductive polymers that will be directly electroplated has attracted the interest of research community. This can be achieved either by intrinsic

4 Boipai B. et.al, "Improvement in Adhesion of Electroless Coating on Plastic Substrates: A Review", Recent Advances in Manufacturing Processes. Springer, 2021, Singapore. https://doi.org/10.1007/978-981-16-3686-8_15

5 "Asia-Pacific metal finishing market – growth, trends, COVID-19 impact and forecasts (2021-2026), Mordor Intelligence, 2021

6 J.G. Drobny, "Application of Fluoropolymer Films", Plastic Design Library, William Andrew, 2020, https://doi.org/10.1016/B978-0-12-816128-9.00001-5

7 R. Gonzalez "A review of thermal spray metallization of polymer based structures", Journal of Thermal Spray Technology 25, 2016, http://dx.doi.org/10.1007/s11666-016-0415-7

8 J. Liebeskind, "Plasma spraying for selective metallization of 3D polymer surfaces" 22nd International Symposium on Plasma Chemistry, July 5-10 2015, Antwerp, Belgium

conductors (i.e conductive polymers) or by extrinsic conductors (i.e. composite conductive polymers) in which the conductivity is due to the inclusion of conductive fillers in the polymer matrix. [9] While the first class of polymers is very expensive and cannot be used in large volume applications, the latter case is a promising technology for commercial applications. In the FreeMe project a variation of this technology is used in the case of sprayable composite resins. The advantage of FreeMe proposed solution is that instead of utilizing a bulk conductive polymer, only a thin layer will be sprayed in the surface of the polymer part making the process less expensive and not creating doubts about the feasibility of recyclability of coated part as it will be later explained.

5. **Cr^{6+} free etching process**. The etching step in PoP has two purposes: to make the surface hydrophilic and to form microscopic holes in the surface of the plastic to provide the anchoring sites for activation and metal deposition. These sites are needed for **adhesion** between the plastic and the metal. The etching or dissolving of the butadiene phase in ABS requires strong oxidizing agents and an acidic medium. The last decade a high R&I effort has been devoted in developing etching process without the use chromic acid. **FreeMe project will upscale such a technology**. Utilizing modelling and DoE, mixtures of REACH compliant acids will be investigated in combination with the main polymers used by the industry today.

Relevant Patent Search and commercial products

The consortium carried out a patent analysis in the field of **toxic-free metallization of plastics**. WIPO, EPO, and PatSnap platforms were used for the identification of the number of patents and other useful information. The targets of the patent search are on one hand to investigate the current technology landscape (Figure 1.2a) and freedom to operate (FTO) study and on the other hand to obtain useful insights facilitating the market uptake of the technologies and the R&D strategy to be followed within the FreeMe project. A detailed patent mapping landscape showcases the patent value, the Top Assignees, the Top Classification Patent Codes (CPC) and other data. An initial **''bottom-up'' approach** was followed starting from the more general concept of "Plating on Plastics" (utilizing similar keywords) and ending up with the "non-toxic plating on plastics". Based on the relevant low number of patents (Table 1.1), it is evident that there is an innovation space for the development of the FreeMe concept. **A first study of these patents showed that there are not direct overlaps with the FreeMe propose technologies.** The patent landscape of the relative "non- toxic plating on plastics" was further analyzed and the Top Assignees Countries (Figure 1.2b) are presented. Among the Top Assignees, China, United states, and Germany have leading positions. The patent mapping information will also enhance the business planning activities by defining the associated geographical market strength and possible IP holders for collaboration and networking. In most cases the research and development activities are associated with the market analysis. Patent mapping analysis will be updated during the project.

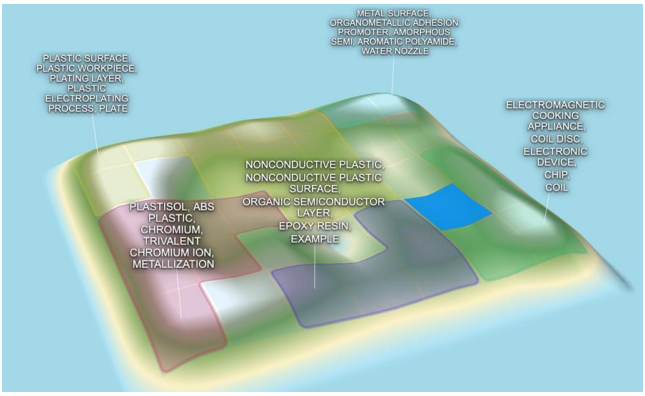


Figure 1.2a: Technology landscape of patent search

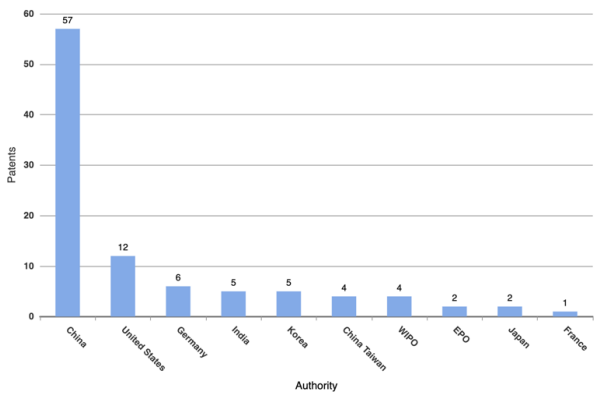


Figure 1.2b: Top assingees countries

Table 1.1: Patent mapping

Patent Mapping	No of Patents	Keywords search on PatSnap
Platting on plastics	26,271	*TAC_ALL:(metallization or metallization or "plating on plastic" or "plating on plastics")*
Electroless plating on plastics	3,793	*TAC_ALL:(metallization or metallization or "plating on plastic" or "plating on plastics") AND TAC_ALL:(electroless or electroplating)*
Toxic-free Electroless plating on plastics	99	*(Metallization OR "plating on plastics" OR "plating on plastic") AND TAC:(electroplating OR electroless) AND TAC:(" hexavalent chromium " OR " chromic acid " OR "Cr^{6+} free" OR "non-toxic" OR "toxic free" OR "toxic-free")*

9 P. Augustyn, "A review on the direct electroplating of polymeric materials", Journal of Materials Science, 56, 2021, https://doi.org/10.1007/s10853-021-06246-w

1.1.4 R&I maturity

In the following table the R&I maturity of the main proposed technologies is presented. The starting and targeted TRL is also provided as an indication of the expected progress in terms of lab to market:

1. Sprayable resin technologies. ***Start TRl: 3*** ***End TRL: 5***

The manufacture of epoxy-based (nano)composite resins with different types of nano-additives and fillers is a mature technology in various industrial sectors, e.g. automotive, aerospace, packaging, etc. [10] **Sprayable resins are attracting increased interest** as they can facilitate continuous processes of thin coatings formation on metals, plastics and other surfaces. However, the viscosity of the epoxy prepolymer to be sprayed and the possible need for use of organic solvenst may in some cases limit the industrial application of spraying technologies. Furthermore, the effect on Ni-salts fillers (i.e $NiCl_2$) on the viscosity and UV curing process, along with the desired property of thin epoxy-based polymeric films, are still under development. FreeMe aims to bring this technology from TRL 3 to 5 and demonstrate the feasibility of the proposed approach. The utilization of **biomass derived platform chemicals**, such as **isosorbide and itaconic acid**, as the starting chemicals to synthesize novel UV curable bio-based epoxy (and epoxy-acrylate) monomers and resins, along with their mixing with Ni Salts will be systematically investigated to evolve from the experimental proof of concept to technology validated in relevant environment (TRL5). Furthermore, recent results on the labs of PoliMi have shown that the reduction of Ni salts particles in resins is feasible and thus the formed nuclei can act as activation sites for the next step of electroless nickel application.

2. Cr^{6+} free etching and Pd free surface activation ***Start TRl: 3*** ***End TRL: 5***

Cnano has studied chromic acid and Pd free technologies in the Greek national funded project 3Dplate. Both ABS, PET and PLA polymers as well as 3D printed resins have been treated in lab scale apparatus with different mixtures of chemicals (i.e. piranha solutions, caustic soda solutions etc) achieving very promising results in terms of etching process and adhesion of the coating in the surface. The novel etching processes were followed by successful Pd free activation of the surface utilizing nickel salts. In the framework of FreeMe project, the technology will be optimized in the laboratory with the contribution of the multi-scale modelling activities and then will be upscaled to the use of pilot line (150 lt volume)

3. In-silico approaches ***Technology transfer***

The proposed methods for multi-scale modelling activities within FreeMe are more than validated in the modelling industry and have been widely used to support research and design activities in the material modelling area. The tools used to perform such modelling activities as NWChem, Gaussian, SIESTA, LAMMPS, Gromacs or COMSOL are also widely validated for both research and commercial activities. However, the specific models to be developed within FreeMe are innovative in terms of the materials used and the targeted applications.

4. Surface inspection methods ***Technology transfer***

IRIS has studied and developed optical methods for inline real-time thickness monitoring of coatings and lamination, compounding, injection/co-injection and electrospray Nano deposition. This activity took place at the OPTINANOPRO project (H2020, GA Nr: 686116). The sensors are based on spectral reflectance and ellipsometry, designed to measure thickness in coating and nano deposition processes. Within FreeMe project, IRIS aims to optimize the technology measuring resins in lab scale and then to upscale it in the pilot line, introducing a real-time acquisition of data.

1.2 Methodology

1.2.1 Concept idea

There are **hundreds of plating on plastics lines** producing millions of chromium or other metals plated parts every day. The vast majority of these lines are using chemical processes based on chromic acid etching and subsequent Pd based catalysed deposition of electroless nickel or copper, technologies that have been developed and designed decades ago for a specific group of plastic substrates (e.g. ABS). However, the plating on plastics market has evolved and now includes a broader range of substrates while huge increases in selective plating demand is also observed. [11] As an example, biobased composite PLA reinforced with hemp or carob, is a promising alternative to ABS when it comes to 3D printing processes. [12] **FreeMe aims to provide a holistic approach for PoP covering a range of plastic substrates (i.e. ABS, PET, PLA, nylon or UV printable Resins)**

The **general concept of the FreeMe project** is presented in Figure 1.2. In the upper part a simplified SoA plating on plastic process is depicted. The **main part of concern is the grey area, where the chromic acid etching and Pd based activation take place**. The FreeMe aspires to replace this process sequence with safer and more sustainable technologies utilizing REACH compliant chemicals and at the same time, meeting the needs for excellent adhesion of coatings.

10

11 D. Wyaness, "Is it time to revisit your Plastic Metallizatio Process?", Products and Finishing, August 2018

12 M. Gali et.al, "A new generation of bio-composite thermoplastic filaments for a more sustainable design of parts manufactured by FDM", Applied Sciences, August 2020

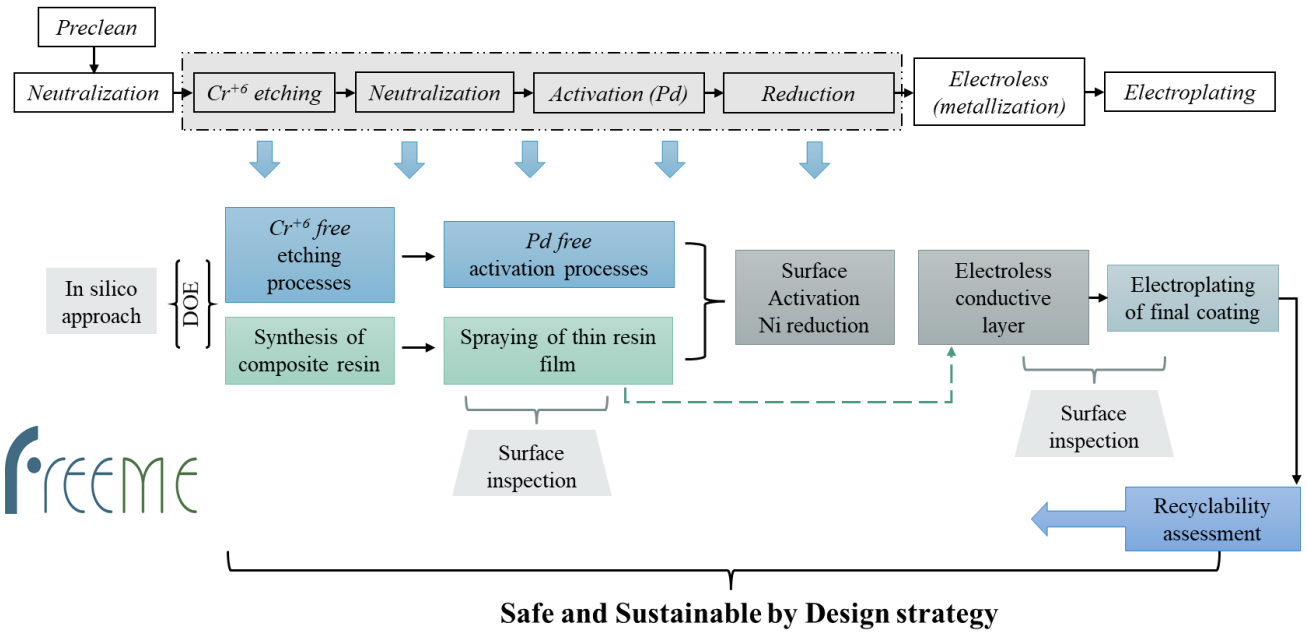


Figure 1.2: FreeMe concept

1.2.2 FreeMe methodology and implementation phases

Following the R&I concept presented in the previous paragraph, in the following the overall methodology is described with references in previous results of the partners, updated literature review and basic models and experimental design and assumptions considered in the project. In each part the main research challenges are analyzed and the projected activities to overcome these are presented.

1.2.2.1 Safe & Sustainable by Design Strategy

FREEME will embrace the Safe Innovation Approach (SIA), which encompasses the concepts of Safe and Sustainable by Design (SSbD) and Regulatory Preparedness (RP), to ensure that the whole R&D process of the new proposed PoP technologies follows a systemic approach. It will employ different tools (e.g., Life Cycle Sustainability Assessment, Safe by Design and Sustainability by Design strategies, toxicology assessment, legal analysis, etc.) to guarantee that the developed products are safe, sustainable, cost-effective, regulatory-compliant, accepted by consumers and, most notably, meet the required technical functionality. FreeMe will apply a safe and sustainable by design approach in the applications. A pre-market design phase to improve product safety, reduce human and environmental risk as well integrating circularity and sustainability principles in line with Europe´s transition to a circular and low-carbon economy will be implemented. Product design will be affected by technical performance, product service, chemical function, circular-by-design and will run in parallel to step-wise safety and sustainability assessment. Thus, by applying SIA tools in the development of FreeMe innovative plating on plastic processes, the consortium is ensuring a faster market uptake after the end of the project, with interested parties having developed a process that is safe, environmentally friendly, competitive and that answers a clear market demand. Moreover, the development of sectoral guidelines and replicability studies will facilitate the knowledge transfer into new products and markets, maximizing the project's expected outcomes and impacts.

1.2.2.2 Metallization through sprayable composite resins

The main concept of this technology is depicted in the following Figure 1.3.

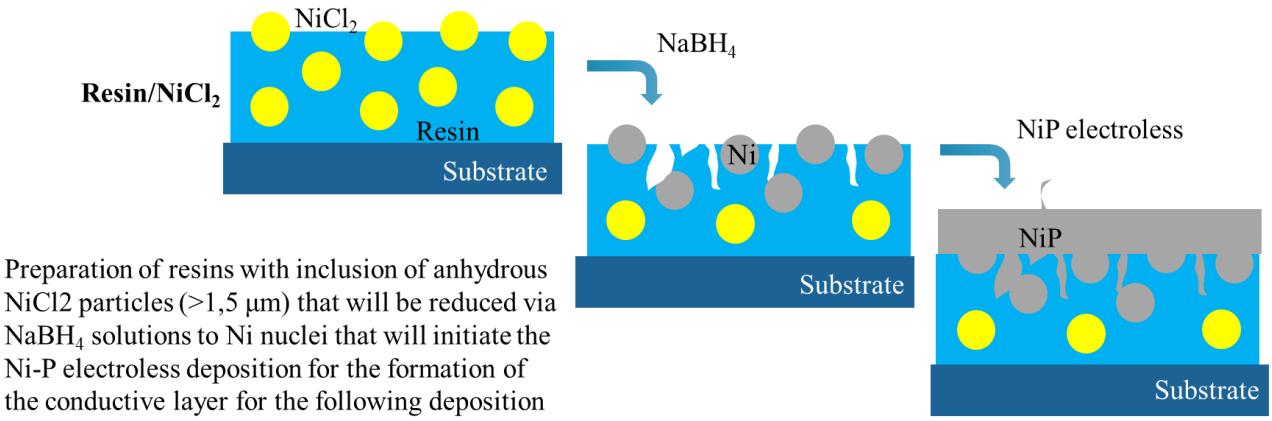


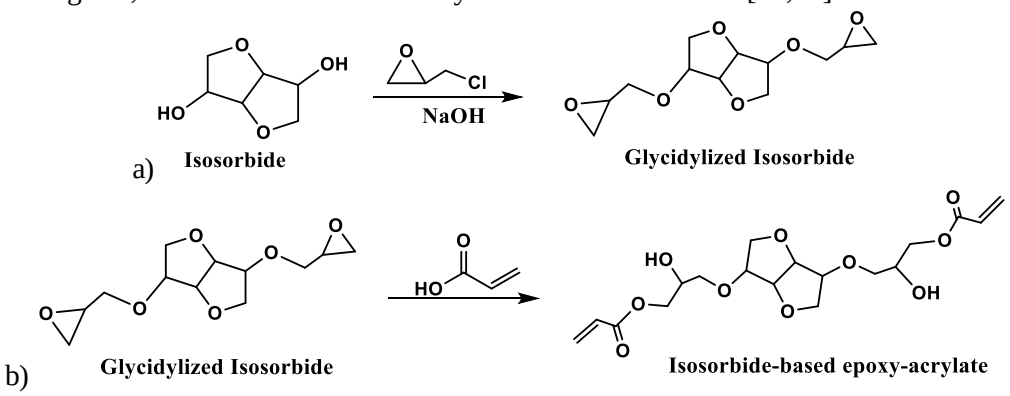
Figure 1.3: Composite sprayable resins approach for plastic metalization

Anhydrous Ni salt (i.e. $NiCl_2$) is pulverized in size smaller than 1,5μm through milling. The powder is mixed in resins mixtures and the resulted composite resin is applied in thin layers (~2 μm) in the polymer surface. Various type of polymeric surfaces will be used (i.e. ABS, PC, PPA, nylon and UV-curable resins) aiming to develop a technology not dependable on the type of the substrate. A key part for this is to achieve excellent adhesion of the sprayable resin in the substrates. Reduction of the nickel salts in the surface of the resin layer is followed with the use of a reducing agent and thus, Ni nuclei that will serve as catalyst for the following electroless step are created.

Main research/technological challenges	FreeMe activities to overcome challenges
Excellent adhesion of Resin in plastic substrate.	Design of resins with good wetting properties based on the presence of -COOH groups, regardless of the substrate surface energy – chemical bonding
Reinforcing particles will be inactive due to the resin layer on their surface	To achieve the reduction of Ni salts to Ni nuclei as catalysts for the electroless step, the hardened resin in their surface should be removed. Mild, chromic acid free etching process will be used.
Not to affect the recyclability of the plastic part	The application of a resin layer in a plastic object might make the recycling more complicated. To avoid this, a very thin (~2μm) layers should be applied.

A. Preparation of composite resins

The Ultraviolet (UV) curing technology has been increasingly used in the last decades as a sustainable way of cross-linking polymerization of monomers/oligomers towards the continuous production of epoxy-based coatings. The aim of FreeMe is to optimize the UV curing process of epoxy or epoxy-acrylate resins towards rigid thin (< 2μm) applied on the above-described substrates (i.e. ABS, PLA, etc.). Importantly, **the conventional bisphenol A (BPA) based epoxy resins will be replaced by biomass-based epoxy resins**, since **BPA is considered as highly toxic and is listed among the Substances of Very High Concern (SVHC) and its use is subjected to authorization**. The proposed green epoxy alternatives will be based on **biomass-derived isosorbide and itaconic acid**, which are considered among the most important emerging bio-based platform chemicals within the concepts of Biorefining [13]. Amongst their diverse application in the polymer, fine chemicals and fuel industry, isosorbide and itaconic acid may be utilized to produce epoxy monomers/oligomers via glycidylation reaction, as shown in Figure 1.4a, for the case of isosorbide [14,15]. The epoxidized bio-based monomers can further react with (meth)acrylic acid to produce the respective isosorbide-based epoxy-acrylates (Figure 1.4b), in analogy to the production of BPA-based epoxy acrylates [16]. Alternatively, isosorbide-based (meth)acrylates may be synthesized by direct esterification of isosorbide by reacting with methacryloyl chloride or methacrylic anhydride and a base catalyst [17], as shown in Figure 1.4c. Similar reaction paths towards the synthesis of itaconic acid-based epoxy or the respective acrylates will be also investigated, in accordance with recently efforts in the literature [18,19].



13 F. Aricò, Isosorbide as biobased platform chemical: Recent advances, Current Opinion in Green and Sustainable Chemistry 2020, 21:82–88

14 C. Musa et. al, "Ultrasound-Assisted Heterogeneous Synthesis of Bio-Based Oligo-Isosorbide Glycidyl Ethers: Towards Greener Epoxy Precursors", Molecules, 24 (2019) 1643.

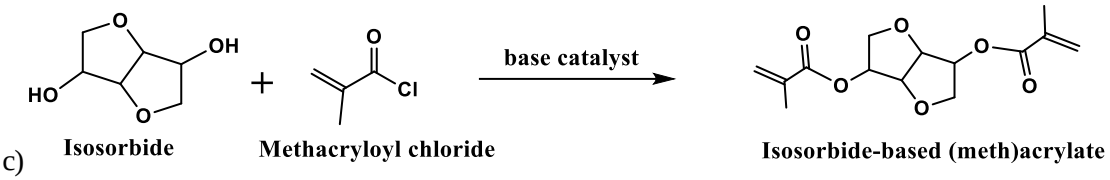
15 S. K. Bobade et al., Bio-based Thermosetting Resins for Future Generation: A Review, Polymer-plastics Technology and Engineering, 2016, Vol. 55 (17), 1863–1896.

16 T.C. Canak et al., "Boron containing UV-curable epoxy acrylate coatings", Progress in Organic Coatings, 77 (2014) 1911–1918.

17 D.N. Lastovickova et al., Resin, cure, and polymer properties of photopolymerizable resins containing bio-derived isosorbide, J Appl Polym Sci. 2021;138:app50574

18 T. Robert et al.,"Itaconic acid – a versatile building block for renewable polyesters with enhanced functionality", Green Chem., 18 (2016) 2922-2934

19 J. Huang et al., "Self-healing, high-performance, and high-biobased-content UV-curable coatings derived from rubber seed oil and itaconic acid", Progress in Organic Coatings, 159 (2021) 106391



c)

Figure 1.4. (a) Glycidylation of biomass-derived isosorbide, **(b)** reaction of epoxidized isosorbide with acrylic acid to epoxy-acrylates, **(c)** direct esterification of isosorbide towards isosorbide (meth)acrylates

UV curing of the bio-based epoxy and acrylate resins can be conducted via the two classical routes, i.e. the **free radical curing for the acrylate resins and the cationic (acidic) curing for the epoxy resins** [20]**.** UV curing mechanism of the epoxy-acrylate resins is based on free radicals which are provided by appropriate photo initiators that break apart into radicals when exposed to UV light, such as the alpha-hydroxy ketone types (i.e. Darocur 1173, Irgacure 184). In order to overcome high viscosity issues that will limit the spray process to be applied in the project, the prepolymer resin can be mixed with mono or multifunctional acrylate monomers that are acting also as reactive diluents, such as tripropyleneglycol diacrylate, trimethylolpropane triacrylate, hexanediol diacrylate, mono-acrylated isosorbide, etc. [21]. In the case of cationic curing of neat (not acrylated) bio-based epoxy resins, different type of photoinitiators will be used, which can be activated by light or heat, forming a cationic "super acid" bearing a proton (H^+) that induces ring opening polymerization reaction (ROP) of the epoxy. Typical initiators can be diaryliodonium or triarylsulfonium salts like Irgacure 250 and Curacure UV6976, respectively [22]. The distinct differences in the mechanism of radical vs. cationic UV curing leads to variations in the required curing time and sensitivity to the presence of oxygen (in radical curing), as well as they affect the properties of the cured resins, such adhesion, shrinkage, resistance to solvents, storage, post-curing, etc. These effects will be investigated towards the optimization and development of fast, reproducible and high quality UV curable bio-based resins.

As the ultimate goal, is to prepare epoxy (acrylate) polymer nanocomposite films with embedded Ni precursors, a thorough study of the incorporation of $NiCl_2$ salts as well as on their effect on the viscosity of the resin mixture and the fast UV curing process will be conducted. A series of characterization techniques will be applied to determine the final dispersion of the Ni sources (also after reduction of $NiCl_2$ by $NaBH_4$) and physicochemical properties of the nanocomposites, such as SEM, TEM, tensile and impact strength, pull off testing (to measure the adhesion strength of the epoxy-acrylate film on the polymeric substrates), TGA/DSC, conductivity, resistance to solvents, etc.

To increase the adhesion with plastic surfaces, both types of proposed resins should have reactive -COOH groups. For this reason, acrylic acid will be used in the case of acrylate epoxy resins at different concentrations for radical polymerization and itaconic acid will be used in the case of glycidylised isosorbide (Figure 1.5). UV-light induced crosslinking epoxy-itaconante resign will be easily produced in the presence of diethoxyacetophenone (DEAP), which is a photoinitiator commonly used in biomaterials. [23] The final resin will be consisted of epoxy and itaconate crosslinked bonds, having also lots of free -OH and -COOH groups (Figure 1.5) that both can increase the adhesion of formed thin layer film with plastic surfaces.

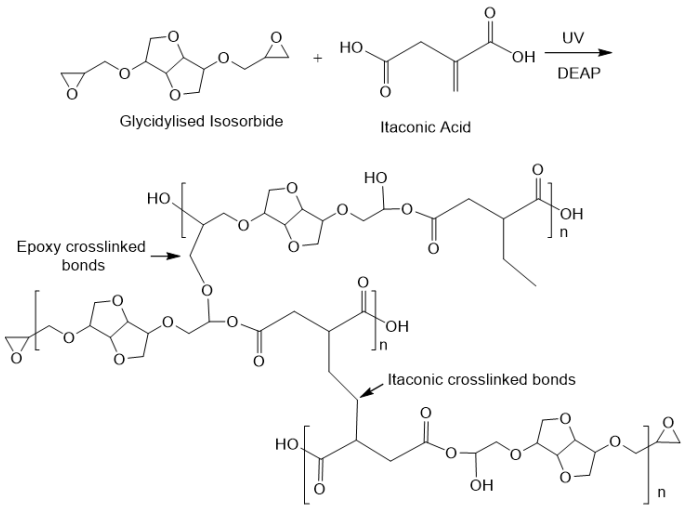


Figure 1.5 Proposed UV curable epoxy-itaconic acid resign.

B. Application of composite resins on plastic surfaces

Prior to the application of the resin the plastic substrates should be pre-treated to achieve good wetting conditions and excellent adhesion. To do so, the surface tension of the resin coating should be less that the surface energy for the substrate. It is found that the surface energy of the substrates should be ~10 dynes/cm higher than the resin to obtain optimal adhesion. [24] Plastics are comprised of long chain non-polar materials with inherently low surface

20 S. Brännström et al., Biobased UV-curable coatings based on itaconic acid, J. Coat. Technol. Res., 14 (4) 851–861, 2017

21 D. Yu et al., "Mono-acrylated isosorbide as a bio-based monomer for the improvement of thermal and mechanical properties of poly(methyl methacrylate)", RSC Adv., 9 (2019) 35532

22 P. Bednarczyk et al., "Photocurable Epoxy Acrylate Coatings Preparation by Dual Cationic and Radical Photocrosslinking", Materials, 14 (2021) 4150

23 T. Robert et.al, . Itaconic acid – a versatile building block for renewable polyesters with enhanced functionality. Green Chem., 2016, 18, 2922–2934.

24 Dr. Stefan Meets, How to Coat Plastics with UV Curable Resin, Cytec Surface Specialties, Radtech Europe 2005 4

energies. [25] However, the presence of -COOH in the resins described previously will ensure the excellent adhesion. The application of the resin in the surface can be achieved **through spaying**. It is important to apply a thin layer (~2μm) so as **not to affect the recyclability of the polymer**. For example, application of such a 2μm layer of resin in an ABS component of a total volume of 1 lt and total surface of 1000cm^2 will result in a resin concentration of less ~2ppm in the whole component. Figure 1.6 depicts SEM images of the surface of an ABS sample in which resin containing $NiCl_2$ have been applied. In higher magnifications $NiCl_2$ particles of ~2μm size are detected. During FreeMe project milling processes will be utilized in order to further decrease the size of the Ni salts particles and achieve a homogenous distribution in the resin. In-silico models will provide guidance on the proper concentration and size of the nickel salts particles for optimum distribution across the surface after the spraying step.

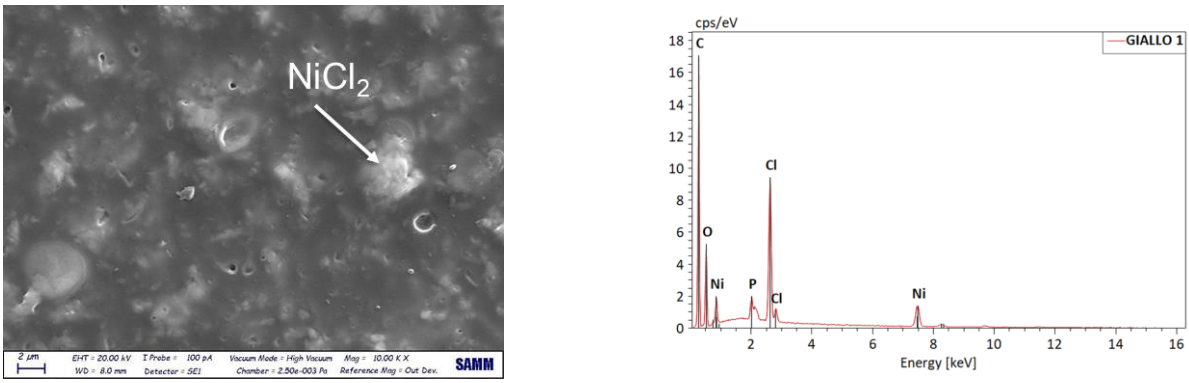


Figure 1.6: SEM image and EDS analysis of spayed resin in the surface of ABS. $NiCl_2$ particulates are detected across the surface

C. Reduction and metallization steps

By utilizing reducing agents (i.e. $NaBH_4$) a dual positive effect can be achieved, On one hand the resin is etched and cavities are formed in the surface while at the same the $NiCl_2$ is reduced to Ni nuclei (Figure 1.7)

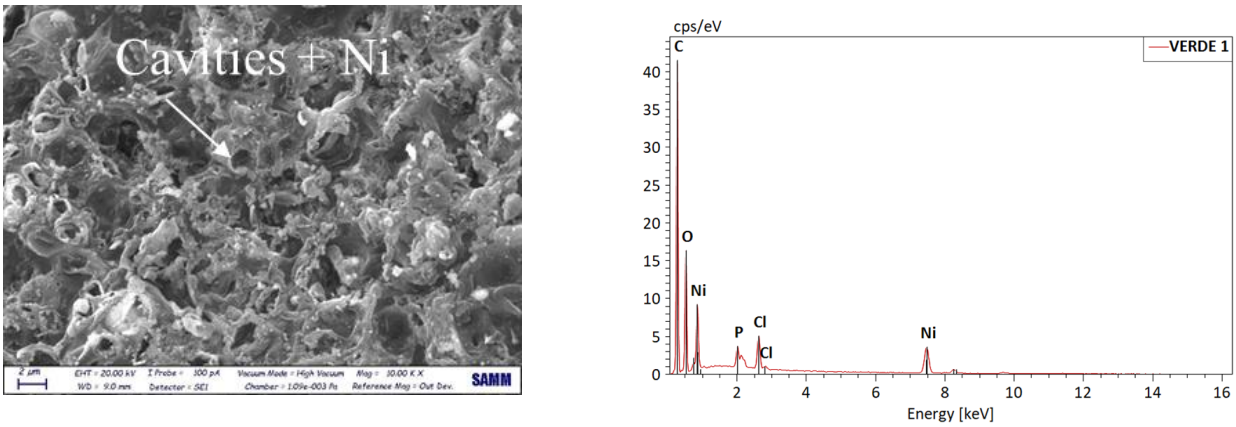


Figure 1.7: SEM image and EDS analysis of activated surface of ABS. Ni nulei are detected across the surface

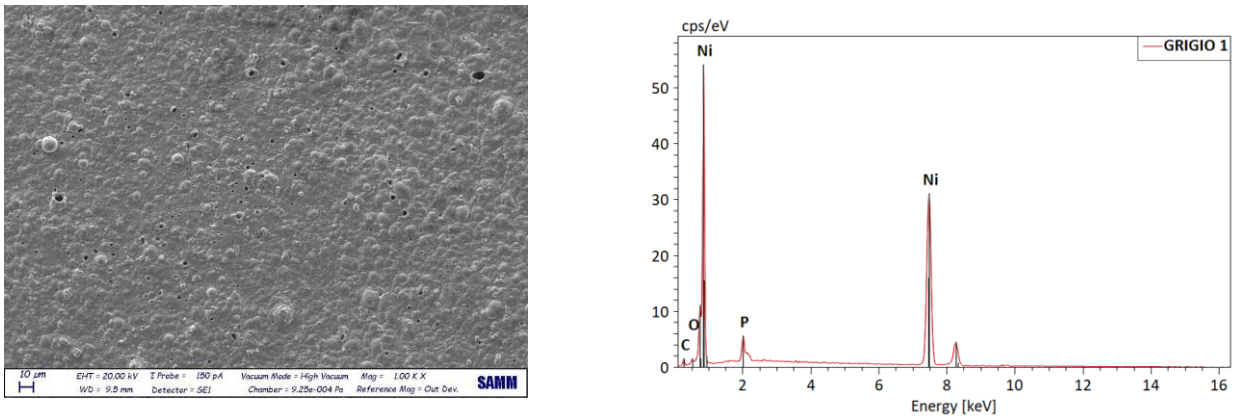


Figure 1.8: SEM image and EDS analysis of metallize surface of ABS

25 Surface Energies of Polymers. The Sabreen Group. Retrieved January 23, 2006, http://www.sabreen.com/energies_polymers.htm

The final steps for the metallization process is to apply a thin Ni-P layer through an electroless Ni bath (Figure 1.8).

In the FreeMe project **optimization of the process** will take place in all the steps of the process. It is important the spayed resin to act as an intermediate layer that will provide both adhesion and the Ni nuclei for the development of the Ni-P layer via the electroless process. The thickness of the layer is of high importance. It should be thick enough to form a continuous layer across the substrate enhancing the adhesion of the system. On the other hand, during the etching/reduction step (Figure 1.7) the $NiCl_2$ salts should be reduced at the maximum possible percentage. This will allow to have a uniform Ni-P layer across the surface.

1.2.2.3 Development of novel toxic free metallization processes

The second alternative method is based on the use of REACH compliant chemicals for **etching of polymeric surfaces** (chromic acid free process) and for the **activation step** (Pd free process). The research activities will cover the treatment of the main industrial used substrates i.e. ABS, PPA, PC as well as 3D printable resins and the emerging bio-based polymer PLA.

Main research/technological challenges	FreeMe activities to overcome challenges
Avoid the use chromic acid in the etching process	Being the core of the research activities for this metallization path, mixtures of acids or alkali will be utilized. Based on promising results, the etching process will be optimized and customized based on the exact type of polymer substrate
Achieve excellent adhesion	A double approach for achieving high adhesion level will be deployed: - Mechanical interlocking due to cavities in the etched surface - Chemical forces due to polar groups created in post-etched plastic surface
Extend the lifetime of baths	Degradation of bath's performance happens due to contaminants from drag out (transfer of chemicals from previous steps of the process) or other sources. In FreeMe project, activities related to process control and contamination reduction techniques will be implemented.

A. Development of novel, Cr^{6+} free surface etching

Chromic acid has been used in a broad range of industrial sectors for more than 50 years. The outstanding technical advantages of Cr^{6+} make it substitution difficult. Various approaches have been made utilizing acidic or alkaline permanganate, phosphoric acid with persulfate, trichloroacetic acid, nitrosyl sulfuric, etc. [26,27]. Out of these a very promising solution turns to be the mixture of sulfuric acid and hydrogen peroxide in various ratios. [28,29]
Indeed, in thermoplastic polymers such as **ABS [poly(acrylonitrile-butadiene-styrene)]**, etching of the surface may be carried out using a "**piranha solution**", a strongly oxidizing, aqueous solution of 1:4 H_2O_2 / H_2SO_4. Piranha solutions are widely used for removing organic residues from surfaces via oxidative degradation. Immersing of the ABS substrate into a piranha solution also leads to the hydroxylation of polybutadiene and polyacrylonitrile parts of the surface. [30] This chemical modification is vital because it **increases the wettability of the surface** due to the presence of highly hydrophilic groups (-COOH) and it **creates holes for subsequent incorporation of nucleation sites (activation)**. Based on the type of plastic substrate he following surface modifications can take place utilizing various mixtures of alkali or acids.

ABS based substrates

Poly-butadiene: $-(CH_2CH{=}CHCH_2)- \xrightarrow{piranha} -(CH_2CH{=}O) + (O{=}CHCH_2)- \xrightarrow{piranha} 2\ -(CH_2COOH)$

Poly-acrylonitrile: $-[CH(C{\equiv}N)]- \xrightarrow{piranha} -[CH(CONH_2)]-$

PET substrate

P-ethylene terepthalate: $-(O(C{=}O)\text{-}Ph\text{-}(CO)OCH_2CH_2)\text{-}_n \xrightarrow{piranha} HCOO\text{-}Ph\text{-}(CO)OH + -CH_2\text{-}CH_2\text{-}OH$

26 J. McCaskie et.al, "New hexavalent Cr free etch for ABS and ABS/PC electroplated plastics", Plating and Surface Finishing, April, 2008
27 D. Akhouri, "A review on the plating process of FDM built parts", Materials Today: Proceedings, 26, 2020, https://doi.org/10.1016/j.matpr.2020.02.461
28 S. Whitmore "Consumable Spacecraft Structures with Integrated, 3-D Printed ABS Thrusters" AIAA 2017-4707, Session: Additive Manufacturing for Propulsion Systems II, 2017, https://doi.org/10.2514/6.2017-4707
29 E. Correa et.al., "Activation, Initiation, and growth of electroless nickel coatings", Taylor & Francis Group, 1st Edition, 2019. eBook ISBN9780429466274
30 X. Tang, C. Bi, C. Han, and B. Zhang, A new palladium-free surface activation process for Ni electroless plating on ABS plastic, Materials Letters 63, 840-842 (2009)

The treated substrate should be rinsed with water and immersed into a sodium bisulfite aqueous solution ($NaHSO_3$), a common industrial reducing agent and antioxidant food additive to remove any piranha residues. [31] The whole etching and cleaning process is a relative fast process that can be easily upscaled.
Concerning **UV-Curable resins (i.e. polyutethane acrylate resins) & bio-based substrates (PLA):** The surface etching of these materials is carried out using an aqueous solution of potassium hydroxide (KOH). [32] The time and cost of the etching process has been optimized via a comparative study. Alkaline KOH or other alkali solutions partially hydrolyze the ester and carbamate bonds of the resin to form a hydrophilic, rough surface which promotes metal adhesion. The resin substrates are subsequently rinsed with water. The whole etching process lasts approximately 30 min.

UV-curable resins

P-acrylate: $–(R^1\text{-}COO\text{-}R^2)– \xrightarrow{OH^-/H_2O} –(R^1\text{-}COOH) + (HO\text{-}R^2)–$

P-urethane: $–(R^3\text{-}NHCOO\text{-}R^4)– \xrightarrow{OH^-/H_2O} –(R^3\text{-}NHCOOH) + (HO\text{-}R^4)–$

PLA substrate

P-lactic acid: $\text{-}(CH_3)CCOOR)\text{-}_n \xrightarrow{OH-/H2O} \text{-}(CH_3)CCOOH + ROH$

B. Development of novel, Pd free activation processes

The surface activation step is a crucial part of the PoP process. This can be achieved by using a nickel acetate/sodium borohydride solution. $[Ni(CH3COO)_2]$ is significantly cheaper than $PdCl_2$ and less toxic than Sn salts. [33] Finely dispersed nucleation sites are generated by the reduction of Ni^{2+} by $NaBH_4$, as follows (Figure 1.9):

$$4\ Ni^{2+} + BH_4^- + 8\ OH^- \rightarrow B(OH)_4^- + 4Ni^0 + 4\ H_2O$$

Metallization of the ABS substrate is completed by immersing the pre-treated substrate into a Ni electroless plating bath containing nickel sulfate as the Ni^{2+} source and sodium hypophosphite ($NaPO_2H_2$) as the reducing agent.[33] The release rate of free Ni(II) cations in the solution is controlled by co-addition of sodium citrate ($Na_3C_6H_5O_7$) as a complexing agent. [34] A fine Ni-P metal homogeneous coating is produced, shown in Figure 1.10.

$$Ni^{2+} + 3\ H_2PO_2^- + 3\ H_2O \rightarrow Ni^0 + 3\ H_2PO_3^- + 2\ H^+ + 2\ H_2$$

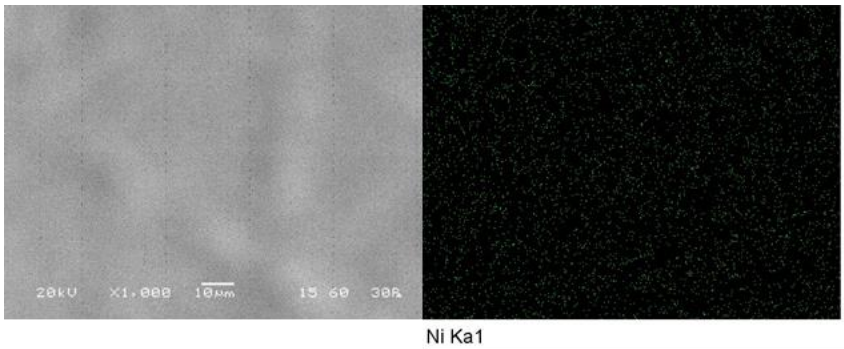


Figure 1.9: EDS mapping activated surface Green dots represents Ni Nuclei

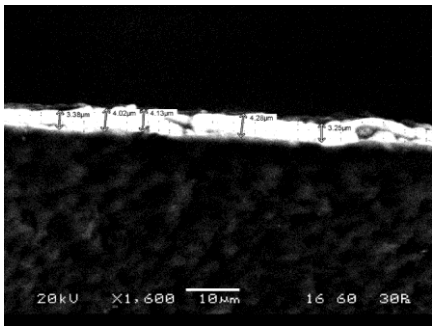


Figure 1.10: Cross section after EN

C. Baths Qualification and maintenance

Spent baths, i.e. baths that can no more be used in the plating process due to low efficiency, are a main source of waste streams of the surface finishing industry. Spent baths are either treated in-house or transferred to off-site waste treatment installations, increasing thus chemical use for treatment, and generating sludges and leading to higher operation costs and environmental/safety risks. Thus, **extension of the lifetime of baths is** of crucial importance for the SSBD strategy of FreeMe technologies. There are a number of factors that leads to the degradation of a bath

31 A. Equbal, N. K. Dixit, and A. K. Sood, Electroless metallization of ABS plastic: a comparative study, International Journal of Rapid Manufacturing 5, 3-4, 255-274 (2016)
32 R. Bernasconi, C. Credi, M. Tironi, M. Levi, and L. Magagnin, Electroless Metallization of Stereolithographic Photocurable Resins for 3D Printing of Functional Microdevices, Journal of The Electrochemical Society **164**, 5, B3059-B3066 (2017)
33 R. Suman, D. Nandan, A. Haleem, S. Bahl, and M. Javaid, Experimental study of electroless plating on acrylonitrile butadiene styrene polymer for obtaining new eco-friendly chromium-free processes, Materials Today: Proceedings **28**, 3, 1575-1579 (2020)
34 V. K. Bulasara, H. Thakuria, R. Uppaluri, and M. K. Purkait, Nickel-ceramic composite membranes: Optimization of hydrazine based electroless plating process parameters, Desalination **275**, 1-3, 243-251 (2011)

performance: (a) depletion of chemical species (b) not balanced bath chemistries (c) contamination from other chemical species (i.e. drag out: transfer of droplets from previous steps to the tank of concern). [35] In the framework of the FreeMe project, qualification protocols for the various types of baths used will be delivered. This will be based on SoA technologies for such protocols as well as measurements for decreasing the baths contamination.

1.2.2.4 Application of final finishing

After the metallization process (i.e. the application of electroless thin coating on the surface of the activated polymer) the final finishing is applied via electroplating method. This final layer is deposited according to the specifications provided by the end users. The most common finishes are decorative chromium (applied through trivalent chromium baths), nickel and nickel alloys, and copper. For specific applications other type of metals can also be plated (i.e. gold, silver). In the FreeMe project the final finishing will be arranged by the three end users.

1.2.2.5 In silico approaches for optimization of FreeMe processes

Main research/technological challenges	FreeMe activities to overcome challenges
Complete modelling of material and processes related to interfaces	Modeling and simulation activities will be selected to answer the key product development questions rather than pure research-oriented modelisation.

The overall goal of the in-silico approach proposed in the FreeMe project is two-fold: First, the in silico activities will target the support of the proposed components design, covering the sprayable composite resins and the etching process. This will be done by developing multi-scale simulations ranging from Ab Initio, Atomistic, Mesoscale and Continuum modelling of the involved materials and processes to provide a proper Design Of Experiments (DOE). Second, such developed models will be used to support the further analysis of the suitability of the proposed processes to specific applications. These modelling activities will be done in parallel and in strong collaboration with the proposed experiments to provide a continuous improvement process of both the developed models and the efficiency of the experimental campaigns.

In the case of the **sprayable composite resins**, the scope of the simulation activities is to understand the capability of the produced resin to adhere to the target materials, analyse the distribution of the Ni nuclei into the target surface and the later efficiency of the final plating process. To do this, ab initio quantum mechanics (QM) tools and classical molecular dynamics (MD) will be used to determine the interfacial interactions and the corresponding adhesion properties of the different analysed resin compounds. These analyses will allow gathering further conclusions on the material experiments and the results of the SEM image and EDS analyses, specially related to adsorption sites and geometry of the resin molecules and the adhesion extent. The different methods proposed for increasing the surface energy of the plastic parts (e.g. surface oxidation of the polymer chains) will also be taken into account through Monte Carlo simulations. Such models will be then combined with coarse-grained and continuous models to analyse the influence of the concentration and size of nickel salts particles within the proposed resin to ensure a suitable distribution of the nuclei across the surface.

Regarding the **plastic surface etching process**, the overall objective of the in silico activities is to support the design and selection of the different researched piranha solutions. To do so, a combination of Chemical, Molecular Dynamics, Monte Carlo and Computational Fluid Dynamics (CFD) models will be used.

1.2.2.6 Development of Inspection technologies

A number of monitoring techniques exist when it comes to characterizing structures during the inspection and quality control in R2R environment for organic solar cells & OLED displays; in the nano- and micro-meter ranges, which have been utilised in the development of the LIPSS fabrication technique, including SEM (scanning electron microscopy) and AFM (atomic force microscopy) [36,37]. Fourier transform analysis of the SEM images have been used to quantify the periodicity of the structures. This approach enables a global value of spatial period to be quickly calculated, rather than estimating line thickness and spacings line by line, and generating a mean value, which is a more laborious process.

However, most commonly used techniques at this dimensional scale are in fact implemented off line and require sample preparation, Optical techniques offer the best opportunity for inline process monitoring, as they are non-contact, non-destructive and can be applied to samples as they are without pre-treatment. According to a review paper

35 W. Wallace, “An introduction to electroplating in engineering and manufacturing” SAT Plating, October 2020.

36 A. Rodriguez, M.C. Morant.Minana, A. Dias-Ponte, A. Martinez-Calderon, M. Gomez-Aranzadi, S.M. Olaizola, Femtosecond laser-induced periodic surface nanostructuring of sputtered platinum thin films, Applied Surface Science 351 (2015) 135-139

37 A. Martinez-Calderon, A. Rodriguez, A. Dias-Ponte, M.C. Morant.Minana, M. Gomez-Aranzadi, S.M. Olaizola, Femtosecond laser fabrication of highly hydrophobic tainless steel surface with hierarchical structures fabricated by combining ordred microstructures and LIPSS, Applied Surface Science 374 (2016) 81-89

by IRIS[38], selected combined optical technologies, such as diffractometry, but also ellipsometry and spectral reflectance provide the opportunity for inline measurement of the thicknesses of micro structured layers.

Main research/technological challenges	FreeMe activities to overcome challenges
Difficulties on measuring coating thickness due to light scattering of particles	The combination of various techniques and optimal calibration off-line of the system will allow to overcome this challenge. The speed and time of the process would already bring a certain idea of this data.
Relation between quality of the layer and process parameters	Correlation between Real-time data gathered and parameters for further processing.

1.2.2.7 Development of Decision Support Tool

The Decision Support Tool will be a solution for industrial end-users to optimally **select and/or re-engineer existing plating production lines** by considering te**chnical, economic, safety, environmental and sustainability constraints** and criteria to maximize efficiencies in terms SSBD. The tool will also propose tuning solutions of the everyday production with respect to the variable input specifications of plastic matrices. The user will be capable of inserting unit operations, materials and products specifications as well as experimental data providing a complete profile of the existing production line. The DST processes the data structuring a dynamic process representation to be optimized considering alternative SSBD technologies (Cr^{6+}etching and spraying), tuning conditions and downstream processing that should be appropriately designed and adapted to the existing lines. For this purpose, the DST exploits in-silico models, experimental data and products/production specifications ensuring SSBD, as prepared in contexts of FreeMe, building a technological database to simulate the overall plating operations of the alternative etching and spraying technologies and test their performance in front of user requirements. Data driven models will correlate experimental data with performance of the plating processes and contribute to fitting and optimizing operating conditions (i.e. method of plating, type of coatings) with respect to user-defined input specifications for plastic matrices (type of polymer, adhesion values, hardness, wear resistance etc.) and the desired end-product specifications. The DST also holds three modules for safety, LCA and LCC analyses that will translate process simulation outcomes into mid- and end-point environmental indicators (e.g. human toxicity, climate change, resources depletion), techno-economic results (CAPEX, OPEX, ROI, etc) and safety issues – based on Common Safety Methods (Article 6 of Directive (EU) 2016/798) – like metals concentrations in the factory, baths' lifetime, Number of Reported Accidents. The overall analysis is based on a dynamic superstructure that holds options for the user-defined operations (e.g. the existing industry) and the candidate SSBD plating options of FreeMe. The optimization framework examines the alternative designs and tuning options (e.g. nozzles diameters, spraying concentrations and flows, type of acids or alkali and concentration, qualification of baths) and offers optimal solutions and guidelines to revamp and adapt an existing technological environment with new technologies ensuring SSBD PoP and following safety, life cycle and sustainability criteria (objectives in optimization).

1.2.2.8 Validation and demonstration activities

A. Automotive: Decorative applications for car components (CRF)

PoP is a technology of high importance for the automotive industry since the '60s. CRF has introduced various plastic components in its automobile production lines which are lighter but equally robust with their metallic counterparts. A variety of these interior (i.e. instrument frames, steering wheel decoration, door handle) and exterior parts (e.g. door handle, bumpers) are made from ABS, PC-ABS and PA polymeric materials. CRF applies various "Cr^{6+} based processes" for both protective and decorative functions depending on each plastic component's position and utility. The chromium coatings typically consist of more than one Cu- and/or Ni-based interlayers, deposited underneath the final Cr layer. However, the process requires a starting etching of the surface using Cr^{6+} solutions and Pd/Sn colloidal solutions for surface activation. CRF is seriously interested in developing Cr^{6+}-free processes for metallization of bio-based, virgin or recycled grade plastic materials.

Figure 1.11:. Interior decorative high contactable components by CRF.

38 E. Bugnicourt, T. Kehoe , M. Latorre, C. Serrano, S. Philippe and M. Schmid, Recent Prospects in the Inline Monitoring of Nanocomposites and Nanocoatings

B. Aerospace: Electromagnetic shielding of 3D printed component (IAI)

The use of electric components and electromagnetic interference (EMI) shielding in the aerospace industry is growing rapidly. This includes antenna components, higher power requirements that need to be electrically isolated by shielding. In addition, particular antenna components require a complicated albeit highly ordered structural arrangement which in turn requires the assembly of multiple individual parts. On top of that, these parts must meet the most common and strict necessity of the aerospace industry, low weight, as low as possible. As a result, aerospace engineers need to shave every possible gram from every component as excess weight accumulates rapidly. To this end, the replacement of even a single metal component with a plastic one could be key to reduce weight and final cost. Moreover, employment of additive manufacturing techniques (by far less expensive and efficient for polymers compared to metals) can significantly reduce the number of the individual parts needed for the final assembly and decrease the manufacturing time of complicated arrays. Therefore, metal plating on plastics is key for the application of a protective and conductive metallic layer to enhance both lifetime and EMI shielding. FreeMe will develop a plating process for 3D-printed antenna depicted in Figure 1.12.

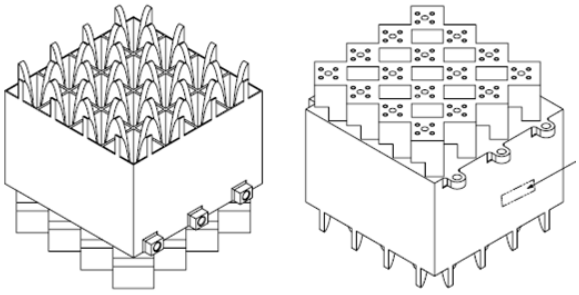


Figure 1.12: Drawings of 3D printed antenna parts

C. Home Appliances: Washing machine frames (ARC)

Home electrical appliances and consumer electronics govern almost every aspect of our lives. The incorporation of inexpensive plastic components can reduce cost and lower weight, therefore enabling manufacturers to offer their goods at more competitive prices. In addition, plastic parts may serve as electricity and heat insulators for certain home appliances such as a washing machine which in turn mitigates health and safety risks. Innovative plating on plastics aims to provide a decorative and protective metal finish on such components. Metal coatings enhance corrosion resistance and improve external appearance, therefore generating added value. For instance, ARC applies nickel, nickel-chromium and multi-layer (nickel-copper-chromium) coatings on the door, the handle and the buttons of washing machines. However, in all cases the metallization process involves etching process with chromic/sulfuric acid and activation with Pd. Shifting to more environmental processes, the company wants to replace this process.

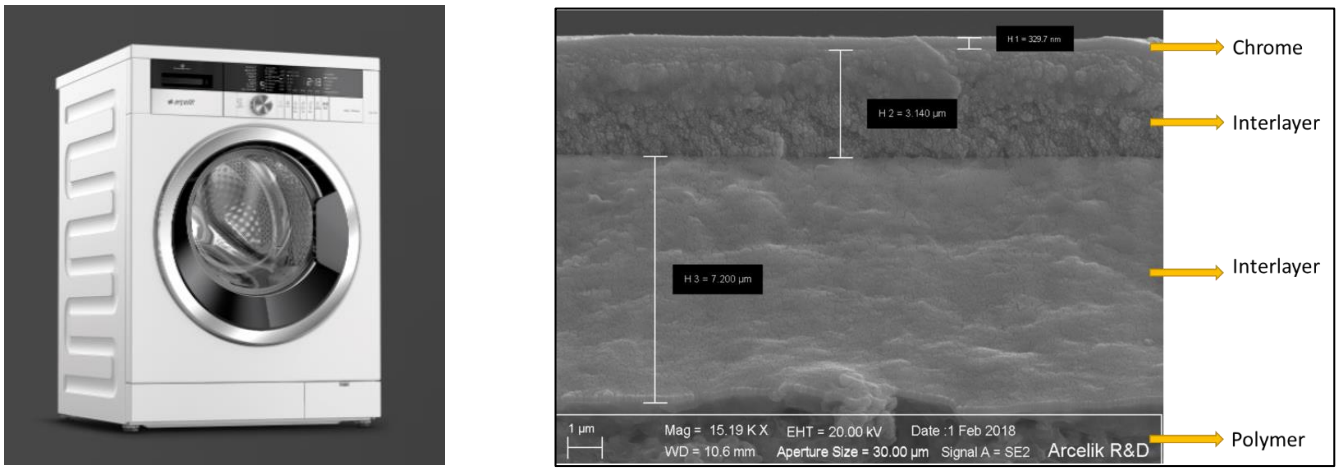


Figure 1.13:. Arcelik's washing machine (left) and SEM image of a multi-layer chrome plating application on ABS.

1.2.2.9 Recyclability of plated plastic components – EoL scenarios

Removal of the metal coatings from plastic substrates is well established technology, especially for coatings applied in the automotive industry, that is part of the dismantling process. [39] In general three methods are applied. 1) abrasive blasting, 2) cryogenic embrittlement and 3) mechanical crushing. **Abrasive blasting**. is the operation of forcibly propelling a stream of abrasive material against a surface under high pressure. **Dry ice-blasting** is a special type of abrasive blasting where dry ice (solidified carbon dioxide) will be accelerated in a pressurized air stream and directed at a surface to clean it. **Cryogenic embrittlement** may also be used to remove metallic coatings from plastic substrates. The recovery system involves feeding metal-plated plastics into a grinder. The grinder contains a pre-crusher which breaks up the parts into smaller pieces and feeds them into the grinder which subsequently reduces the scrap from half-inch pieces to coarse powder. A magnetic drum separator removes the pieces of metal that have been

39 E.van Westing et.al, "Recycling of metals from coatings", Materials innovation institute M2i, November 2013

split off during grinding. The final step contains going through a cryogenic feeder, where liquid nitrogen cools the scrap to almost freezing, creating stresses which cause additional metal to splinter off. Finally, metal-plated plastics, could also be separated using **mechanical crushing**. Coatings could be easily stripped when the items are subjected to shear force and impulsive force. Concerning the repair of partially damaged (scratched) metal-plated plastics, **brush plating**, a portable method of electroplating without the use of an immersion tank, will be tested and employed. In the framework of FreeMe project, the recyclability of the plated plastic parts will be validated utilizing the SoA recycling methods.

1.2.3 Relevant research and potential synergies

The FreeMe project will either be based on results of previous project or will create synergies with such projects. The following table presents these projects and analyse the links with the FreeMe project.

Table 1.2: National and International projects correlating with FreeME

Project	Link to FreeMe
PureNano (821431, H2020)	**Gaser, PoliMi, Cnano, Gaser** The project focuses on the development of an in-house system for plating baths purification of plating lines. Cnano, ISQ, Gaser and PoliMi, partners of the project, will transfer this technology to TreeMe, targeting to a more sustainable and environmentally friendly lines.
3DPlate (National)	**Cnano** is already providing a range of 3D printing services using environmentally benign and sustainable solutions to artists and designers. Importantly, collaboration with the famous Greek artist Costas Gaïtis has already been established and Cnano is currently producing Ni metal coatings for some of the artist's resin-based medium-size sculptures.
PROCETS (686135, H2020)	**PoliMi Cnano** has a major role in the project targeting the delivery of a sustainable electroplating method for applying hard protective coatings.
LightMe (814552, H2020)	**PoliMi** are collaborating in the OITB LightMe that is dealing with the upscale of production processes of lightweight alloy composites. LightMe is a test bed that will interact with FreeMe targeting the optimization of lightweight metal surfaces.
INFUCOMP (FP7, 233926)	**IAI** The project focused on the development of simulation-based solutions for the industrial manufacture of large infusion composite parts, covering all popular Liquid Resin Infusion (LRI) methods used in the Aerospace industry. IAI was one of two aircraft manufacturers that were participating in the project.
DIAGONAL (953152, H2020)	**ICCRAM, Cnano** DIAGONAL addresses nanotechnology safety issues by developing SbD tools and guidelines. ICCRAM is the coordinator of the project and contributes to activities related to modelling, risk assessment and sustainability. ISQ will perform exposure assessments to the various industrial demonstrators. IRIS is involved as a pilot line. Cnano is an industrial case for data generation and output validation.
SafeMarble (co-funded, EU & National)	**AUTH** is involved in the project related to the improvement of the properties of commercial protective agents/ epoxy polymers using (nano)additive materials
OLEDSOLAR (820789, H2020)	**IRIS** brings its knowledge for the development of in-line monitoring techniques for the OLED and thin film and organic photovoltaic industries.
NANOGENTOOLS (691095, H2020)	**ICCRAM** has a major role in the project targeting the development and implementation of a new generation of nanosafety assessment tools.

1.2.4 Inter-disciplinary approach

It is obvious that an interdisciplinary approach is necessary to deliver the project's objectives, covering aspects of chemistry, physics, modelling, engineering, data driven algorithms and sustainability. The FreeMe consortium comprises R&I experts in the above disciplines, as well as industrial partners in the PoP field. The first interdisciplinary dimension (IDD1) stems from the collaboration between AUTH (organic chemistry – polymers) with PolyMi (materials chemistry – plating on plastics) to deliver the biobased sprayable resin. IDD2 connects the above world-leading research institutions with two chemical engineering oriented SMEs (Gaser, Cnano) whose expertise lies in PoP and nanotechnology to produce high-quality PoP coatings. This will be guaranteed by IDD3, running in parallel with IDD2, which brings together the disciplines required for the in-line monitoring (physics and spectroscopy, IRIS) and the in-silico optimization (computational modelling, IDENER) of the PoP processes. IDD4 will establish a two-way communication between sustainably experts (ICCRAM) with all partners of the consortium to adopt SSbD principles, perform necessary LCAs and ensure compliance with REACH directives. IDD5 encloses three different and highly competitive industrial sectors (i.e. automotive, aerospace and home electric appliances) that will validate FreeMe's disruptive techonology. Finally, IDD6 concerns the Dissemination, Commercialization and Exploitation of the project results (Exelisis).

1.2.5 Gender considerations

Within the FreeMe project, special consideration will be paid regarding the gender dimension in the research and innovation process. Identification of gender differences in the working environment is crucial for the implementation of the necessary measures to ensure the health and safety of the researchers and workers. Utilization of Plating on Plastics process in FreeMe requires the careful handling of chemical substances. Sex and gender analysis will be performed during the implementation of the project and key factors are already identified from the proposal stage. Under this scope, the magnitude and duration of an employee's exposure to chemical substances can be gender-depended, as social and occupational roles of women and men might differ. Women were mistakenly considered to undertake less hazardous tasks and might not be properly included in the risk assessment of specific procedures, or not received the proper size of personal protective equipment (PPE). [40] FreeMe will ensure that all researches will be equally approached in terms of personal safety, with appropriate PPE that will ensure the protection and reduce the risk of exposure to Nickel and corrosive substances, such as H_2SO_4/H_2O_2 and KOH. Another aspect that will be examined is the impact of exposure to the above-mentioned chemicals on both sexes. Biological characteristics differ between women and men, leading to different health effects of toxic chemicals. In particular, Nickel can induce skin allergies, contact dermatitis and hand eczema, which are statistically more common in women. [41] Additionally, it is more likely to be stored in female tissues and bioaccumulate. What is more, special attention is required to female workers during childbearing years and pregnancy. Women are affected dramatically during this period by even lower doses of chemicals. Finally, data on occupational carcinogens have presented an increase in cancer rates of female workers in chemicals, raising concerns about the different cellular response to oxidative stress among sexes. [42] Partners of FreeMe will adopt a responsible approach regarding the gender dimension on health and safety matters.

1.2.6 Open science Practices

Adopting Open Science Practices in EU-funded research and innovation has been a fundamental and unnegotiable part of research ethics in these programs, ensuring transparency and accessibility to public funding outcomes. There is a number of benefits to be taken into account, from providing information to the society about the research and innovation process and its particular results to assuring the general public and in particular European citizens and taxpayers of a evidently transparent, well-guided and efficient R&D policy on a European level. From the perspective of academics and researchers, those open science practices can have significant impacts on the ability of our society to innovate, surpassing barriers and rendering research outcomes and technological developments findable, accessible, interoperable and reusable (**FAIR**). In the last decades, adoption of open science practices has benefited in terms of quality, efficiency, transparency and social acceptance of research in Europe, in addition to making scientific progress faster and with more integrity. FreeMe is adopting and strongly supporting Open Science practices to the maximum extent. In particular, scientific publications (papers) produced within the project will be published as open access (gold or green access), in full compliance with the Horizon Europe guidelines. In addition to that, an Open Access Data Repository like Zenodo (https://zenodo.org/) will be used in order to store and maintain publicly available all open access material (as it will be also mentioned in the next section). Zenodo, upon acknowledgement of the HE grant, is automatically connected with OpenAire (https://www.openaire.eu/) and with the Open Research Europe platform (https://open-research-europe.ec.europa.eu/browse), adding more visibility to the project public results. The goal is to guarantee the validity, the integrity and the reproducibility of the results, while also facilitating any derivative or relevant works that will be free to use, integrate and/or build on FreeMe's public results. FreeMe is also aiming to invest quite a lot in citizen engagement as part of its open science practices. The public will be engaged through various communication and dissemination channels (see section 2.2) where an integral part of this approach is to also receive feedback and integrate the general public into the research and development decision making. Actions like visits to local educational institutions, creating public polls, and engagement with consumers of end-products will be taken into consideration on a central level. Furthermore, a key aspect of open science adoption in FreeMe is its Safe and Sustainable by Design approach (SSbD). The SSbD assessments involve open science concepts by default and as part of FreeMe's exploitation planning, as they aim to the applicability of their designs and guiding principles to other applications by extension or adoption of this work. Work containing SSbD criteria, best practices and relevant results will be released as open as possible to ensure boosting to the project's commercial exploitation but also to support open science principles and facilitate the above mentioned benefits. As a matter of fact, this is a key aspect of the project and a critical factor for its successful implementation and impact maximization. Moreover, the academic partners (RTDs) of FreeMe have a long history of performing research activities using open science practices and integrating state-of-the-art methodologies and advancements in educational modules for students and/or researchers. This approach will be further boosted in FreeMe. Within the project's dissemination, communication and exploitation measures, partners will integrate ideas and novel methodologies to educational activities, hire and train young

[40] http://gender-chemicals.org/chemical-safety-at-work#_ftn1

[41] https://www.undp.org/content/dam/aplaws/publication/en/publications/environment-energy/www-ee-library/chemicals-management/chemicals-and-gender/2011%20Chemical&Gender.pdf

[42] Ali I, Högberg J, Hsieh JH, Auerbach S, Korhonen A, Stenius U, Silins I. Gender differences in cancer susceptibility: role of oxidative stress. Carcinogenesis. 2016 Oct;37(10):985-992. doi: 10.1093/carcin/bgw076. Epub 2016 Jul 31. PMID: 27481070; PMCID: PMC6276900.

researchers, creating a highly skilled specialized workforce and motivating young researcher to further explore these advancements. At the same time, (in detail in section 2.2) 3 workshops will be held to promote the advancements of FreeMe and contribute to its open science activities , openly engaging with the community and relevant stakeholders.

1.2.7 Research data management

To manage the research data generated, collected, and processed during FreeMe multi-disciplinary project requires careful consideration. Different types of data will require unique data governance procedures, including technical and open-science policy requirements for data collection, management, preservation and sharing. A data management plan (DMP) defining the governance of each types of data, including standards, formats, requirements and responsibilities will be delivered at M6 in the corresponding deliverable and will be updated with the progress of the project. In the following the main aspects of this DMP as presented as a draft version. FreeMe in general will use common and non-proprietary data formats to enable interoperability within the project across other data sets. The project will need to consider the governance of each of the different datasets in relation to legal and policy issues. IPR issues will be raised in relation to the ownership of data supplied as well as the ownership of data in the FreeMe project. The partners will use the CA as an initial plan to manage the ownership of the data generated within the project. Issues related to research ethics, privacy and legislation will also be considered. The DMP will include which data should be collected, obtaining informed consent and governing how the data should be processed, stored and shared. The DMP will evaluate the extent to which the data can be made openly accessible to other researchers and stakeholders via an open access repository. FreeMe will make use of the Open Research Europe platform to freely share publicly available research data as described in the previous section. To aid the preservation and re-use of the FreeMe data, the Zenodo repository connected to the OpenAIRE platform will be used ensuring the integrity and replicability of publicly available project data and results at the early stage of the project. The consortium will follow standardized guidelines in relation to authoring appropriate and useful metadata incorporating them into the DMP. Standardized formats for data sharing, and non-proprietary formats where possible – for example, .txt in respect of interview transcripts originally produced in Microsoft Word. As such, will avoid access barriers related to proprietary software will be avoided. DMP will include metadata information regarding all relevant datasets, addressing the aforementioned descriptive information (e.g. ownership, type, origin, purpose, availability, reusability, relation with project objectives and WPs etc.) and all FAIR aspects of the enlisted datasets, while the DMP will also take into consideration all GDPR guidelines. Finally, the DMP will align with FreeMe's exploitation plan aiming to best accommodate the (post-project) data exploitation. The DMP will use these as a starting point to outline which partners can exploit the data, how the data can be exploited and also find possible additional ways in which data of FreeMe could be exploited beyond the initial confines of the project – e.g., using data for new and unforeseen purposes.

Table 1.3: *Draft DMP*

Core information to be provided in detail within the DMP	Draft research data management plan provision of metadata
Types of data/research outputs	Data on: Naming, definition, description, type, purpose and connection with WP. All this metadata to be publicly available
Findability of data/research outputs:	All data are to be FAIR, with the exceptions imposed only by particular commercialization interests, when properly justified and in compliance with the guidelines of the Horizon Europe. All this metadata to be publicly available.
Accessibility of data/research outputs	
Interoperability of data/research outputs	
Reusability of data/research outputs	
Curation and storage/ preservation costs	Data on: Storage, preservation, curation, associated costs, quality protocols. All this metadata to be publicly available.

Table 1.4: *Draft data generating processes assessment*

WP	Processes	Type of data	Purpose
1	Specs and Requirements	Texts, tables numeric ranges,	SSbD, material and process development
2	Sustainability, recyclability, tech. design	Designs, data basis	SSbD, material and process development and optimization, tech. development
3	Resin development	Numeric tables, images, raw data from instruments, texts	Assessment of experimental protocols, feeding process simulation
4	Metallization development		
5	Process modelling	Data basis, numeric, models	Process optimization, SSbD
6	Product validation	Images, videos, Numeric, Comparative report	End-user exploitation, certification, specs and requirements
7	Standardization, DST	Models, Process reports	SSbD, policymaking, Market uptake

2. Impact

2.1 Project's pathways towards impact

2.1.1 FREEME Key Impact Pathways

The FreeMe project is dealing with a **real industrial need** in the field of surface finishing and more specifically in the Plating on Plastics process: The elimination of use of chromic acid and palladium. The project objectives are in full accordance with expected outcome of the project that is the development of new metal coating systems produced without the use of toxic substances (i.e. Cr^{6+}) and PGMs (i.e. Pd). It is important to state that heavy metals or other toxicants release in the environment is mainly happened from the **plating installations** where these compounds are in a dissolved state (i.e. ions) through wastewater and sludges, rather than from the coatings, which are in a solid stable state, of the plated objects. [43] More specifically, the FreeMe project's results will contribute towards the main expected outcome and impact set by the EC as follow:

Safe & Sustainable by Design approach

FreeMe will develop two alternative coating solutions for PoP applications. As mentioned, the current practice is based on the **use of chromic acid (Cr^{6+}) and Pd**. The new technologies will provide the surface finishing industry with **non-toxic processes, free of the use of critical raw materials** (i.e. Palladium) enhancing thus the safety in the occupational environment and the environment as well as the overall resilience and sustainability of the process. Both technologies are based on REACH compliant chemicals and no listed compounds (i.e. in the list of SVCH) will be utilized. For example, **formulation of new UV-curable resins utilizing bio-based chemical platforms** (i.e. isosorbide and itaconic acid) will take place instead of the use the Bisphenol-A compound that is recognized as a toxic for reproduction and endocrine-disrupting substance that may cause adverse effects to people's health and to the environment and is listed in SVCH. Additional, mixtures of common industrial used acids or alkali will be used in the second path instead of the chromic acid. Elimination of Cr^{6+} from the PoP will be a significant innovation towards safer and more sustainable surface finishing industry.

Scale: The **global market** of **chromic acid** was estimated at ~600 million € during 2020, with ~8% to be used in plastic etching processes. [44] Based on the CrO_3 price (i.e. 22€/kg) this is translated to ~2.100 tones per year. **Europe** has a share of ~25%, thus, **525 tones of Chromic acid are consumed in Europe annually for PoP purposes**.
Significance: Reduction of chromic acid etching process usage will have a **tremendous impact on the PoP industry** in terms of environmental and occupational health issues. **Cr^{6+} has been proven to be a toxic and carcinogenic compound** affecting not only the plating workers but also the citizens around the plating plants (i.e. the famous real case of Erin Brockovich against the Pacific Gas & Electric Company that operated chromic acid processes). Exposure to Cr^{6+} results in excess risk of lung or other type of cancers [45] Wide adoption of SSBD technologies of FreeMe will significantly reduce such incidents in the European PoP in the long term, targeting in zero deaths after 30 years.

New Materials Development with equal or enhance properties compared to SoA

The POP is the main technology that can provide decorative and functional finishing in plastic surfaces. Practically most POP applications concern decorative finishes, while a small part is for engineering purposes in which properties such as hardness, wear and corrosion resistance or EMI shielding. In all the cases, **of primary importance is the excellent adhesion of the electroless coating in the surface of the plastic**. Low adhesion will lead to pre-mature failure of the component in all cases. Thus, all studies towards the direction of novel PoP technologies focuses mainly on this property. The FreeMe will provide technologies that will have superior adhesion as well as other benefits.
In the case of s**prayable resins,** containing Ni salts free reactive -COOH groups that acts as anchoring groups to plastic surfaces enhancing thus the adhesion of the resin to the substrate will be inserted. In the following steps the activation of Ni sites and the electroless plating (formation of Ni-P layer) will take place. The adhesion of the latter layer is of crucial importance for the coating's performance. **Its excellent adhesion is ensured based on the distribution of the Ni nuclei that are mechanically entrapped in the resin matrix.** Another benefit of the proposed technology is that PoP becomes a **substrate-independent process** in terms that after the deposition of the resin thin film in the substrate surface, the following steps involve the sprayed resin and not the substrate. This could be of extremely high industrial importance as it can be **the basis for metallization of various types of surfaces**. On the other hand, the adhesion of the coatings prepared through the green etching process is expected to as effective as the current chromic acid etching route, based on double mechanism. At micro level **mechanical interlocking** of the metal film on the etched polymer surface takes place, while at molecular level **chemical forces** (i.e. Van der Waals attraction of valence bonds of metal with polar groups of plastic resulted from the etched solution) are developed.
Scale: **The global market for Plating on Plastics was estimated to be ~580 million €** (2020) while the **Europe share was ~190 million €**. There is a strong increasing demand for the technology as it combines lightweight, low cost and high aesthetic, especially in applications in automotive and aerospace industry, where the reduction of weight

43 V. Vhahangwele, "Environmental Contamination of Heavy Metals", 2017, DOI:10.5772/intechopen.76082
44 «Global Chromic Acid sales market report 2020", 360 Research Reports, 2020
45 "Final Report for Hexevalent Chromium", ECHA/2011/01-SR-11, November 2013

has multiple beneficial impacts (i.e. reduction of fuel consumption, increased autonomy of electric cars etc).
Significance: The participation of three major large enterprises in the FreeMe project provide a high innovation potential for the adoption of technology in the market. **Stellantis (CRF)** is the the sixth-largest automaker in the world having in its portfolio brands such as Fiat, Jeep, Alfa Romeo, Opel, Citroen, Maserati etc, being first in Europe with 854.000 cars registered during the first quarter of 2021. **Arcelik** is active in more than 100 countries worldwide operating 15 production plants in Europe, Asia and Africa offering products under its own brand names (i.e. Beko, Grundig, Artic etc) or acting as a Tier 1 for other brands for hundreds of millions of spare parts. Finally, **Israel Aerospace Industry** operates in a high niche market, developing technologies for the aerospace industry.

Materials Modelling

FreeMe activities in the modelling area are innovative especially regarding the usage of multi-scale modelling and simulation to the overall spray application of resins for supporting the metallic plating of the plastic materials. The usage of a multi-scale approach in such multi-step processes is challenging and aligned with the goals of the European Material Modelling Council (EMMC). The need for improved linking and coupling of models to describe all relevant phenomena is already raised by the EMMC and is addressed in FreeMe by the proper specification of the modelling and simulation needs and the interfaces between the different scale activities. This will be reinforced by adopting the MODA (MOdelling DAta generalisation) to reach a standardised description of the required materials models and linked activities, and by also adopting the European Materials Modelling Ontology (EMMO) for the project activities definition. This will, in parallel, maximise the impact of the modelling activities beyond the project scope, by allowing future usage and linking of the developed models, potentially within the EMMC targeted European Marketplace. The usage of AI-based surrogate models to facilitate the interlinkage of models with the application allows it, is also a relevant innovation that can find applications in many other material-modelling related activities.
Scale: The field of materials modelling spans across a wide range of disciplines, i.e. computational chemistry, continuum mechanics, thermodynamics etc. A recent study, contacted by the EMMC-CSA and MarketPlace EC funded projects, resulted that the estimated materials modelling market size is 340 million €. [46]
Significance: Modelling is widely used to enhance research efficiency, contributing to reduction of costs and time by identifying alternative, more cost-effective materials, and processes. Based on statistic data [47] it is estimated that the cost saving utilizing targeted materials modelling in the framework of FreeMe project (i.e. concerning the development of sprayable resins and the etching process for increasing adhesion) is ~ 60% of the budget allocated in WP3 and WP4, meaning ~1.000.000€ for the duration of the project.

Market replication of new technologies

The EC defines the meaning of market replication Innovation Actions as those aiming to support innovative advancements to reach the market after successful demonstration but not application/deployment. This is due to several obstacles that can be encountered (see next subsection). Therefore, it is often highly beneficial to make use of tools and practices that facilitate the market replication, or, in other words, that help innovation make it in the market. FreeMe is designed to utilize them, aiming to maximize its impact and boost the market uptake of its results. The consortium's work in SSBD will produce not only sets of criteria, guides and safe methodologies for the processes of focus, but also standardization results, via integration of SSBD findings into protocols and guidelines, useful for derivative and relevant works. Those results will essentially be combined within a bigger set of guidelines and good practices that will result from the project and will constitute one of its main outcomes. They will be widely communicated but also disseminated, targeting policymakers and stakeholders of the field, aiming for the integration of these protocols and good practices into cluster efforts, widely used by field-related stakeholders. In addition, FreeMe will develop a DST, with the aim of demonstrating the technologies of the project, with the systematic integration of SSBD considerations. This means the DST will give interested users -that will not have to be by any means programming experts- the opportunity to simulate virtually their plating production lines, examining their optimal selection and/or re-engineering according to FreeMe's results, while, automatically, the tool will be taking into consideration in its mathematical computations all SSBD aspects (sustainability, recyclability, safety, cost) and including those considerations in the results. The tool will be highly customizable, aiming to offer high potential for future works to build upon it adding derivative or relevant technologies. Moreover, FreeMe is also employing additional, more traditional demonstration and facilitation of technology adoption measures like training activities and modules. Workshops and training activities will be organized within the project, with all this material remaining public after the project's completion, aiming to simplify workforce upskilling for interested technology users (particularly targeting SMEs). All the above target to support market replication of FreeMe, in strong connection with the project's exploitation and business planning.
Scale: According to recent studies, [48] around the European surface finishing industry employs about 900,000

46 G. Goldbeck et.al., "Materials Modelling Software Market", 2019
47 G. Goldbeck, C.Court, "The economic impact of materials modelling", EMMC 2016
48 P. Leisner, C. Larson, L.P. Nielsen, S. Müller, W. Hansal, J.-C. Puippe & A. Zielonka (2020) Global competitiveness of European surface technology. Part 1: competitiveness and the relationship between European manufacturing and surface finishing sectors, Transactions of the IMF, 98:6, 279-287

people recording an 3.6% average growth rate. Almost one third of it is done in the form of subcontracting. And with the majority of enterprises being under 10 employees small. At the same time, data indicates a clear correlation between company size and automation, resulting to smaller companies having very law automation rating.
Significance: FreeMe is aiming to surpass socio-economic barriers inhibiting innovation uptake by SMEs, by promoting its market replicability tools in the market. Easy to use guidelines and standardization protocols, combined with free training modules will support the adoption of FreeMe's good practices by small enterprises looking for a competitive edge. The DST will act as an enabling platform to build on and further develop, for automation and digitalization even by non-experts and without requiring large investments. FreeMe's business planning will specifically target SMEs in the field and aim to offer SSBD adoption and upskilling, while supporting the achievement of sustainable economic and technological advancements.

FreeMe contribution to the wider, long-term impacts as set by the destination

FreeMe project's result will have a contribution in the wider impact set by the EC in a different lever as follow:

- ✓ Resilient, sustainable and secure (critical) raw materials value chains for EU industrial ecosystems, in support of the twin green and digital transformations.

FreeMe project will have a significant contribution towards this impact. FreeMe targets to eliminate the use of Cr^{6+} substance from PoP processes, moving to greener technologies in the surface finishing industry. Of similar importance is the target for replacement of Pd used as activator (catalyst) during the activation process, prior to the metallization step. Pd is a critical raw material and EC is depended on imports, mainly from Russia. Successful replacement of Pd will ensure the value chain and supply of materials for the EC sector.

- ✓ New sustainable-by-design materials with enhanced functionalities and applications in a wide range of industrial processes and consumer products.

PoP is an important sector for many industries, mostly for automotive, bus also for other sectors including consumers goods (i.e. home appliances, fashion etc). FreeMe will invest in materials modelling approach to enhance the most important property of coatings on plastic, that is the adhesion.

- ✓ Leadership in producing materials that provide solutions for clean, toxic/pollutant free environment, decarbonising industry, and safeguarding civil infrastructures.

Concerning this impact, FreeMe is also expected to have a major contribution. Surface finishing industry is an important sector for the EC and the manufacturing industry. Specifically, PoP shows a significant growth for reasons already analysed in the previous paragraphs (i.e. lightweight of plastic products). By developing a green, toxic-free solution, EC plating sector will be favourably positions against competition from Asia (i.e. China, India). Additionally, elimination of Cr6+ usage will give a relief on society (i.e, plating workers, and neighbouring areas to plating facilities) leading to significant decrease of environmental pollution and cancer incidents.

- ✓ Leadership in circular economy that strengthens cross-sectorial cooperation along the value chain and enable SMEs to transform their activities and business models.

FreeMe will be developed as a technology harmonized with the circular economy concept, ensuring the recyclability of plated components with SoA technologies.

- ✓ Increased adoption of key digital and enabling technologies in industrial value chains and strategic sectors, paying particular attention to SMEs and start-ups.

The advancements in materials modelling as well as the development of a Decision Support Tool for SSbD application of FreeMe technologies in the industry, including SME, will have a significant impact towards digitalization and faster adoption of technologies.

2.1.2 Requirements & barriers and mitigation approach

The adoption of a new technology is inextricably linked with the environment in which it will be deployed. Thus, besides technological issues, political, economic, social, legal, and environmental factors are very important and should be thoroughly considered so as to achieve the maximum potential of exploitation of the results. In the following PESTLE analysis, the most important requirements and barriers related to these factors are mentioned. These are categorized either as positive (opportunities of the adoption of the FreeMe technologies) or negative (barriers that should be surpassed for the adoption of the technologies.

Table 2.1: *FREEME PESTEL analysis*

POSITIVE for achieving impact	***NEGATIVE for achieving impact***
Political	
European policies towards greener and sustainable industries (Green Deal): The EC policy favors the development and adoption of new Green Technologies providing multiple opportunities for further upscaling and commercialization of the technology such as easier	Strong advocacy from industrial lobbies to postpone the ban of Cr^{6+} use: Large industrial consortia are putting pressure for extension of ECHA authorization for using hexavalent chromium. However, the main reason for this is that currently there is no vial economic and

access to funding, favorable regulatory framework.	efficient technical solution. If FreeMe achieve its objectives, industry will adopt it as it will be attractive (REACH compliant, low cost and of high quality).
Economic	
Increase demand for PoP, especially from automotive and aerospace industry: PoP technology can provide properties of metals in the surface of lightweight plastics, making it extremely attractive ton sectors where reduction of weight is of crucial importance. The presence of two major companies in these sectors (CRF, IAI) in the consortium is indicative of this fact.	Need for investements: FreeMe project will bring the technology to TRL 5. To further upscale it and enhance its robustness for industrial applications further investments will be needed. As explained under the political factors, the favorable policies applied by the EC will allow easier access to funding both from public sources and private investements.
Social	
Strong requirement for safe operational environment in the plating industry: Workers in the surface finishing industry have been suffered from exposure to highly toxic and carcinogenic Cr^{6+}. FreeMe technologies will relieve the industry from such substances as it is based on safer, REACH compliant chemicals. Thus, it is expected that it will have the social acceptance	Public perception about environmental and health hazards risen from the surface finishing industry: This perception has been created due to several incidents related to polluted areas next to plating shops and severe negative impact in health of workers and inhabitants. Extensive communication activities will be taken in order to raise the awareness of public about FreeMe
Technological	
Advances in Materials Modelling: It is remarkable that the introduction of AI modelling and advance simulation models have given an impetus to materials development leading to significant savings in resources (costs, effort and time). It is expected that the actions undertaken in the framework of FreeMe towards this direction will further facilitate the upscaling process.	Competitive technologies to FreeMe: There are several competitive or emerging technologies for PoP as already mentioned in paragraph 1.1. However, most of them have a completely different approach compare to what is currently applied in the industry. FreeMe may have a higher lever of adoptions as it will be based on present experience of surface finishing sector.
Environmental	
FreeMe technologies is in accordance with trends for greener technologies: Thus, in general terms it is expected to have an overall beneficial impact. In order to enhance this, special focus will be given in the extension of baths lifetime as well as in the assessment of recyclability of coated objects so as FreeMe to be consider a real circular technology.	Generation of wastes: Even though FreeMe is based on green technologies, as a wet-based technology, generates wastewater from the spent baths and rinsing processes. Since the main pre-treatment solutions used in the FreeMe process are non-toxic their treatment as wastes is relative easy and in accordance with SoA technologies for industrial treatment of acids and Alkali.
Legal	
FreeMe is harmonized with the general regulatory framework (i.e. REACH, Directive 94/62/EC and Directive 2019/904 for plastic waste recycling compliance.	There is no identified legal based negative aspect of the FreeMe technology

2.2 Measures to maximise impact - Dissemination, exploitation and communication

2.2.1 Draft Dissemination and communication plan

The Dissemination and Communication Plan of the FreeMe project will include all relevant activities within the project and will provide the guidelines for the dissemination and communication to be followed by all partners. It contains the **methodology and the tools** to be used for the project's needs, as well as a detailed timeline to adhere to. It is a working document and thus the draft present plan presented here will be updated at M6 in details, while further updates will take place as opportunities arise during the project. As the main target of the dissemination and communication plan is to raise awareness of the project and its results and thus, enhance the exploitation capacity of these results, it is important to define the target groups and the tools that these groups will be reached. In the following tables the description of the target groups and the main dissemination and communication activities are presented.

Table 2.2: *FreeMe Dissemination and Communication target groups*

Category	*Description of Target group*
SMEs	SMEs across the supply chain of PoP. This included SME active in chemicals supply, plastic surface treatment, manufacturing of plastic components, end users and recycling companies
Large Enterprises	Large enterprises are important as their active involvement in the exploitation of results

(LE)	may have a major impact (economy of scales). End user (i.e. automotive, aerospace, home appliance) as well as companies having vertical productions and in-house PoP.
Industrial associations (IA)	A main role of the industrial association is to inform their members about new sustainable technologies that can be applied by them. Active involvement of such Association (i.e. the European Committee for Surface Treatment – CETS and its members) will be a main target during the implementation of the dissemination plan.
Regional innovation networks/accelerators (RIN)	Existing innovation hubs interested in expanding existing local support to European cross-border collaborations. Ecosystems that are related to polymers and surface finishing such as Open Innovation Test Beds.
Partners of Clusters (CL)	Participants in other projects of this call or of relevant EC projects, and of CSA topic HORIZON-CL4-2021-RESILIENCE-01-08
Broader innovation community (BIC)	Community interested in best practices for advancing the EU polymer sector. High tech SMEs interested in upscaling new materials and applying this technology in other fields.
R&D communities	Universities and research centres, participants in related fields, research societies, interested in the PoP sector, researchers working on end-use application sectors.
Regulatory & Public Authorities (RPA)	Ministries, organisations, regulatory authorities, public initiatives, standardization and certification bodies
	Communication Target group
General public (GP)	General Public and citizens affected directly (e.g. residents of neighbouring to PoP facilities) or in-directly by the use of chromic acid in industry.
Young people and students (YPS)	Students of primary, secondary schools and university who will become the new generation of users and may acts as ambassadors to promote the concept of sustainable consumption into their families. Young researchers looking to work on inspiring topics.
NGOs	Non-profit organization, Worker unions and other stakeholders sensitized in issues related to environmental pollution, occupational safety and sustainable growh.

Table 2.3: *FreeME Dissemination and Communication tools*

Dissem.activity	*Target audience**	*Expected Impact*	*Measure/Target (KPIs)*
		Consortium activities	
Project Website	All	Awareness of the FreeMe project and its scope	- 50 visits per day after year 3 - Receive 5 emails via website / week
Printed material	EU, BIC, GP, RC	Attract interest of industry, including SME and scientific community	- Development of 10 printed concepts - Distribution of 5.000 leaflets in events -Express of interest, 250 stakeholders
Presence in social media (SM)	GP, EU, BIC, CL, UPS,NGO	Attract interest of industry and connect with stakeholder pages and accounts in SM	- 1000 connections in LinkedIn account by the end of the project - weekly posts in SM accounts
Videos posted in website, YouTube	GP, IN, AN, HT, EU, RC	Attract industry and general public by showing the capacity of FreeMe	- 1 infographic video at M6 - 2 videos presenting the FreeMe - 3 videos of demonstrators
Press Releases	GP, RC, IN, AN, RC, PM	Awareness of the FreeMe project results and capacities	- 2 issues per year in 3 different languages (German, Italian, Greek,)
Conferences, exhibition & matchmaking events	EU, BIC, GP, RC	Oral presentations in conferences to policy makers. Participation in pitching/B2B events.	- Participation to at least 20 relevant international events/year - Participation to at least 4 relevant national events/year/country
Participation in events	RC, BIC, INAN, HT, EU, RNA, PM	Liaison with stakeholders and present the FreeMe Ecosystem	- Participation if 30 events (fairs, conferences, etc) during the project, having booths is at least 5 of them
FreeMe conference	RNA, BIC, IN, AN, HT, EU, RC, PM	Establishment of a biannual conference in the field of PoP post project	- 1 international conference at year 4 - 3 invited speakers, 20 presentations - 300 attendees
Workshops/ training	HT, RC, BIC	Awareness of FreeMe and success stories	- 3 workshops in international events - 50 attendees/workshop
Online Journal	HT, EU, RNA, IN, AN, PM RC	Presentation success stories to stakeholders	- 6 issues per year after Year 4 of the project

Citizen's engagement	GP, YPS	Raise awareness of new PoP technologies and impacts	- 3 Open labs events, 1 Public event in mall, - 2 events organized by end users

Conferences and other related events to FreeME	
Conferences on technology development	European Conference on Antennas and Propagation, EuCAP, SAT Plating Exhibits at Satellite 2020 Conference and Exhibition, Conferences and Meetings on Plastics, Paints and Synthetic Materials, Plastics Conference 2021, EUROMAT**,** European Advanced Materials Congress (EAMC), Conference on Mechanical and Aerospace Engineering (ICMAE), NANOSMAT, Plast Eurasia, Automotive Plastics Conference, **E**uropean plastics converters
Conferences on innovation management	Innovation Roundtable® Summit, Academy for Design Innovation Management Conference, Innovation, Management and Technology Conference, Innovation Events and Open Innovation Conferences, R&D Global Innovation Forum, Angel Investment Day – HunBAN.
Trade fairs	Global Automotive Materials Congress, Intelligent Traffic and Transportation (ICITT), Industrial Technologies, HANNOVER MESSE, International Plastic Fair, Plastic Trade Shows.

Journals related with TreatMe			
Journal	Impact factor	Journal	Impact factor
Journals on Materials and structures			
Engineering Sciences	8.843	Engineering Sciences	8.843
Journal of Chemical Education	1.758	Journal of Chemical Education	1.758
Surface and Coatings Technology	4.158	Surface and Coatings Technology	4.158
Business Management related Journals			
Creativity & Innovation Management	2.015	Creativity & Innovation Management	2.015

2.2.2 IPR management

The IPR management of the FreeMe project will be based on three pillars: Documentation of agreements, direct actions for the management of the IP, and necessary interactive activities. The two legal documents defining the main aspects of the IPR management are the Grant Agreement (GA) signed between the EC and the partners and the Consortium Agreement (CA) signed among the partners. In the latter, all aspects related to the **background knowledge** of the partners that are necessary for the implementation of the project as well as for the post-project exploitation of the results will be defined in detail. In general, it is already agreed that access to the background knowledge, IPRs, and research data will be royalty-free during the project implementation, while the post-project exploitation it will be done under fair conditions.

The general IP management actions will be undertaken by Incotec. The ownership of the **foreground knowledge** will be depicted in the final "**Result Ownership List" (ROL)** and will be decided between the partners consensually based on which partners have been involved in the generation of these results. The communication between the IPR manager and the consortium through appropriate meetings and questionnaires will define efficiently the foreground knowledge. In general, the owner of the IP belongs to the partner that develops it. Joint ownership will be accepted in the case that the results have been jointly generated, however in this case the joint owners will agree on the allocation and terms of exploitation of this joint ownership in communication with the Incotec and the Coordinator. The partners will be fully responsible for the protection of their foreground ownership in respect to their own legitimate interests, as well as the legitimate interests of the other partners. Already the partners have agreed that results with a promising commercial or industrial exploitation capacity will be protected. All the IPR updates as well as the Key Results and their IP routes presented in the business case will be included in the PEDR deliverables in terms of Task 8.3.

Regarding the interactive activities with the consortium, an internal IPR seminar will be organized for the presentation of the different IP protection types and exploitation routes for each Key Results in the first year (M6) as well as a final IPR seminar on M46 for the presentation of the ROL. Other additional activities for the knowledge management support and the patent application process will be a patent mapping analysis on M12 for the FreeMe technological landscape by defining the number of patents, the Top Authorities, and the Patent Classification Codes (CPC) codes. All these activities will be also communicated through the ''Management board and IPR issues'' annual meetings between the Exploitation, Dissemination, IPR and Technical manager.

2.2.3 Draft Exploitation plan

The Draft Plan for the Exploitation of results is based on (a) Expected Exploitable Results, ownership and market addressed (b) Market analysis & customers segmentation, (c) Expected market positioning and (d) Exploitation routes

A. Expected Exploitable Results – Ownership – Markets/stakholders addressed

The following table briefly presents the expected exploitable results of the project, ownership and in which markets or stakeholders will affect.

Table 2.4: *Expected exploitable results*

Expected Exploitable Result	Initial ownership split	Market/stakeholders
Bio-based resins containing Ni precursors (salts) for use in PoP processes	AUTH, Gaser, PoliMi	Companies in the field of plating on plastics (i.e. suppliers of materials, plating shops) AND research community
A plating on plastic process based on sprayable resins	Gaser, Polimi	Companies in the field of plating on plastics (i.e. suppliers, plating shops) AND end users: automotive, aerospace, home appliance AND research community
A toxic-free process (Cr6+,& Pd free) for plating on plastics	Cnano, Gaser	
A system based on spectroscopy principles for inspection of PoP process	IRIS	Companies applying PoP processes as contractors or in-house
Multiscale models for materials modelling in the field of PoP	IDE	Companies in the field of plating on plastics AND research community
A Decision support tool facilitating the adoption of SSbD FreeMe technologies	EXE, IDE	Partners of FreeMe, AND PoP industry including SMEs.

B. Market analysis and customer segmentation

The global electroplating Market was estimated to be ~ 13 billion € while the European share was ~5 billions for 2019. Figure 2.1 provides an integrated view of the European electroplating market. [49] **The PoP accounts for the ~4% of the total plating and is ~190 million€/year**

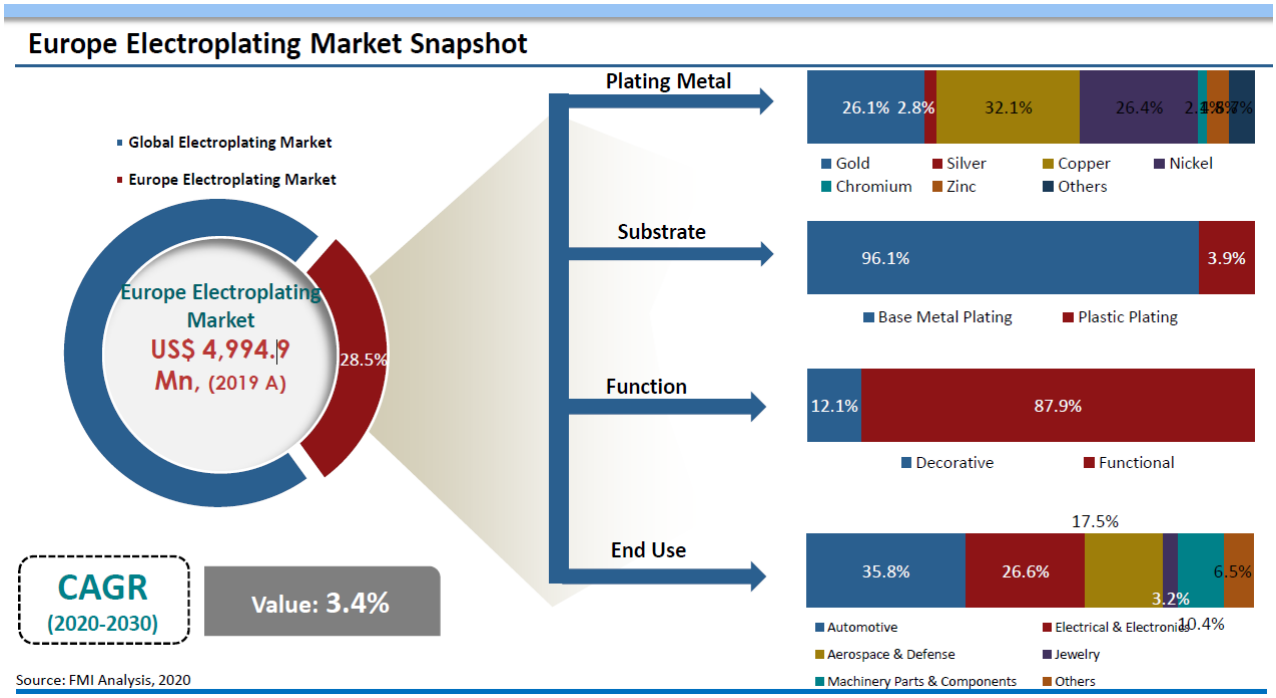


Figure 2.1:. European Electroplating Market.

As shown in the general electroplating sector will show a CAGR of 3.4% howerv when it comes to **Plating on Plastics** the grow is expected to be significantly higher, (**CAGR 7.9%**). [50] The need for lightweight and high-performance products in several applications (automotive, aerospace, consumer goods, arts, design, fashion) is a key driver for the POP market growth. The combination of the metallic surface properties with no changes on the composites structure along with the customer's demand for the enhanced aesthetic appeal of the product will drive positively the POP market [51]. However, the regulatory framework (EPA, OSHA, REACH) has worked as a constrain basically for the hexavalent chromium plating process that causes environmental issues. The hexavalent chromium is under REACH authorization in the European Union. Based on the final finishing the POP process is segmented into Precious metals (the higher revenues with the lower volume), Copper, Nickel, Chromium, and others.

49 "Global Electroplating Maket 2020-2030" Future Market Inshight, 2020
50 "Plating on Plastic Market Size by Finish, Plastic, Application, Industry", Global Market Insights, 2020
51 "Plating on plastic market size, share and trends analysis", Grand View Research, 2017

In the last years' Nickel is widely used in the POP processes for automotive parts such as door handles and wheels for decorative purposes of the interior and exterior area of the vehicles. Moreover, the addition of resins on the nickel coating offers properties benefits such as corrosion resistance. The **automotive POP application** in 2020 was estimated at **383,3 million EUR** driven by global manufacturers such as Stellantis, partner of the consortium, willingness to achieve **lower CO2 emissions and reduction on vehicle weight** [52]. In profit terms of this market, the first place is owned by Europe, followed by North America. Germany, UK, France, Italy, Russia, and Spain hold the largest share of the European market. More specifically, the **European POP market** was calculated at 1906 million EUR in 2019 and will reach **274,8 million EUR by 2027**. In France, it was estimated at 30 million EUR and is expected to grow at 8,47% CAGR from 2019-2027 as well as in the UK. Some European plating on plastics market players and possible investors for the FreeMe project are Atotech, Dow Chemicals, Chromoplastica Srl, Galva Decoparts Ltd, Coventya, and Bolta Werke GmbH. A more detailed market analysis will be included on the M6 exploitation dissemination and communication plan with updates and the addition of other market details.

C. Expected market positioning

Upon successful implementation of the project the FreeMe technologies is expected to enjoy a favourable positioning in the PoP market as they are REACH compliant processes, ensuring occupation safety and environmental sustainability. The presence of three major companies, word leader in their sector (Stellantis, Arcelic and Israel Aerospace Industry) with a total annual revenue reaching the 160 billion € is significant of the potential market appeal of the FreeMe processes. It should be highlighted that the 2 different PoP routes delivered in FreeMe will have a production cost similar or even lower to the current PoP practice. The elimination of the use of CrO_3 (~22€/kg) and of Pd (~55.000/kg) and the introduction of common industrial acids or alkali or of bio-based resins that have a remarkable lower cost will lead to a reduction of the materials costs. Additionally, savings will result from the fact that companies operating chromic acid procedures, will not have to apply for REACH authorizations and the accompanying compliance cost in the production sites. On the other hand, the additional investments necessary for the application of the technologies (i.e. spraying chamber and components, inspection system) in production installations are expected to be reasonable, not surpassing the ~150-200.000€ per installation. During the project implementation a detailed cost/benefit analysis will be conducted by the partners for adoption of technologies in their installations (i.e. Gaser, Cnano, CRF, Arcelic and IAI).

D. Exploitation routes

Each partner with established sales network and distribution channels will approach the respective market in the framework of the dissemination and exploitation plan. Each industrial partner will identify the total number of potential customers/collaborators based on their current clientele and positioning in the market. Similarly, the research oriented partners (i.e universities) will identify the most suitable exploitation routes for their results. The

Table 2.5: *Partners exploitation routes*

Partner	IP protection and handling	Exploitation route
Gaser	Trade secret, licensing of technologies	Gaser will adopt the new technologies in its premises providing PoP services
PoliMi	Open access respecting partners IPRs	Further research activities including new type of polymeric materials
Cnano	Patent and licencing of technology	Adoption of technology for high-tech applications
AUTH	Open access respecting partners IPRs In the future spin-off company	Research for transfering of bio-based composite resin technologies to new applications
Idener	Open access	Collaboration with EMMC and other partners for development of new modelling software
IRIS	Patent of system	Integration of monitoring systems in PoP facilities
ICCRAM	Open access of SSbD protocols	Further research activities and consultancy in the field
EXE	Open access of DST	Provision of DST tool and collaboration with FreeMe partners for widening the adoption of technology
INC	N/A	Provision of market services in the field of PoP
CRF	Licencing of technology	Close collaboration with the technology developers of FreeMe in order to be the early adopters of the PoP technologies in their products
ARC	Licencing of technology	
IAI	Licencing of technology	

2.3 Summary

SPECIFIC NEEDS	EXPECTED RESULTS	DEC MEASURES
1. *Elimination of toxic and carcinogenic Cr^{6+} due to environmental & workers' health issues from PoP sector* 2. *Replacement of Pd (CRM) from the plating industry* 3. *Increase of adhesion of PoP regardless the substrate material* 4. *Increase use of metal plated lightweight plastic components in automotive and aerospace industry*	1. *Development a sprayable bio-based resin-based process for PoP regardless the plastic substrate* 2. *Development of a Cr^{6+}, Pd-free wet chemistry PoP process customized on plastic substrate* 3. *A monitoring system based on spectroscopy principles, for inspection of surfaces at various steps of FreeMe production process* 4. *Development of multiscale models for materials modelling related to plastic surface treatment and adhesion enhancement* 5. *A Decision support tool facilitating the adoption of the SSbD FreeMe technologies by the PoP industry.*	**Exploitation activities:** 1. *Dedicated business planning for PoP to prepare industrial partners for reaching the market:* 2. *Patent application for the two approaches* 3. *IPR agreements inside and outside the consortium* **Dissemination activities:** 1. *Scientific community: 12 Scientific publications, 25 oral presentations in conferences, 2 workshops and 2 trainings* 2. *Industrial community: 6 participations in Fairs and partnering events, 6 Publications, 2 trainings* 3. *Standardization, Regulatory bodies: PoliMi, CRF, IAI, Gaser and Cnano, active in liaison with these groups will interact for promotion of new technologies* **Communication Activities***: Various events such as a shopping mall show, open labs (i.e. PoliMi, Gaser, Cnano), events organized in the premises of end users*

TARGET GROUPS	OUTCOMES	IMPACTS
Industry: 1. *Plating shops (contractors) active in the PoP currently utilizing Cr^{6+} processes* 2. *In-house installation for PoP* 3. *Materials suppliers for plating industry* 4. *Manufacturers of plastic components* 5. *OEMs in the field of automotive, aerospace, home appliances. In future applications fashion industry* **Scientific community**: In the field of green chemistry and plating on plastic. **Society**: Workers in the PoP industry, citizens in the areas that PoP facilities are located	1. *Gaser and Cnano apply the technologies (sprayable resin route and Cr^{6+},Pf-free wet route, respectively) to their premises* 2. *CRF, IAI & ARC become the first adopters of FreeMe technologies in their plated components* 3. *Wide utilization of the DST web-platform from potential adopters of the technology, mainly plating on plastics shops and in-house facilities* 4. *Enhance SSbD approaches based on the first industrial adoption of the technology*	**Scientific:** New breakthrough results on plating on polymeric substrates. Advancements in materials modelling. **Economic**: Secure supply chains by elimination of Pd use from PoP. **Technological**: New improved processes and products in the growing markets of PoP, automotive, aerospace and consumer goods. Widening the technology application to new markets such as fashion, electronics. Increase of coatings quality due to better inspection **Societal**: Significant reduction of Cr^{6+} environmental pollution, reduction of occupation cancer incidents and elimination in the future 2) Maintenance and/or creation of high tech jobs within EU

3. Quality and efficiency of the implementation

3.1 Work plan and resources

In line with the objectives that FreeME must achieve in these 48 months project, the work to be accomplished has been structured in nine work packages (***Figure 3.1***). Each WP has been sub-divided into tasks. WP leaders and task leaders have been assigned based on their core expertise and contribution in the specific tasks, with main purpose to lead and monitor the progress. The reporting and monitoring of the progress and results is based on a set of Deliverables (Table 3.1c) and Milestones (Table 3.1d). The Deliverables have been scheduled to provide an integrated reporting of the work implemented in WPs and relevant Tasks. Some of them will serve as living documents (i.e. the PEDRC). Milestones are positioned across the 48 months with the purpose to monitor the achievement of specific goals of the project that will allow the smooth continuation.

Table 3.1a: List of work packages

WP No	Work Package Title	Lead Par/nt No	Lead Par/nt Short Name	PMs	Start Month	End month
WP1	*Project's Requirements & Specifications*	*1*	*Gaser*	*42*	*1*	*6*
WP2	*Safe & Sustainable by Design Strategy*	*7*	*ICCRAM*	*100*	*1*	*48*
WP3	*Sprayable resin process*	*2*	*PoliMi*	*100*	*5*	*24*
WP4	*Cr^{6+} & Pd free metallization process*	*3*	*Cnano*	*68*	*6*	*48*
WP5	*Development of supporting technologies*	*5*	*IDE*	*82*	*1*	*48*
WP6	*Validation of technologies in end user's test cases*	*10*	*CRF*	*100*	*6*	*48*
WP7	*Roadmap to market deployment*	*2*	*PoliMi*	*76*	*1*	*42*
WP8	*Extroversion and Exploitation activities*	*8*	*EXE*	*104*	*1*	*48*
WP9	*Project management*	*1*	*Gaser*	*53*	*1*	*48*
			Total	**725**		

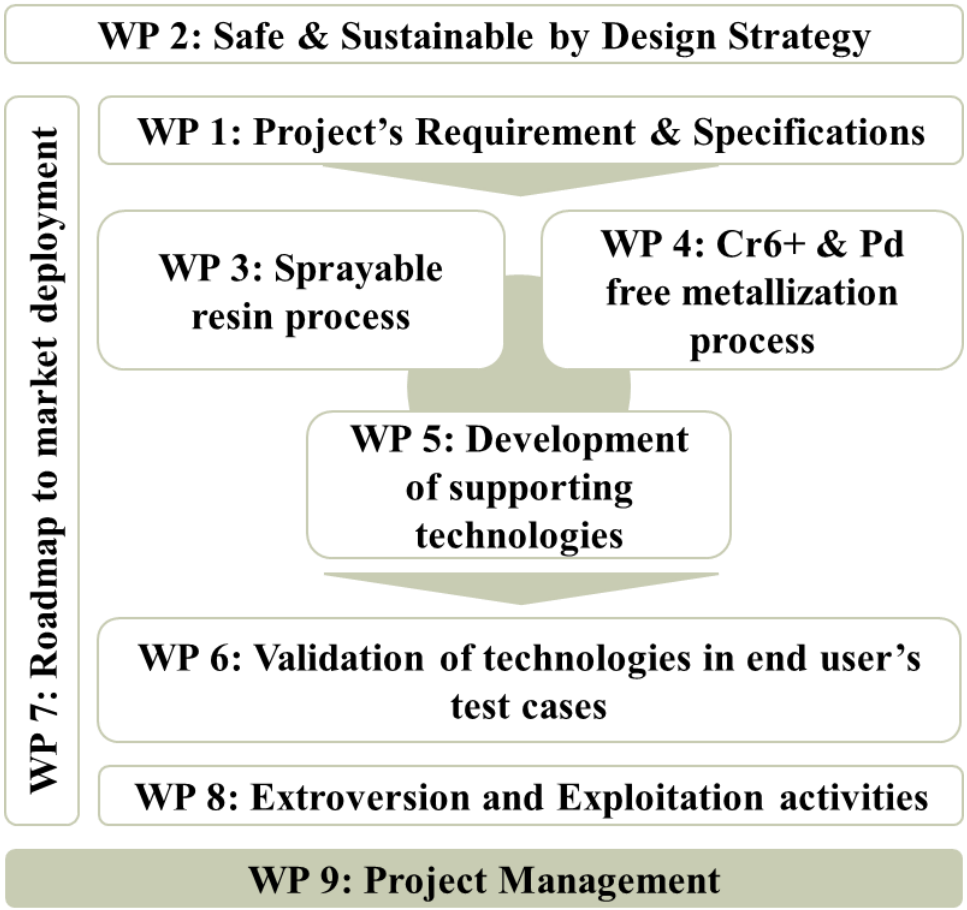


Figure 3.1:. FreeMe PERT diagram.

Figure 3.1: *Gantt chart of FreeMe project.*

	Leader	Start	End
WP1: Project's Requirments & Specifications	**Gaser**	**1**	**6**
T1.1: REACH &RoHS compliance	Gaser	1	6
T1.2: Specifications on methods and equipment	PoliMi	1	6
T1.3: Requirements of end users' test cases	CRF	1	6
WP2: Safe & Sustainable by Desing Strategy	**ICCRAM**	**1**	**48**
T2.1: Identification of options for technologies delivery	ICCRAM	1	36
T2.2: Assessment of environmental sustainability	ICCRAM	12	48
T2.2: Human and enviroment safety assessment	ICCRAM	12	48
T2.3: Application of the SSBD strategy	ICCRAM	24	48
T2.4: Recyclability and Repairability assessment	PoliMi	12	48
WP3: Sprayable resins process	**PoliMi**	**6**	**36**
T3.1: Development and optimization of composite resins	AUTH	6	36
T3.2: Application of resins layers in polymer substrates	PoliMi	8	36
T3.3: Metallization process of the resin layer	Gazer	8	36
T3.4: Baths qualification process	Gaser	14	36
WP4: Cr6+ & Pd free metallization process	**Cnano**	**6**	**36**
T4.1: Development of Cr+6 free etching process	Cnano	6	36
T4.2: Development of Pd free activation process	Cnano	6	36
T4.3: Metallization process of the polymeric activated surfaces	Gazer	8	36
WP5: Development of supporting technologies	**IDE**	**6**	**48**
T5.1: Materials and process in-silico modelling	IDE	6	36
T5.3: Development of in-line optical inspection system for coatings	IRIS	6	48
WP6: Validation of technologies in end user's test cases	**CRF**	**28**	**48**
T6.1: Decorative automotive parts	CRF	28	48
T6.2: EMI shielding on antennas	IAI	28	48
T6.3: Applications of washing machines parts	ARC	28	48
WP7: Roadmap to market deployment	**PoliMi**	**1**	**48**
T7.1: Implementation of standardization activities	PoliMi	1	48
T7.2: Development of DST	IDE	1	48
WP8: Extroversion and Exploitation activities	**EXE**	**1**	**48**
T8.1: Dissemination & Communication	EXE	1	48
T8.2: Exploitation and business cases	INC	1	48
T8.3: Open science and Research Data Management	PoliMi	1	48
T8.4: Training Activities	EXE	1	48
WP9: Project management	**Gazer**	**1**	**48**
T7.1: Administrative & Financial Management	Gazer	1	48
T7.2: Technical Management	PoliMi	1	48

Table 3.1 b: Description of each work package

Work Package Number			1								**Lead Beneficiary**	Gaser
Work Package Title			**Project's Requirements & Specifications**									
Nr.	1	2	3	4	5	6	7	8	9	10	11	12
Name	Gaser	PoliMi	Cnano	AUTH	IDE	IRIS	ICCRAM	EXE	INC	ARC	IAI	CRF
PMs												
Start Month		1									**End Month**	6

Objectives

WP 1 has the following specific objectives:

1. To harmonize the R&I towards REACH compliance chemicals
2. To provide specification on the methods and the equipment that will be utilized in the project
3. To define the requirements of the end users in terms of materials, properties and standards

Description of Work

Task 1.1: REACH & RoHS compliance (Task Leader: Gaser), M01-06
Supported by: All partners

As a leading industrial partner in the surface and finishing industry, with a strong position in the Plating on Plastics field, Gaser will lead the compliance task. In collaboration with the other partners the proposed chemicals for the development of the technologies will be listed and analyzed. No chemical under the restriction lists of REACH (SVHCs) as well as RoHS will be selected, while data from the REACH (e.g. risks associated with exposure to the selected chemical throughout the lifecycle of the targeted products, including waste and disposal will be reported. Interaction with Task 2.1 will take place.

Task 1.2 Specifications on methods and equipment (Task Leader: PoliMi), M01-06
Supported by: All partners

In this task the specifications for the methods and the equipment that will be utilized during the project will be provided. **Flow charts and diagrams of raw materials, processes and wastes** will be developed, including quantities and potential exposure risks based on data of Task 1.1. Precaution measures for the proper handling of materials and equipment will be also delivered. Finally, dimensioning of lines will be made in order to cover the needs of the end users' applications based on data provided by task 1.3 will be made.

Task 1.3 Requirements of end users' test cases (Task Leader: CRF), M01-06
Supported by: All partners

In this task the specification and requirements of the end user's applications will be collected and reported. This will include the type of polymeric substrates used, designs of the objects, current pre-treatment and metallization processes, desired properties (e.g. surface hardness, wear and corrosion resistance, aesthetics) will be provided together with the final surface finishing preferences (i.e. the type of the final electroplated metal after the metallization process). Additionally, the QC protocols and standards followed will be also provided. Strong interactions with Task 2.1 will take place.

Deliverables

D1.1	Report on specifications. The deliverable will contain specifications and requirements as collected in tasks 1.1 and 1.2 and designs and data provided by end users concerning the test cases as well as the standards that will be utilized for the validation of them (Task 1.3). **Delivery: M06** by **GASER**

Work Package Number			2								**Lead Beneficiary**	ICCRAM
Work Package Title			**Safe & Sustainable by Design Strategy**									
Nr.	1	2	3	4	5	6	7	8	9	10	11	12
Name	Gaser	PoliMi	Cnano	AUTH	IDE	IRIS	ICCRAM	EXE	INC	ARC	IAI	CRF
PMs												
Start Month		1									**End Month**	48

Objectives

WP 2 has the following specific objectives:

1. To reduce human and environmental risk avoiding known substances of concern through the assessment of potential safety impacts of prototype candidates.
2. To determine the sustainability performance of new PoP processes through life cycle against various dimensions
3. To assess the recyclability of the coated plastics

Description of Work

Task 2.1: Identification of options for technology delivery (Task Leader: ICCRAM), M01-36
Supported by: All partners
During this task, **the design team defined in the consortium** (product designers, material engineers, chemists, toxicologists, and sustainability experts, designers and business developers from end uses), will consider the functionalities that the FreeMe technologies will deliver in the final products. Based on the specifications provided in WP1 a minimum level of performance will be defined to be the baseline for the development of technologies targeting to prolong the products lifecycle by increasing durability and repairability. Another target of this task will be to reduce the complexity of product's designs based on the progress of the technical work in WP3, 4 and 5 and thus, to easy the processes of recycling, repairing or reduce the raw materials used, fostering the Green Deal targets.

Task 2.2: Assessment of environmental sustainability (Task Leader: ICCRAM), M12-48
Supported by: All partners
The task aims at supporting the sustainable synthesis and implementation of the proposed PoP processes. Life Cycle Assessment (LCA), Life Cycle Costing (LCC) and Social Life Cycle Assessment (S-LCA), will be performed to compare:
- Sustainability by Design strategies: Based on inputs collected (e.g. materials and energy use, costs, working conditions), the materials and process alternatives leading to the best sustainability performances for the same functionality will be identified to support sustainable-by-design of the metallization technologies.
- Market implementation scenarios: Depending on the application envisaged, potential impacts will be compared to benchmarking technologies, to understand the benefits and potential trade-offs of the deployment of the newly proposed technologies (e.g. higher safety for workers but increased carbon footprint).
The LCA, LCC and S-LCA evaluations will follow existing standards (e.g. the Product Environmental Footprint) and use the same scope (e.g. compared scenarios, system boundaries, functional unit). Based on the results analysis, recommendations for the Sustainable by Design and the future large-scale deployment of the proposed PoP processes will be drawn.

Task 2.3: Human and environmental safety assessment (Task Leader: ICCRAM), M12-48
Supported by: All partners
This task will provide evaluations that demonstrate the reduced toxicity of the new processes developed and materials involved. A set of protocols will be applied in order to measure their potential risk in comparison with SoA equivalent processes. Toxicological potential against human health will be determined employing appropriate cellular models adequate to expected exposure routes for in vitro cytotoxicity and cell viability assays. The ecotoxicity and potential danger for the environment of the new catalysts will be also assessed analysing their impact in the bioluminescent marine bacterium Vibrio fischeri, in the model fungus Saccharomyces cerevisiae, as well as in a set of microbial exoenzymes responsible for different biological functions from eukaryotic and prokaryotic organisms. The proposed toxicological tests will be performed employing samples obtained at different process stages.

Task 2.4: Application of the Sustainability and Safe-by-Design strategy (Task Leader: ICCRAM), M24-48
Supported by: All partners
This task will implement a SSbD innovation approach to ensure that the resulting FREEME's processes are safe, both for humans and the environment, and sustainable along their life cycle at an early stage, aligned to the EU chemicals strategy for sustainability, including:
(i) Safe-by-Design (SbD) strategies, which focus on a) Safe-by-Material Design –SbMD- strategies (that will allow reduce potential risk based on the physicochemical and (eco)toxicological properties of the materials; and b) Safe-by-Process Design -SbPD- strategies, based on the relevant factors during the processing of the materials in terms of potential exposure to hazardous substances, and
(ii) Sustainability-by-Design (SusbD) strategies that can increase the environmental, economic and social sustainability performances of the novel PoP processes.
An assessment of the safety and sustainability performances throughout the processes and product's life cycle will be done for both types of strategies. All the evaluated indicators will be integrated (e.g. via multi-criteria decision analysis) resulting into a SSbD index for each FreeMe process. This will allow the identification of the most safe and sustainable alternatives at the design stage, further translated into recommendations for the implementation of SSbD strategies

Task 2.5: Recyclability and Repairability assessment (Task Leader: PoliMI), M12-48
Supported by: Gaser, Cnano,
Removal of the metal coatings from plastic substrates for recycling will be investigated by employing one of the following methods: 1) abrasive blasting, 2) cryogenic embrittlement and 3) mechanical crushing. **Abrasive**

blasting. is the operation of forcibly propelling a stream of abrasive material against a surface under high pressure. **Dry ice-blasting** is a special type of abrasive blasting where dry ice (solidified carbon dioxide) will be accelerated in a pressurized air stream and directed at a surface in order to clean it. **Cryogenic embrittlement** may also be used to remove metallic coatings from plastic substrates. The recovery system involves feeding metal-plated plastics into a grinder. The grinder contains a pre-crusher which breaks up the parts into smaller pieces and feeds them into the grinder which subsequently reduces the scrap from half-inch pieces to coarse powder. A magnetic drum separator removes the pieces of metal that have been split off during grinding. The final step contains going through a cryogenic feeder, where liquid nitrogen cools the scrap to almost freezing, creating stresses which cause additional metal to splinter off. Finally, metal-plated plastics, for instance electroplated Cu, Ni and Cr on ABS, could also be separated using **mechanical crushing**. Coatings could be easily stripped when the items are subjected to shear force and impulsive force. Concerning the repair of partially damaged (scratched) metal-plated plastics, **brush plating**, a portable method of electroplating without the use of an immersion tank, will be tested and employed.

Deliverables	
D2.1	Report on the outcome of Task 2.1 with recommendations for SSbD for products and processes. **Delivery: M18** by **ICCRAM** and **update at M36.(D2.2)**
D2.3	Initial report containing the outcome on assessment of potential impacts and environmental sustainability (Task 2.2 & 2.4). **Delivery** by **ICCRAM** and **updated at M48**

Work Package Number	3										**Lead Beneficiary**	PoliMi
Work Package Title	**Sprayable Resins process**											
Nr.	1	2	3	4	5	6	7	8	9	10	11	12
Name	Gaser	PoliMi	Cnano	AUTH	IDE	IRIS	ICCR	EXE	INC	ARC	IAI	CRF
PMs												
Start Month	6										**End Month**	36

Objectives

WP 3 has the following specific objectives:

1. To develop appropriate formulations for composite resins containing Ni salts
2. To optimize the application of sprayable composite resins in polymer surfaces resulting to excellent adhesion
3. To investigate suitable surface treatment methods of hardened resins to facilitate the following metallization step
4. To optimize the activation and/or metallization processes for the two types of resins
5. To achieve excellent adhesion of metallic coatings on plastics

Description of Work

Task 3.1: Development and optimization of composite resins (Task Leader: AUTH), M06-36
Supported by: PoliMi, Gaser, end users

UV curing of epoxy and epoxy-acrylate resins that will be synthesized from biomass-derived platform chemicals, i.e. isosorbide and itaconic acid, as well as of their (nano)composites with nickel salts will be studied and optimized towards the formation of rigid thin (< 2μm) polymeric films on polymeric substrates (i.e. ABS, PC, etc.). The prepolymer resins will react with mono or multifunctional acrylate monomers, such as tripropyleneglycol diacrylate, trimethylolpropane triacrylate, hexanediol diacrylate, etc, in the presence of appropriate photo initiators, such as the alpha-hydroxy ketone types (e.g. Darocur 1173, Irgacure 184 etc.) in the case of epoxy-acrylates (free radical mechanism) or diaryliodonium and triarylsulfonium salts (e.g. Irgacure 250, Curacure UV6976) for the acidic UV curing mechanism of epoxy resins. The Ni/epoxy-acrylate composite resins mixture will be characterized for their viscosity (targeting to low values of ca. < 200 cP, that will enable solventless spraying application and ability to perform fast UV curing, while the final cross-linked (nano)composites will be characterized a series of techniques, such as SEM, TEM, tensile and impact strength, pull off testing (to measure the adhesion strength of the epoxy-acrylate film on the polymeric substrates), TGA/DSC, conductivity, resistance to solvents, etc.

Task 3.2 Application of resins layers in polymer substrates (Task Leader: PoliMI), M08-36
Supported by: Gaser, end users

A conventional spray gun will be used for the formation of the resin layer on the polymer surface. The whole process will be conducted inside a customized, well ventilated and filtered cabinet to enclose and isolate hazardous fumes and airborne particles. The “pot life” of the composite resin (the amount of time the resin will remain in a liquid state) will be optimized by varying temperature, mixing solvent and viscosity of the resin). The spraying pressure will be varied between 1 and 5 bar to optimize the spraying velocity and as a result the duration of the overall process. Different spraying nozzles (shape and size) and different spraying directions (angled or perpendicular) will also be considered to obtain a homogeneous resin layer.

Task 3.3 Metallization process of the resin layer (Task Leader: Gaser), M08-36
Supported by: All partners

Subtask 3.3.1: Metallization of the composite resin layer initially requires the generation of metal nucleation sites. To this end, samples will be immersed into an aqueous $NaBH_4$ solution to reduce Ni from the cationic Ni^{2+} to the metallic Ni^0 state and to roughen the resin surface. A Design of Experiments (DoE) will be implemented to optimize the activation rate and the roughness of the surface, considering the following parameters: (i) $NaBH_4$ concentration (5 to 50 g/L), (ii) immersion time (1 – 30 min) and (iii) temperature (20 – 50 °C). The resulting dispersion of the Ni metal sites and the surface roughness will be validated by SEM and EDX mapping.

Subtask 3.3.2: Metallization will be completed by immersing the activated samples in a Ni-P electroless aqueous bath containing nickel sulfate [$Ni(SO_4)_2$] as the Ni^{2+} source, sodium hypophosphite ($NaPO_2H_2$) as the reducing agent, sodium citrate ($Na_3C_6H_5O_7$) as the complexing agent to control the release rate of free Ni^{2+} cations in solution and an ammonium chloride/hydroxide buffer to adjust the pH. The electroless plating conditions, i.e. pH, time and temperature will be optimized depending on the surface area of the sample and the targeted layer thickness. Based on the obtained results, a finer adjustment of the above parameters will follow. The morphological and structural properties of the Ni-P layer will be investigated via microscopy (optical digital microscopy, SEM and EDX mapping) and spectroscopy (FT-IR, XRF and XPS). Metal-polymer adhesion will be evaluated via standard "peel" or "pull off" tests. The final metal finishing will be produced via pulse current plating, according to the specifications of the end users (e.g. nickel, nickel alloys, copper, precious metals).

The development of this task will be split in two phases: in the 1st phase, lab-scale experiments will be carried out (baths up to 3 L) whereas in the 2nd phase, scaling up of the process will be explored using Cnano's pilot line (baths up to 150 L).

Task 3.4 Baths qualification process (Task Leader: Gaser), M14-36
Supported by: All partners

Baths qualification protocols will be developed in this task as a main part of the SSBD approach. Several methods and techniques will be applied to monitor and adjust in real-time baths parameters as well as to avoid condamintation that have an important effect on the plating process and on the lifetime of the baths. Chemical analysis will be carried out to determine chemical composition of the baths (titration, elementary analysis for organic species). ICP (Inductively Coupled Plasma) spectroscopy will also be used for determination of metals content in the baths.

Deliverables	
D3.1	Intermediate report containing the progress on the activities of WP3 **Delivery: M18** by **PoliMi** .
D3.2	Report including the resins' formulation and optimized methods for application on polymer surfaces **Delivery: M36** by **AUTH**
D3.3	Report on the optimized activation and metallization processes of composite sprayed resins **Delivery: M36** by **Gaser**

Work Package Number	4										**Lead Beneficiary**	Cnano
Work Package Title	**Cr^{6+} and Pd free metallization process**											
Nr.	1	2	3	4	5	6	7	8	9	10	11	12
Name	Gaser	PoliMi	Cnano	AUTH	IDE	IRIS	ICCR	EXE	INC	ARC	IAI	CRF
PMs												
Start Month		6									**End Month**	36

Objectives

WP 4 has the following specific objectives:

1. To develop and optimize a Cr6+ free process, based on acids or alkali mixtures
2. To replace Pd during the surface activation process with more abundant materials
3. To perform metallization of polymers based on the previous process
4. To achieve excellent adhesion of metallic coatings on plastics

Description of Work

Task 4.1: Development of Cr^{6+} free etching process (Task Leader: Cnano), M06-36
Supported by: All partners

ABS, PET, PLA and UV-curable resins will be used as polymeric surfaces for electroless metal plating. Surface preparation will begin with surface degreasing during which the samples will be immersed into a soap solution under sonication. Afterwards, Cr^{6+}-free surface etching will be carried out to create cavities and to install polar groups on the polymeric surface to enhance adhesion. ABS and PET substrates will be immersed into a strongly oxidizing aqueous solution of 30% wt H_2O_2 / H_2SO_4 ("piranha solution"). PLA and UV-cured resins will be etched by immersing the samples in highly alkaline solutions of KOH or NaOH. The effect on surface properties of H_2O_2 / H_2SO_4 ratio (from 1:1 to 1:7), [OH^-] concentration (1 – 5 M) and immersion time (30 s to 2 h) will be investigated,

i.e. roughness (measured by optical microscopy and SEM), hydrophilicity (measured by contact angle analysis), chemical composition (measured by EDX and FTIR). Finally, all surfaces will be neutralized to remove remnants of the etching solutions by rinsing with DI water and/or immersing in an antioxidant solution, such as sodium bisulfite ($NaHSO_3$). The combined comparative studies will furnish an environmentally benign, time- and cost-efficient etching method.

Task 4.2 Development of Pd free activation process (Task Leader: Cnano), M06-36
Supported by: All partners
Pd and Sn free surface activation will be held via immersing the samples into a nickel acetate [$Ni(CH_3COO)_2$] solution and optimizing the concentration (1 - 10 g/L) and the immersion time (10 – 120 min). Afterwards, surface adsorbed cationic Ni^{2+} species will be reduced to metallic Ni^0 nucleation sites by immersing the samples into an aqueous $NaBH_4$ solution. $NaBH_4$ is also expected to further roughen the polymeric surface. Although the $NaBH_4$ reduction step resembles the one described in Subtask 3.3.1, the chemical environment around Ni^{2+} is now different. Therefore, an independent DoE will be implemented to probe the effect of $NaBH_4$ concentration, immersion time and temperature on the resulting dispersion of Ni^0 metal sites (measured by EDX mapping), surface roughness (measured by SEM) and hydrophilicity (measured by contact angle analysis).

Task 4.3 Metallization process of polymeric activated surfaces (Task Leader: Gaser), M08-36
Supported by: All partners
Metallization will be completed by immersing the activated samples in a Ni-P electroless aqueous bath. Optimization of the electroless plating conditions (i.e. pH, time and temperature) and validation of the resulting Ni-P layer will be caried out following the protocols of subtask 3.3.2. Development will be again split in two phases, i.e. lab-scale experiments using baths up to 1 L(1st phase) and upscale using baths up to 150 L (2nd phase). As above, the final metal finishing will be provided via electroplating based on the end user's needs.

Deliverables	
D4.1	Intermediate report containing the progress on the activities of WP4 **Delivery: M18** by **Cnano** .
D4.2	Report on the development of a Cr6+free etching process and Pd-free activation step for plastic surfaces metallization including characterization of structures and properties. **Delivery: M36** by **Cnano**

Work Package Number			5							**Lead Beneficiary**		IDE
Work Package Title			**Development of supporting technologies**									
Nr.	1	2	3	4	5	6	7	8	9	10	11	12
Name	Gaser	PoliMi	Cnano	AUTH	IDE	IRIS	ICCR	EXE	INC	ARC	IAI	CRF
PMs												
Start Month		6									**End Month**	48

Objectives

WP 5 has the following specific objectives:
1. To support the development of sprayable resin and chromic acid free etching with in-silico techniques based on multiscale modelling
2. To optimize an optical inspection system for monitoring of PoP process.
WP leader: IDE

Description of Work

Task 5.1: Materials and process in-silico modelling (Task Leader: IDE), M06-36
Supported by: All partners
The objective of this task is to perform the multi-scale modelling activities required to boost the design process of the proposed sprayable composite resins. For this purpose, IDE, will work closely with the rest of the material developers in the project towards the proper modeling of the required materials and interfaces. The specific scale and type of models to be developed will be selected so as to fill the gaps in the information provided by the performed experiments and to optimise their design, minimising the required tests. The selected models will be implemented using a combination of ab-initio quantum (QM) and Coarse-Grained (CG) models which will be simulated using molecular dynamics (MD), Monte Carlo (MC) and FEM techniques to unlock multi-scale modelling advantages. Moreover, the developed models will aim to support the process scale-up by providing key information on their applicability for different applications. The work of this task will be split in two subtasks, T5.1.1 and T5.1.2, which will cover the modeling required for the sprayable composite resins and the

Task 5.2 Development of in-line optical inspection system for coatings (Task Leader: IRIS), M06-48
Supported by: All partners

An optical system for inspection of surface, based on spectroscopy principles will be delivered for high-speed inline monitoring of, on the one hand, the coating application processes and, on the other hand, to assess the quality of organic deposits on a range of substrates. The selected optical system will be assembled and pre-calibrated in the lab scale on a relevant set of samples for obtaining respectively coatings and deposition in non-continuous modes. It is expected that material dispersed in the resin coatings (i.e. Ni salts or Ni NPs) may lead to scattering, therefore a specific calibration may be needed for the two concluded different formulations as opposed to the absolute values that are expected in case of unfilled coatings or dense layer. In this latter case of deposition, the speed and time of the process would already bring a certain idea of this data. The overall performance of the system will be validated during a certain period of time at pilot scale.

Deliverables

D5.1	Intermediate report containing the progress on the activities of WP5). **Delivery: M18** by **IDE** and updated at **M36 (D5.2)** by **IRIS**
D5.3	This deliverable will consist of the implemented multi-scale models. Such models, the data required for replication and the data produced through simulation will be openly published accordingly to the DMP. Whenever the data requires protection due to the targeted commercialisation, the models will be accompanied by synthetic data (Task 5.1). **Delivery: M48** by **IDE**
D5.4	This report will provide the conclusions drawn through the project on the suitability of different multi-scale models for supporting the design of PoP processes. Moreover, it will contain an analysis of the optimisation of the design process achieved through the DoE produced in base of such models (Task 5.1). **Delivery: M48** by **IDE**.
D5.5	Development, installation and validation of in-line optical inspection system (Task 5.3). **Delivery: M48** by **IRIS**

Work Package Number			6							**Lead Beneficiary**		CRF
Work Package Title			**Validation of technologies in end user's test cases**									
Nr.	1	2	3	4	5	6	7	8	9	10	11	12
Name	Gaser	PoliMi	Cnano	AUTH	IDE	IRIS	ICCR	EXE	INC	ARC	IAI	CRF
PMs												
Start Month		28									**End Month**	48

Objectives

WP 6 has the following specific objectives:

1. To demonstrate the sprayable resin route for metallization of plastic surface in real applications
2. To demonstrate the Cr6+ and Pd free metallization process on real applications
3. To proceed in the qualification of coatings through industrial QC protocols of end users

Description of Work

Task 6.1: Decorative automotive parts (Task Leader: CRF), M28-48
Supported by: All partners

The deployment of this taks will be split in two phases. Firstly, tests on non-definitive samples (e.g. flat plaques) will be carried out to ensure quality of the coatings and possibly finely tune the environmentally benign novel technologies, developed in FreeMe. Secondly, will deliver various components for both interior and exterior use, made mainly of ABS and PC-ABS. Both developed ways will be applied (sprayable-resins and Cr^{6+}-free etching) for the metallization step, depending on the size and the geometry of the parts. Afterwards the layer electroplated layered structured of Ni and trivalent Cr coatings will be followed. CRF will validate the quality of the produced metallic coatings by performing in-house tests, according to automotive standards and international norms (e.g., ISO, ASTM) and compared the obtained results to benchmark performance. These tests include mechanical and physicochemical properties of the coatings such as adhesion, brightness, anti-scratch, impact and UV resistance..

Task 6.2: EMI shielding on antennas (Task Leader: IAI), M28-48
Supported by: All partners

IAI will manufacture and deliver to Cnano 3D-printed antenna parts (Figure 1.12.). Cnano will apply a metallic coating following the wet metallization route. Sprayable resin rout will not be investigated due to the complexity of the antennas' structure. The application of a Ni-P thin layer will be carried out, following the optimization process described in Subtask 3.3.2. The final metal finishing will be provided by pulse current electroplating depending on the needs of IAI. First row transition metals (e.g. Ni and Cu) will be initially used to achieve high conductivity and related EMI shielding values (e.g. 10^{-1} S/cm and – 40 dB, respectively). Alternatively, a silver interlayer will be applied to promote conductivity. The morphological (optical microscopy, SEM) and the physicochemical properties of the metallized antennas will be investigated by Cnano (EDX mapping, XRF). The electrical properties of the metallized antenna will be investigated by IAI. Special tests regarding mechanical strength and wear resistance in

extreme conditions will be also conducted by IAI

Task 6.3: Applications on washing machine parts (Task Leader: ARC), M28-48
Supported by: All partners
With the FreeMe project, ARC aims to eliminate chromium from the plating process and develop alternative, less hazardous routes while maintaining the top quality of the final metal finishes. ARC will provide specific plastic parts of their washing machine production line (e.g. dials and buttons) to PoliMi, Cnano and Gaser for plating via the project's innovative technologies. Both developed routes will be followed for these parts. ARC will serve as the end user validating the quality of the metal coatings. To this end, adhesion, corrosion and aging tests will be carried out to measure properties such as adhesion (100% based on ASTM D3359), thickness (varying between 0.5 and 10 µm, depending on the application), brightness (based on ASTM D523), humidity resistance test (100% adhesion after 10 days based on ASTM D2247), salt spray test (based on ISO 9227), UV resistance (< 10% brightness loss after 10 days exposure based on ASTM D4587) and thermal cycle tests. Samples should not show any signs of staining, matting, cracking, bubbling of peeling after the tests.

Deliverables	
D6.1	Report on the results on the demonstration activities. The report will include the PoP process for the demonstration, characterization profiles of the coatings and QC tests of end users **Delivery: M48** by **CRF**
D6.2	Public report on the demonstration activities that will include non-confidential results targeting on the functionalities of new coating solution and their wider impact **Delivery: M48** by **PoliMI**

Work Package Number	7										**Lead Beneficiary**	PoliMi
Work Package Title			**Roadmap to market deployment**									
Nr.	1	2	3	4	5	6	7	8	9	10	11	12
Name	Gaser	PoliMi	Cnano	AUTH	IDE	IRIS	ICCR	EXE	INC	ARC	IAI	CRF
PMs												
Start Month		1									**End Month**	48

Objectives

WP07 has the following specific objectives:
1. To facilitate the acceptance and utilization by the market of the developed processes and materials
2. To provide benchmark information for the activities of other WPs, ensuring compatibility with standards
3. To use standardization system as a tool for dissemination of the project results and interaction with market
4. To develop a decision support tool facilitating the SSBD PoP technology by the industry

Description of Work

Task 7.1: Implementation of standardization activities (Task Leader: PoliMi), M01-48
Supported by: All partners
By M6 a benchmark analysis of standard and regulations will be performed, based on the needs of the projected research activities and the expected results. Contribution to the ongoing and future standardization developments will be followed after M7, allowing the project to interact with the related standardization technical committees, assessing they type of communication and data share (monitoring their information, attending to TC meetings, establishing formal liaisons, organizing joint events, etc.). FreeMe will also contribute to new standards developments in the field of surface treatment and PoP, promoting the inclusion of the outcomes of the project in new or future standards that can be easily used by the European or international industry, increasing this way the impact of the project. Based on the progress of the standardization sectors at the moment, this can involve providing information, participating in ongoing works, submitting technical manuscripts and promoting the development of new standard documents.

Task 7.2: Development of DST (Task Leader: EXE), M01-48
Supported by: All partners
The core of DST will consist of a dynamic superstructure, whose elements enable representation of technologies and processing steps, their connectivities as well as materials flows, concentrations, and products specifications. In-silico models of the individual phenomena are integrated to simulate the different processing steps of the superstructure. Experimental data (from project or user-defined) will systematically be processed for data-driven analysis of flows, concentrations and specifications (on matrices and products) of the superstructure with respect to outputs in operations, safety demands, technologies qualification and products quality control standards. The LCA module holds an environmental factors database for KPIs analysis. The LCC module includes fixed (by project) and incoming (by user) economic data on technologies, chemicals, operating and capital costs that are processed to

calibrate cost models and systematically incorporate techno-economic analysis in the optimization procedure. The safety module will be structured by risk factors, methods and descriptive data for risks evaluation and assessment of the simulated operations (based on user-defined inputs) and will offer lists of options to mitigate hazards across the design process consulting industrial users. The overall tool will be constructed as an optimization problem to be programmed with GAMS modelling environment and will hold a user-friendly interface (GUI) to enable interaction with the user. The DST will be provided as a stand-alone tool to use for decision making on the selection and design of best practices for plating on plastics.

Deliverables	
D7.1	Report containing the activities undertaken by the consortium towards the standardization of new processes and liaison with corresponding stakeholders, **Delivery: 48,** by **PoliMi**
D7.2	Delivery of DST ready to be used by the industry **Delivery: 48**, by **EXE**

Work Package Number			8								**Lead Beneficiary**	EXE
Work Package Title			**Extroversion and Exploitation activities**									
Nr.	1	2	3	4	5	6	7	8	9	10	11	12
Name	Gaser	PoliMi	Cnano	AUTH	IDE	IRIS	ICCR	EXE	INC	ARC	IAI	CRF
PMs												
Start Month		1									**End Month**	48

Objectives

WP 8 has the following specific objectives:

1. To coordinate of the communication and dissemination activities for the project´s outcomes and results targeting the identified stakeholders and target groups and maximizing, thus, the project´s visibility.
2. To define and implement the exploitation strategy based on the key results identified.
3. To conduct IPR management seminars to jointly seek ownerships and joint exploitation potentials.
4. To carry out training activities responding to the needs and specific requests of identified stakeholders.

Description of Work

Task 8.1: Dissemination & Communication (Task Leader: EXE), M01-48
Supported by: All partners

The FreeMe dissemination strategy will ensure an efficient project visibility and that communication actions are measurable, in order to divulge the project. The strategy will also provide recommended channels for scientific dissemination, guidelines, graphics, and templates to allow a coordinated dissemination of results and news. Dissemination tools (website, brochures, newsletters, posters, videos, press releases etc.) EXE will work closely with all partners to identify, develop and publish dissemination materials. The target is to reach multiple stakeholders, raise their awareness, and encourage the engagement of specialized and general audiences. More specifically, it will include (A) the development of a project identity/brand (logos, colours, font sets, banners, templates etc.), Q&A documents, press releases, infographics, flyers and similar resources to support partners in publicizing the project (B) the establishment of a project website from M3 containing information on the project (weekly progress updates). Social media channels including professional networks (e.g. LinkedIn) and public/ social network (e.g. Facebook, Twitter) will be established at M1 to facilitate the project´s promotion. The preliminary D&C plan has been included in Section 2.2 and a deliverable will be submitted in M6, this will be updated for each reporting period.

Task 8.2: Exploitation and business cases (Task Leader: INC), M01-48
Supported by: All partner

The exploitation and commercialization strategy will comprise different phases such as product identification, market analysis, preparation of product launch and strategic alliances. The business cases will be analysed versus market landscape of each result, definition of values proposition, market competition and value creation (value change, customer relationship, etc.). A financial analysis will be performed to evaluate the profitability of the revenue model, understand future cash flow and highlight the need for additional funding or external investment to reach the first stage. Focus will be given on marketing activities to support the future exploitation of the FreeMe results, to assist in raising the awareness and interests of potential investors to contribute to the financing and growth of the project results, post-project training, dissemination campaigns to the European industry through major exhibitions and shows, additional demonstration work with industry, communication with policy makers and standardization bodies, etc. Knowledge seminars will be organized for the identification, distribution, and representation of knowledge generated. Finally, Freedom to Operate (FTO) assessment will be conducted for the technologies and the results arising of the project to avoid the infringement of third-party intellectual property rights, and to plan the necessary mitigation strategies.

Task 8.3: Open Science and Research Data Management (Task Leader: PoliMi), M01-48
Supported by: All partners

A concrete and sustainable IPR strategy will be designed following the exploitation strategy towards the Open Science framework. Crucial to this effort will be the development of a Data Management Plan (DMP) that will incorporate the technical and business knowledge produced during the course of the project. It is foreseen that in the first version of DMP will be delivered in M6 and will updated for each reporting period. PoliMi will take responsibility for the collection, management, and sharing of the research data, as well as for the day-to-day quality assessment. The associated data files will be captured using Office software and will be analyzed using appropriate analytic tools. Finally, data will be maintained in an open XML format to enable open re-use of the data. Over the course of the project, data will be collected and entered into databases as described in Section 1.2. Project results and Public Deliverables will be posted on the project website within one month. The site will be established using an appropriate content management system (Drupal or Joomla, etc.) so that data users can participate in adding site content over time, making the site self-sustaining. For preservation, the coordinator will supply periodic copies of the data to "repository". Data available and cited in publications will follow Open Access principles.

Task 8.4: Training Activities (Task Leader: EXE), M01-48
Supported by: All partners

The objective of this task is organizing and carrying out a number of training/dissemination workshops. For this purpose, a set of materials, like training manuals and guidelines will be developed. A database of videos and photos will be created, with the assistance of all partners, to facilitate training. The training workshops will be organized for two different targets, internal training (aimed at staff of the project partners) and external training. At least two (2) international and two (2) national training/dissemination workshops are planned. A final workshop with at least 50 participants will be implemented in M35. EXE will introduce a training module related to the project results into their standard training packages for plating shops. A final event will be organized, in order to gather all the industry stakeholders from several sectors and present the main results and expected gains of its deployment for the targeted markets, providing for a faster acceptance and integration of the project results.

Deliverables	
D8.1	A report containing the Plan for the Dissemination Exploitation of Results and Communication activities (PEDRC) including training activities **Delivery:M6** by **EXE.** The deliverable will be updated at **M18 (D8.3)**, **M36 (D8.4)** and **M48 (D8.5) by EXE**
D8.2	Data Management Plan **Delivery:M6** by **PoliMi**
D8.6	Report containing the dissemination material and the open training material developed by the partners **Delivery:M48** by **EXE.**

Work Package Number	9										**Lead Beneficiary**	Gaser
Work Package Title			**Project management**									
Nr.	1	2	3	4	5	6	7	8	9	10	11	12
Name	Gaser	PoliMi	Cnano	AUTH	IDE	IRIS	ICCR	EXE	INC	ARC	IAI	CRF
PMs												
Start Month		1									**End Month**	48

Objectives

WP09 has the following specific objectives:

1. Establishment and follow-up of Consortium Agreement and implementation of Grant Agreement
2. To overview the financial management. Responsibility for the communication within the consortium.
3. Controlling of project advance and time schedules. Review of milestones and deliverables.
4. Organization of regular reports and generation of special mid-term reports & summary reports.

Description of Work

A management team has been defined for the proper monitoring of the project. This consists of:
Mr Enrico Galliani from Gaser that will be the Project Manager (PM). Together with his management team will be responsible for the coordination activities and the relationship with the EC and the FreeMe partners.
Prof. Luca Magagnin from PoliMi will be the Technical Manager following the research and development activities of the technical WPs and communicating with WP and Task leaders.
Mrs Laura Zacares from Incote will be the Innovation Manger responsible for monitoring of the Innovation Strategy of the Project
Dr Ioanna Katsavou from Exelisis will be the Dissemination & Communication Manager.

Task 9.1: Administrative and financial project management (Task Leader: Gaser), M01-48

Supported by: All partners
Gaser's Management and Coordination Tasks include: * Communicating between consortium and EC, including the provision of legal and financial information about the partners during the GA and CA preparation phase. * Administrative, financial, legal, and ethical management of the consortium and the resolution of contractual issues during the project. * Manage and define IPR issues in order to submit by the end of the project the Innovation Radar. * Preparation and post-processing of EC reviews from the consortium side, including support in the implementation of any recommendations from the EC and reviewers, with the support of the WP leaders. * Coordination of payments and the distribution of finances; keeping of records and financial accounts, collation of financial statements. *Day-to-day communication with FreeMe project partners, organizing meetings (including the kick-off and six-monthly meetings) and circulating minutes. * Ensuring gender equality and Open Science, collating of the deliverables and milestone reports, and collaborating in strong ties with partners to effectively prepare progress and periodic reports and timely submit deliverables.

Task 9.2 Technical project management (Task Leader: PoliMi), M01-48
Supported by: All partners
Prof. Luca Magagnin from PoliMi in collaboration with the project coordinator Mr. Franzosi and each WP leader will monitor the technical delivery of all Work Packages applying a day to day management of technical project activities and coordinate activities between Work Package Leaders. A risk log and milestone progress reports will be developed and maintained. The work package leaders will report directly Prof. Magagnin and will be responsible to coordinate the work package specific technical activities and review and manage the according deliverables, milestones and risks. The deliverables will follow an internal peer review process where a partner will be assigned to evaluate the quality of the deliverable versus its objective. Each deliverable will be ready as a draft version one month before the submission and the partners will have time to comment in order to ensure the quality. May results be available for open access publications the papers will be circulated 40 days before its submission in order to receive approval by the partners.

Deliverables

The reporting of this WP will be done in the contractual periodic reports.

Table 3.1c: *List of Deliverables*

Deliv. Nr.	Deliverable name	WP Nr.	Lead Partner	Type	Diss. level	Deliv. date
D1.1	Report on specifications.	1	Gaser	R	SEN	6
D2.1	Recommendations for SSbD products and processes A	2	ICCRAM	R	SEN	18
D2.2	Recommendations for SSbD products and processes B	2	ICCRAM	R	SEN	36
D2.3	Report on assessment of potential impacts and environmental sustainability	2	ICCRAM	R	SEN	48
D3.1	Intermediate report of progress of the activities of WP3	3	PoliMi	R	SEN	18
D3.2	Report including the resins' formulation and optimized methods for application on polymer surfaces	3	AUTH	R	SEN	36
D3.3	Report on the optimized activation and metallization processes of sprayed resins	3	Gaser	R	SEN	36
D4.1	Intermediate report of progress of the activities of WP4	4	Cnano	R	SEN	18
D4.2	Report on toxic-free wet process for metallization of plastic surfaces	4	Cnano	R	SEN	36
D5.1	First report of progress of the activities of WP5	5	IDE	R	SEN	18
D5.2	Second report of progress of the activities of WP5	5	IRIS	R	SEN	36
D5.3	Multi-scale models for environmentally friendly PoP	5	IDE	OTH	SEN	48
D5.4	Guidelines on in silico models usage for the design of new PoP processes	5	IDE	R	SEN	48
D5.5	Development, installation and validation of in-line optical inspection system	5	IRIS	DEM	SEN	48
D6.1	Report on the results on the demonstration activities.	6	CRF	R	SEN	48
D6.2	Public Report on demonstration activities	6	PoliMi	R	PU	48
D7.1	Report on standardization activities	7	PoliMi	R	PU	48
D7.2	DST webplatform	7	EXE	OTH	PU	48
D8.1	First PEDRC	8	EXE	R	SEN	6
D8.2	Data management Plan	8	PoliMi	R	PU	6
D8.3	Second PEDRC	8	EXE	R	SEN	18
D8.4	Third PEDRC	8	EXE	R	SEN	36

D8.5	Final PEDRC	8	EXE	R	SEN	48
D8.6	Dissemination and training material	8	EXE	OTH	PU	36

Milestones of FREEME

The Project Milestones have been targeted planned in order to be used as a valuable tool for evaluating critical performance and determining whether or not the project is on schedule, these additionally have been projected to cover all reporting periods and to be balanced split between them. Milestone reviews will be held at each meeting of the PSC six monthly and a critical review of the outlook for effective exploitation of results will be monitored by the partners. Successful achievement of the targets specified in the milestones will ensure a smooth project implementation and the on-time planning.

Table 3.1d: *List of FREEME milestones*

No	*Milestone name*	*WP*	*Date*	*Means of verification*
MS01	Kick off meeting	1	1	Meeting minutes approved by partners
MS02	Definition of REACH compliance baths chemistries	2	5	Chemical to be used in baths and Safety Data Sheets approved by the consortium
MS03	Integrated safe- and sustainable-by-design approach	3	10	Common evaluation framework applicable to the case studies
MS04	First successful UV-curing of bio-based composite resin in lab	3	14	Report on resin formulation, UV-curing protocol and characterization of hardened resin
MS05	Utilization of spray apparatus for application of resin in substrates	3	19	Video from application of resins in the plastic samples
MS06	Successful activation utilizing Pd-Free baths	4	25	Characterization via SEM and EDS measurement of Ni nuclei and Ni-P laye
MS07	Achievement of adhesion value of 4,9 N/25mm	3-4	29	Via peel-test configuration, coatings of both routes will be studied
MS08	Delivery of demonstrators in Cnano, Gaser and PoliMi premises	6	33	Delivery notes from Arc, CRF and IAI
MS09	Recommendations for the implementation of SSbD strategies presented in workshop	3	36	Report on the materials presented and minutes of meeting from the round table discussion
MS10	First successful application of resin based POP process on automotive part	6	40	Full characterization profile of the coating based on end user's QC protocols
MS11	First use of beta version of DST by external user	8	42	Report on the feedback from applicants
MS12	Open event for public engagement & Training workshop for industry	9	47	Press release on media for the implementation of the events and interviews from attendees

Risks and Migration Measures

All necessary risks and migration measures have been dully foreseen, categorized in terms of managerial, technical and Business/ Exploitation and dissemination. Management related risks will be evaluated and monitored by the PC. In more details, the table below shows the main risks identified, their probability of occurrence, these risk table is intended to be reviewed and the status will be evaluated during each project meeting.

Table 3.1e: *Critical risks for implementation (where L: Likelihood, I: Impact)*

Description of risk	WP(s)	Proposed risk-mitigation measures
Management Risks		
Management issues **L**: low. **I**: high	All	The proposed working groups that will be chaired by the PM and TM, have relevant representatives of the most relevant disciplines to assure the required leadership.
Technical Risks		
Lack of data from the test cases to model the sustainability performances **L**: medium **I**: medium	2	Bilateral meetings with relevant partners to fill data gaps, use of literature data and sensitivity analysis on key uncertain parameters to understand the influence on the results
UV-cured resin layers result to limited Ni nuclei and thus, low quality metallization **L**: medium **I**: high	3	Instead of using anhydrous salts (i.e., $NiCl_2$), mixing of Ni nanoparticles with the resin will be tested. Following, an etching process will be applied to expose Ni NPs to the surface which will act as Ni nuclei for the electroless deposition.

Insufficient etching process via the wet route based on piranha sol. **L**: medium. **I**: medium	4	This risk has been foreseen and is specifically connected to the complexity of wet chemistry of the etching process. Use of stronger mixtures (i.e. KMnO4/H2SO4) will be utilized.
Pd-free activation process results in limited Ni nuclei **L**: medium **I**: medium	4	A revising of both etching and activation step will be made as they are directly connected. Utilization of more dense Ni solution or of copper salts solution.
Required models too complex or computationally intensive **P**: medium **I**: medium	5	Targeting of the modeling activities. Usage of HPC/Cloud services if needed. Usage of surrogate models when required
Final finishing coatings are of low quality **P**: low **I**: medium	3-6	The quality of the final finishing is based on the quality of the metallization step (application of Ni-P layer). Usually this defect is solved by application of Ni-P layers of higher thickness (with a low impact on the duration of the process)
Results on demonstration activities do not correspond to lab scale results **P**: high **I**: low	6	During demonstration activities, larger scale components will be used, having as a result the need for a more robust operation and control of parameters. The presences of Gaser in the consortium that has huge experience in plating process will be valuable.
Difficulty in matching user defined input data with data required by DST. **P**: medium **I**: low	7	Missing input data will be completed by experimental (generated by the project) and data-driven modelling outputs.
Industrial implementation/Business/ Exploitation and dissemination risks		
Internal disputes regarding IPR and exploitation routes **P**: low **I**: high	WP8	CA will define the IPR issues, while at the end of the project the Results Ownership List will be established based on Horizon Europe principles for ownership of results

Summary of staff effort

The consortium set up by FreeMe has a clear vision of its objective and therefore is committed to the right allocation of the financial resources. There is an appropriate budget allocated to R&D work, communication, dissemination, IP and knowledge management and business planning activities for impact. Considering that there are 12 partners in the project, and the competitive 48-month work plan, an EC Contribution of **4.834.592,00 €** is requested.

Table 3.1f: *FREEME Summary of staff effort*

	Name	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	WP9	Total PMs
1.	**Gaser**	**7**	5	18	21	2	13	4	3	**28**	**101**
2.	**PoliMi**	4	11	**31**	0	2	9	3	10	6	**76**
3.	**Cnano**	4	5	0	**35**	3	15	6	5	2	**75**
4.	**AUTH**	2	2	39	0	3	4	3	2	2	**57**
5.	**IDENER**	2	5	3	3	**40**	2	10	3	2	**70**
6.	**IRIS**	3	2	2	2	25	7	2	2	2	**47**
7.	**ICCRAM**	2	**54**	0	0	2	2	2	2	2	**66**
8.	**EXE**	1	4	0	0	0	2	**25**	**30**	2	**64**
9.	**INC**	3	3	0	0	0	5	13	40	2	**66**
10.	**CRF**	6	2	2	3	3	**15**	3	3	2	**39**
11.	**Arcelic**	4	4	3	2	1	15	3	2	2	**36**
12.	**IAI**	4	3	2	2	1	11	2	2	1	**28**
Total PMs		**42**	**100**	**100**	**68**	**82**	**100**	**76**	**104**	**53**	**725**

In the below Figures the other directs costs are analysed for the partners that exceeds the 15% of other direct costs compared to the personnel costs.

Table 3.1h: *''Purchase costs' items (travel and subsistence, equipment and other goods, works and services) and 'Other costs categories' items (e.g. internally invoiced goods and services)*

3. Cnano	Cost (€)	Justification
Equipment	14.000€	Peel test for adhesion measurements: 14.000€
Remaining purchase costs (<xx% of pers.	48.000€	

Total	**62.000€**	

4. AUTH	**Cost (€)**	**Justification**
Equipment	25.000€	Reactor for monomer synthesis/polymerization and parts of it: 20.000€ Components of analytical instruments: 5.000€
Other goods, works and services	14.000€	Consumables and raw material (chemical reagents/gases): 8.000€ Services for analytical measurements: 6.000€
Remaining purchase costs (<xx% of pers.	28.000€	
Total	**67.000€**	

6. IRIS	**Cost (€)**	**Justification**
Other goods, works and services	53.000€	Cost for the manufacturing and integration of sensor (optical component, mechanical supports etc): 53.0000€
Remaining purchase costs (<xx% of pers.	18.000€	
Total		

7. ICCRAM	**Cost (€)**	**Justification**
Other goods, works and services	15.000€	Toxicology Lab consumables: commercial cell lines, cell culture reagents and consumables; microbiology reagents and consumables; buffers and plastic and glass material for laboratory; kits and consumables for toxicity and field assays: 15.000€
Remaining purchase costs (<xx% of pers.	33.250€	
Total	**48.250€**	

Table 3.1i: 'Other costs categories' items (e.g. internally invoiced goods and services)

Participant Number/Short Name		
7. ICCRAM	**Cost (€)**	**Justification**
Internally invoiced goods and services	5.000€	The R+D+i service - Technological Park - of the University of Burgos passed on the researchers, based on usage, all expenses associated with their teams, among which are: NMR, FT-IR, elemental analyzer, electron microscopy scanning, ICP-Masses, High-resolution masses, RAMAN, etc.

3.2 Capacity of participants and consortium as a whole

3.2.1 Consortium as a whole

FreeMe presents a highly interconnected network of partners that will act towards the sustainable development of three different end user applications, which have been selected due to the fact that they are representing a wide spectrum of sectors interested in the project results. The partners are distributed across 5 Countries (Italy, Spain, Greece, Israel, and Turkey). The partners will have a unique role in the consortium and synergies will be created between them across the project implementation. The consortium is characterized as highly industrial driven having in its partnership **6 SMEs including the coordinator**, **3 Large Enterprises and 3 Universities**. More specifically:

Gaser: An SME in the north of Italy member of a group of companies acting in the field of surface finishing. Gaser has long tradition in development of proprietary technologies via in-house research activities. It has a long-standing collaboration with PoliMi and has collaborated also closely with Cnano. It role will be vertical spanning from synthesis of chemical baths and up to application of the FreeMe technologies and validation of coatings.

Cnano is Greek a SME that is well know on the optimization of industrial surface treatment processes, offering multiple solutions on every day troubleshoots of plating facilities and develop customized plating methods. The company has been working the last 5 years on 3D Printing developing PoP protocols and plating baths formulations. Cnano is active also in the art & design sector, collaborating with famous artists (i.e. Costas Gaitis) for applying coatings in artefacts. Finally, Cnano has research agreements with Gaser and PoliMi.

IDE is an IT company offering solutions to materials modelling, member of EMMC. It has collaborated with Gaser, PoliMi and Cnano in the framework of PureNano project providing optimization models. Through this collaboration IDE has developed knowledge in the demanding field of surface finishing, something that will useful for the FreeMe implementation.
IRIS has long experience in monitoring technologies. It develops monitoring and inspection systems for process QC and IoT-Industry 4.0 concepts and its systems are integrated in various production facilities, mainly related to plastic processing.
EXE is an engineering company developing customizable solutions, such as decision support tools, for industrial end-users. EXE has a strong background in process design, simulation and mathematical programing that are properly integrated towards the construction of optimization tools for decision-making and support. For this purpose, EXE utilizes modelling environments, like GAMS software, for the development of DSTs, which offer user-friendly interfaces and multiple-purpose services.
INC is a consultancy company with experience in capitalizing research results, developing business plans and commercialization roadmaps. It will have an important role in the project providing guidelines for in-time actions related to market adoption.
The three large enterprises are very well positioned in their fields. A more analytical description is in paragraph 3.2.2. Their role in the project will be of high importance providing specification and validating the technologies based on industrial standards.

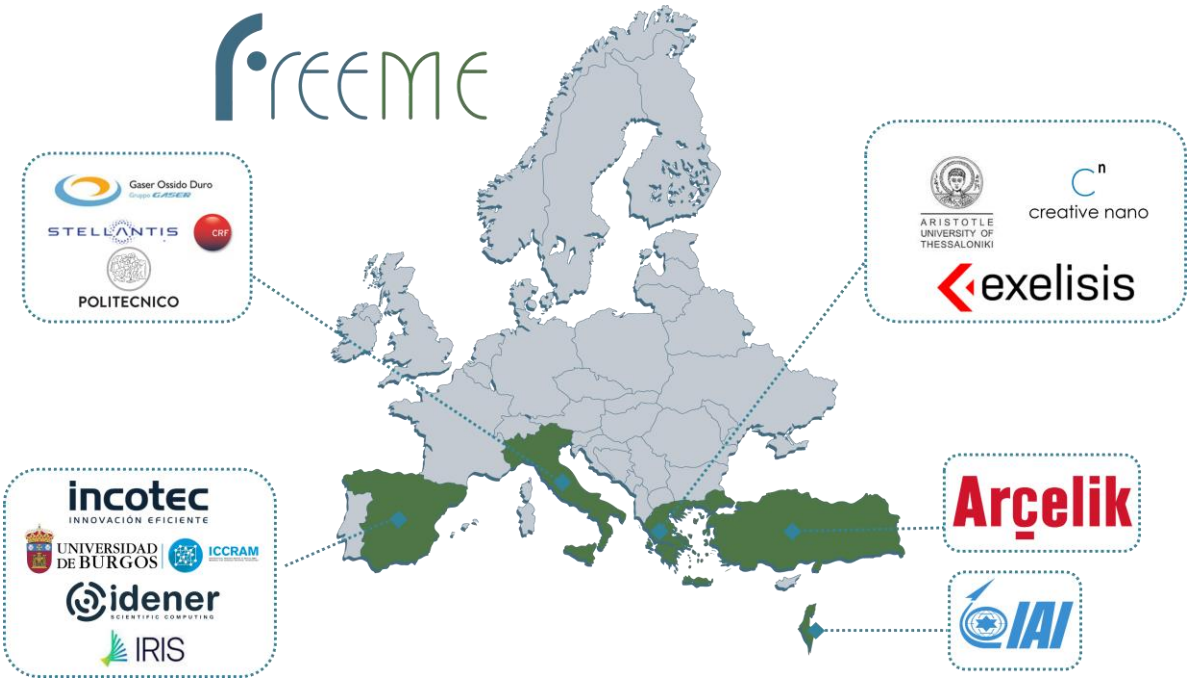


Figure 3.2: FreeMe partnership

The three Universities participating in the project are well established in their corresponding fields:
PoliMi has a more than 50 years tradition in the surface and finishing industry and is point of reference for plating process across Europe and more importantly in the North Italy that provide research services and consultancy to companies. It has collaborated in the past with most of the partners of the consortium in previous EC projects. Its role is crucial for the development of the technologies.
AUTH's team has a world recognition on developing bio-based polymers and resins with more than 250 publications in the relevant field the last 10 years. It has strong experience in collaboration with the industry providing research activities design of new chemical formulations.
Finally, **ICCRAM** has developed a high expertise in the field of safety and environmental assessment. It coordinates a number of relevant projects, is member in EC initiatives about safety and sustainability. It will lead the relevant activities in the project.

The nature of the complementary interrelations within the work structure of the proposed work, arises from the interdisciplinary contributions, described in sections 1.2 and 3.2 In Figure 3.3, the key expertise for project implementation brought by each partner is presented. The consortium covers all aspects for the development of the technologies having main contributors and partners with supplementary knowledge and expertise.

Main Area of Expertise Secondary Area of Expertise	GASER	PoliMi	Cnano	AUTH	IDE	IRIS	ICCRAM	EXE	INC	CRF	ARC	IAI
Plating on Plastics												
Polymers and resins												
Safe by design (toxicology)												
Development of treatment baths												
Monitoring Systems												
Material Characterization and Validation												
Recyclability assessment												
End user on Automotive												
End user on Home Appliances												
End user on Aerospace												
Decision Support Tool												
Ecodesign, LCA & LCC												
Exploitation, Business plan, D&C												
Dissemination and Communication												

Figure 3.3: Consortium complementarity table

3.2.2 Access to critical infrastructure

The consortium members conduct innovative research in their own sub-disciplines for which they are recognized. They have all the necessary expertise and equipment to carry out their tasks within FreeMe. The table below presents the infrastructure and important instrumentation brought in the project by the partners. The consortium has an autonomous capacity to implement this project.

Table 3.3: *Partners Infrastructures*

Partner	Infrastructure
Gaser, Cnano, PoliMi	Plating baths up to 1000 L, Pilot Lines of Plating baths (1-120L)
PoliMi, AUTH	Chemistry and analytical laboratories, Polymer production and Compounding, reactors for most relevant polymer/composites production technologies present in the industry
Cnano	Characterization facilities (FTIR, UV-Vis spectrophotometer, hardness, contact angle, DSL
IRIS	Spectroscopic Laboratory, Optical photonics-based technologies and PAT systems, Electronics, mechanical and optical workshops and laboratories
AUTH	Polymer production and Compounding (reactors representing the most relevant polymer/plastics/composites production technologies present in the industry)
PoliMi, AUTH	SEM, XRD, XPS, profilometry, Raman, AFM, NMR, Spectrophotometers, gonioreflectometer, X Ray Fluorescence (XRF and micro-XRF) X Ray Diffraction (Micro XRD Spectroscopy)
CRF, IAI, ARC	Products validation: aging and reliability, Reliability chambers: UV , ° C, %RH cycling, salt spray, cyclic corrosion, CASS test Mechanical tests: abrasion, adhesion, stone chipping, dynamometer, Charpy, Izod
ICCRAM, IRIS, IDE	Toxicology testing, chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes).
IRIS, EXE IDE, ICCRAM	Big data storage (software/modelling tools), Matlab and Python toolboxes, GAMS
ICCRAM	LCA software Simapro 9.2, LCA,

3.2.4 Role, adequacy and resources of each partner

FreeMe has been designed with a clear set of objectives surrounded by the Safe and Sustainable Plating on Plastics Industry, aligned with the Work Programme topic description and the Destination and directly transferred to the work structure as the primary and only objectives of each work package. The consortium has been built with distinct roles to be undertaken but ensuring complementarities will be exploited and that the final outcome will be the delivery of the exact project objective. Each partner has evaluated their role and assessed in detail the work to be carried out, the lead as well as the supporting roles to be undertaken, and the requirements arising by the above. The contribution requested by the EC is 4.834.592,00 € € for a total of 12 partners, which can be considered as profitable ratio of investment over expected impact. As an indication of the high potential of the proposed work should be considered the ambitious post-project exploitation planning. End-users will have to undergo a 5 years technology uptake plan followed by a certification acquirement procedure that comes with costs in the range of millions that will be own investments, the SMEs will additionally apply a request to the Horizon Results Platform

(HRP) building a bridge with private investors.

3.2.5 Industry involvement and commercialization potential.

FreeMe is a highly industrial driven project led by an SME (Gaser) belonging to a group of companies (Gruppo Gaser) with established presence and commercial networks in the field of surface finishing, including plating on plastic processes. The commercialization potential of FreeMe project is consider as significantly high as it deals with a real industrial need that seeks an urgent solution. Three large enterprises, point of reference in their sectors, have joined the consortium to assist with their industry driven innovation philosophy the fast uptake of the results. Stellantis (CRF) is a top automotive group of companies having in its portfolio world recognized brands. Arcelik, is a major manufacturer and provide of home appliance and other consumer goods while Israel Aerospace Industry is a world recognized innovation in the field of aerospace. All three companies are currently operating either in-house or via their subcontractors chromium acid etching and Pd based activation steps. Moving to safer processes (both from the side of occupation health and environmental protection as well as from the side of securing supply chains via exclusion of Pd use) will give them a significant competitive advantage for the highly aware consumers.

Their role in the project will be of high importance for the development of the technologies. They will provide industrial based specifications and high standards, while they will follow and properly guide the research activities towards achieving thes standards.

Finally, it should be highlighted that in FreeMe consortium the whole supply chain of PoP is formed, covering materials suppliers (i.e Gaser), plating shops (i.e. Cnano and Gaser) and end users (i.e. CRF, ARC and IAI). This will allow a holistic view of industrial spec and requirements across the value chain, providing valuable input in the research activities.

This proposal version was submitted by **Enrico GALLIANI** on **23/09/2021 16:36:21** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

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